

UNIVERSIDADE ANHEMBI MORUMBI

VANESSA GONÇALVES DE ALMEIDA ROSA

**ART, DESIGN AND MATHEMATICS:
IMAGES, ALGORITHMS AND CULTURAL DIVERSITY**

Arte, Design e Matemática: Imagens, Algoritmos e Diversidade Cultural

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Design. Approved by the following Examining Board:

Prof. Dr. Gilberto dos Santos Prado

Advisor — Universidade Anhembi Morumbi

Prof. Dr. Hermes Renato Hildebrand

Pontifícia Universidade Católica de São Paulo

Profa. Dra. Priscila Almeida Cunha Arantes

Universidade Anhembi Morumbi

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TRANSLATOR'S NOTE

This is a complete English translation of the master's dissertation *Arte, Design e Matemática: Imagens, Algoritmos e Diversidade Cultural*, presented by Vanessa Gonçalves de Almeida Rosa to the Graduate Programme in Design at Universidade Anhembi Morumbi, São Paulo, in January 2018, under the supervision of Prof. Gilberto Prado. The translation was prepared in 2026.

A few conventions adopted throughout:

Quotations. In the original, quotations from English-language sources appear in the body in the author's own Portuguese rendering ("tradução livre da autora"), with the original English given in a footnote. In this edition those quotations have been restored to their original English wording in the body text, and the corresponding translation footnotes have been omitted. Quotations from French sources (notably Yaovi Akakpo) have been translated into English from the author's Portuguese rendering, checked against the French originals, which are retained in the footnotes as in the original. Substantive footnotes — those that carry content rather than translations — have been kept and translated, and all notes have been renumbered consecutively.

Figures. All 58 figures are reproduced from the original document, in their original positions within the text. Captions have been translated; source credits are kept as in the original.

Figure references in the Introduction. In the original, the opening paragraphs of the Introduction refer to "Fig. 1" and "Fig. 2" where the figures described are, by the numbering of the List of Figures, Fig. 2 (the painting *Padrões*) and Fig. 3 (the CSDT screenshot); Fig. 1 is the "sona" drawing accompanying the epigraph. These in-text references have been silently corrected to match the figure numbering.

Terminology. "Matematização do mundo" is rendered as "mathematization of the world"; "caixinhas de infinito" as "little infinity boxes"; "curva-de-espelho" as "mirror curve", following Paulus Gerdes's own English usage. Portuguese, French and German titles of works are given as in the original, with English translations where the author provided them.

Abstract. The original dissertation includes both a Portuguese *Resumo* and an English *Abstract* written by the author. The author's own English abstract is reproduced here unaltered.

ABOUT THE AUTHOR

Vanessa Rosa completed her undergraduate degree in Art History at UERJ in 2012, having spent an exchange period at Université Paris 8. She has worked as a visual artist since 2010, taking part in group and solo exhibitions as well as a variety of projects concerned chiefly with public art and multimedia, in South America, North America, Europe and Africa. She also works as an illustrator and art director at Editora Viajante do Tempo, of which she is one of the owners.

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CDD 306

DEDICATION

*I dedicate this work to my sister, Diana Rosa,
who is always present even in her absence*

ACKNOWLEDGEMENTS

To my advisor Gilbertto Prado, my parents Regis and Regina, my brother-in-law Angelo D'Elia, my project partner Verônica Natividade, and my dear friends who made this undertaking possible with all their support, especially Diana Stephan, Leonardo Guimarães and Tatiana Cordoniz. Thanks are also due to Anhembi Morumbi, for granting the institutional scholarship that made this work possible. And, of course, to the Brazilian people, who make this kind of research viable.

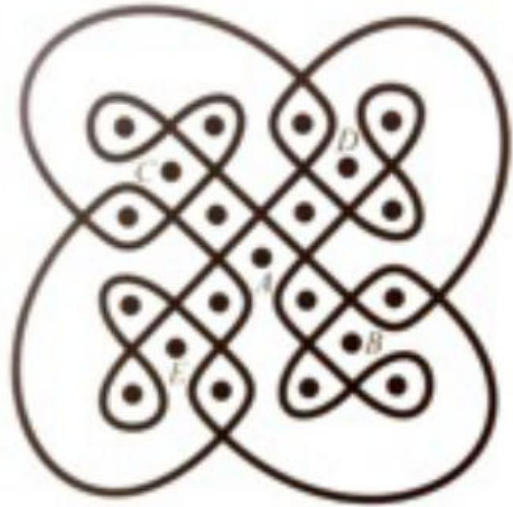


Fig. 1: "Sona" drawing. Source: GERDES, 1999, p. 169

EPIGRAPH

"Sambálu, the rabbit (positioned at point B), discovered a salt mine (point A). Immediately, the lion (point C), the jaguar (point D) and the hyena (point E) came to demand possession and affirm the rights of the strong. Then the rabbit quickly made a fence to isolate the mine from all usurpers [in fact, only from B can one go to point A, without going beyond the line that represents the fence]. The fable affirms the inviolable rights of the weak." (GERDES, 1999, p. 163)*

* Cokwe tale represented in Figure 1 (GERDES, 1999, p. 169)

ABSTRACT

[The author's own English abstract, reproduced from the original. — Trans.]

This dissertation discusses the issue of algorithms mainly from their characteristics as a representational model, in other words, tool to create images. It's an artist-oriented approach, how it is possible to express oneself through algorithms, but also designer-oriented, how one can project with them. First, we try to understand the specificity of algorithmic images within the history of representation, why they mean a paradigm shift in relation to previous representational models, their place within the congruence of art history, or rather, history of images, with the history of science. Second, it is assumed that it is essential to discuss trends in imagery and technology development through an approach that focuses on cultural diversity, considering such approach a political necessity to comment and, as far as possible, to influence the changes that are occurring globally. Therefore, a combination of several fields of knowledge, articulated around concepts such as "representational models", "algorithms", "mathematization of the world" and "cultural diversity" are presented. The research is also based on empirical experience, narrated in the second section of the last chapter with a case study. This research hopes to contribute to the expansion of studies on design, algorithmic processes of image and object creation that are based on a broad and multicultural view of mathematics. This is an ethical and historical contribution to such studies.

Keywords:

Design. Art. Mathematics. Algorithms. Cultural Diversity

CONTENTS

List of Figures

List of Abbreviations and Acronyms

Introduction

1. Models of Representation and the History of Science and Culture

1.1 Symbolic Forms, Art History and the History of Science

1.2 Thinking Linear Perspective beyond the Eurocentric Narrative

1.3 Mathematics and Cultural Diversity

2. Symbolic Forms and Algorithms — the Contemporary Image

2.1 The World of Software

2.2 The Algorithmic Image, Design and Models of Production

2.3 Algorithms and Pedagogical Proposals Oriented towards Multiculturality

3. Exercises in Algorithmic Design

3.1 Research and Applications of Ethnomathematics: Paulus Gerdes and the "Sona"

3.2 Case Study: Infinity Game — Artistically Experimenting with Multicultural Algorithmic Design

Final Considerations

Bibliographic References

Page numbers of the original Portuguese edition have been omitted, as pagination differs in this translation.

LIST OF FIGURES

Fig. 1: "Sona" drawing. Source: GERDES, 1999, p. 169

Fig. 2: *Padrões* (Patterns), canvas by Vanessa Rosa, 2017, acrylic and spray paint on canvas. Source: Personal collection of Vanessa Rosa

Fig. 3: CSDT developed by Ron Eglash. Source: <https://csdt.rpi.edu/projects/>

Fig. 4: Piero della Francesca (c. 1415–1492), panel *The Flagellation of Christ* (c. 1455–1460). Source: Wiki Commons

Fig. 5: Fresco by Pietro Perugino (c. 1448–1523), *Delivery of the Keys*, 1481–1482, Sistine Chapel, Rome. Source: Wikimedia Commons

Fig. 6: Symbols by date of introduction into the field of mathematics. Source: Wiki Commons

Fig. 7: Pablo Picasso, *Portrait of Ambroise Vollard*, oil on canvas, 65 × 92 cm, Pushkin Museum, Moscow, Russia. Source: Wiki Commons

Fig. 8: Alberti's winged eye, 16th century. Source: <http://www.thewingedeye.eu/>

Fig. 9: Lattice window, part of the conservatory of Amer Fort, city of Amer, India, built in the 16th century. Source: Pinterest

Fig. 10: Ivan Sutherland and the Sketchpad. Source: Wiki Commons

Fig. 11: The pixel grid in the Photoshop program. Source: Wiki Commons

Fig. 12: Photo of London's City Hall, a project by Foster and Partners. Source: Foster and Partners

Fig. 13: On the left, a representation of the carbon molecule known as 'buckminsterfullerene'. In the centre, a 'sepak raga' ball. On the right, a representation of the weaving technique. Source: <https://csdt.rpi.edu/culture/legacy/african/hex/intro.html>

Fig. 14: Main characters of the game 'Never Alone'. Source: E-Line Media

Fig. 15: The game 'Never Alone', graphics based on the traditional Alaskan technique known as 'scrimshaw'. Source: E-Line Media

Fig. 16: Geometric algorithm of the 'chased chicken' type in execution. Source: GERDES, 2010, p. 35–38

Fig. 17: Geometric algorithm of the 'chased chicken' type, completed. Source: GERDES, 2010, p. 38

Fig. 18: Images of the "sona" drawing representing friendship and the images of animals that are built from it by adding a few lines. Source: GERDES, 2010, p. 33–34

Fig. 19–21: "Sona" drawings following different algorithms. Each represents a story, action or animal/object. Source: GERDES, 1999, p. 162–166

Fig. 22: Representation of small double mirrors placed transversally between the points of the grid. Source: GERDES, 2010, p. 45

Fig. 23: Representation of the 'chased chicken' algorithm following the logic of mirror reflection. The rectangle enclosing the grid would also be mirrored. Source: GERDES, 2010, p. 44

Fig. 24: Representation of the 'chased chicken' algorithm with its mirror curves. Source: GERDES, 2010, p. 45

Fig. 25–26: Drawings similar to the "sona" in various cultures. Source: GERDES, 2000, 2010

Fig. 27: Drawings of the 'generalised mirror-curve' type and the positioning of their mirrors. Source: GERDES, 2010, p. 60–61

Fig. 28: Path of the line of the 'chased chicken' drawing being numbered and turned into a table. Source: GERDES, 2010, p. 50–51

Fig. 29: Representation of the count of the sums of the small squares applying modulo 8, and representation of how a modulo 4 count works. Source: GERDES, 2010, p. 54–55

Fig. 30: Representation of the count of a 'chased chicken' algorithm using modulo 4. Source: GERDES, 2010, p. 56–57

Fig. 31: Tamil drawing numbered with modulo 2. Light squares are placed for the number 0 and dark ones for 1, forming a pattern. Source: GERDES, 2010, p. 64

Fig. 32: Drawing mixing the types of mirror curve, creating a new kind of pattern. Source: GERDES, 2010, p. 65–66

Fig. 33: Representation of the properties of 'Lunda-designs'. Source: GERDES, 2010, p. 72–73

Fig. 34: Two mirror-curve drawings form the same 'Lunda-design'. Source: GERDES, 2010, p. 74

Fig. 35: Generation of fractals from Lunda-designs and mosaics. Source: GERDES, 2010, p. 84, 90

Fig. 36: Representation of the colour arrangement of Liki-designs. Source: GERDES, 2010, p. 94

Fig. 37: "Sona" drawing of a turtle and possible colourings for each of its constituent cycles. Source: GERDES, 2010, p. 99–100

Fig. 38: Representations of Liki-designs and the simulation of their multiplication following the rules of matrices. Source: GERDES, 2010, p. 112–113

Fig. 39: Hexagonal, polyhedral, circular-symmetric, three-colour Lunda-designs. Source: GERDES, 1999, p. 197–200

Fig. 40: Laser-cut stencils developed by Verônica's students. Source: Personal archive of Verônica Natividade

Fig. 41: Development of the fractal pattern from the Grasshopper code. On the right, two other stencils made for the project. Source: Personal archive of Verônica Natividade

Fig. 42: Photos of the final mural painting and its process. Source: Personal archive of Verônica Natividade

Fig. 43: Screenshot of the album on the social network Pinterest, showing part of the image references used in the project. Source: Personal archive of Vanessa Rosa

Fig. 44: Fractal logic of the Ethiopian Cross as presented in Ron Eglash's book *African Fractals*. Source: EGLASH, 1999, p. 136

Fig. 45: First pattern studies made in Rhinoceros 3D. One can see a mixture of patterns inspired by Christian, Islamic, Han (China), Ashanti (Ghana), Hindu (India), Cokwe (Angola) and Jewish cultures, and a playful nod to an illustration of an alien. Source: Personal archive of Vanessa Rosa

Fig. 46: Interaction of a pattern inspired by Ashanti culture with a Cokwe pattern. Source: Personal archive of Vanessa Rosa

Fig. 47: Fractals developed from a clover-type pattern. Source: Personal archive of Vanessa Rosa

Fig. 48: Fractals developed from an Ethiopian-cross-type pattern. Source: Personal archive of Vanessa Rosa

Fig. 49: Layout of a little infinity box, using the algorithmic design based on the Ethiopian cross. Source: Personal archive of Vanessa Rosa

Fig. 50: Little infinity box, design inspired by the Ethiopian Cross. Source: Wenqi Li

Fig. 51–52: Animations by Wenqi Li made in Processing. Each figure presents 4 different compositions that needed to be placed together in the line of code to ease the entry of multiple images into the Madmapper software. Source: Personal archive of Vanessa Rosa

Fig. 53: Installation of the "Infinity Game" exhibition. Source: Personal archive of Vanessa Rosa

Fig. 54: Detail of the installation of the "Infinity Game" exhibition, painting with video mapping. Source: Personal archive of Vanessa Rosa

Fig. 55: Layout of the installation composed of paintings on canvas. Source: Personal archive of Vanessa Rosa

Fig. 56: Light tests for the "Little Infinity Boxes". Source: Personal archive of Vanessa Rosa

Fig. 57: Demographic changes in the USA. Source: <https://newrepublic.com/article/120370/five-graphics-show-why-post-white-america-already-here>

Fig. 58: Little infinity box, design inspired by the Ethiopian Cross. Source: Personal archive of Vanessa Rosa

LIST OF ABBREVIATIONS AND ACRONYMS

BIM – Building Information Modelling

CAD – Computer Aided Design

CAM – Computer Aided Manufacture

CNC – Computer Numeric Control

CPS – Cyber-Physical Systems

CSDT – Culturally Situated Design Tools

FEA – Finite Elements Analysis

FEM – Finite Element Method

NURBS – Non-uniform rational basis spline

Pixel – Picture element

PLC – Programmable logic controllers

TCP – Transmission Control Protocol

INTRODUCTION



Fig. 2: *Padrões* (Patterns), canvas by Vanessa Rosa, 2017, acrylic and spray paint on canvas.
Source: Personal collection of Vanessa Rosa

In Fig. 2 there is a black-and-white symbol that stands out among the countless reddish patterns. It is called Funtunfunefu Denkyemfunefu, and it belongs to the set of pictograms known as Adinkra, a tradition of the Ashanti peoples of West Africa, found mainly in present-day Ghana (LARKIN; GÁ, 2009). It represents two Siamese crocodiles who share a stomach, and yet fight each other in order to feed. Although the food necessarily has the same destination regardless of which head swallows it, the crocodiles cannot live in peace. The Funtunfunefu signifies "unity in diversity"; it is a symbol for reflecting on the indispensability and, at the same time, the difficulty of democracy (Ibidem, p. 50). Fig. 2 is a photograph of a painting by the author of this dissertation, and it was chosen to represent geometric art traditions from various cultures, presenting Portuguese, Islamic and Chinese patterns alongside patterns from different parts of Africa, composed in a

deliberately confusing manner. The centrality of the Funtunfunefu is a reminder of how complex, and frequently far from peaceful, the process of exchange between cultural groups across the globe is.

This same symbol was used by the American ethnomathematician Ron Eglash as an exercise in the presentation of one of his CSDTs (Culturally Situated Design Tools). In the following image (Fig. 3), the Funtunfunefu can be seen rendered through an algorithmic description. Algorithms are nothing more than finite sequences of instructions; one could say that a cooking recipe is an algorithm. Nowadays, however, to speak of algorithms is usually to speak of their essential role in the data-processing model of any computational system. To the extent that our daily activities are increasingly mediated by technologies based on mathematical algorithms, reflecting on algorithmic processes becomes relevant to every field of knowledge.

CSDTs are software tools accessible online that allow students to create visual simulations from the mathematical principles of diverse geometric arts, mainly those originating from cultural groups frequently regarded as primitive. Ethnomathematics is, roughly speaking, the study of mathematical ideas and practices situated in their cultural context; Eglash thus moves from ethnomathematics towards what he calls ethnocomputing (EGLASH et al., 2006, p. 347). The result is a set of expressive computational media that open new opportunities to explore relations between identity and culture, and to understand the cultural construction of mathematics and computing. "Computational ethnography helps us make our own assumptions visible; we begin to see that some 'self evident' aspects of math or technology are actually choices that could have been otherwise." (Ibidem, p. 350). The aim of Eglash and his collaborators is to create a culturally enriched computational medium that offers students new opportunities for the construction of their own identities, through contact with tools and critical perspectives on the social and technical domains, as well as on the interrelations between the two (Ibidem, p. 352).¹



Fig. 3: CSDT developed by Ron Eglash. Source: <https://csdt.rpi.edu/projects/>

1. To learn more about the characteristics of the CSDTs and the assessment of their implementation in schools in the USA in recent years, the reading of "Standardizing Culturally Situated Design Tools" by Christina Bert, Keshia King, Aiyana Brown and Deanna Watkins, 2009, is recommended; see bibliography.

The research of the ethnomathematicians Ron Eglash and Paulus Gerdes is addressed in this work in recognition of their contribution towards connecting cultural diversity with present-day technological possibilities, especially in terms of algorithmic processes of image creation. Paulus Gerdes studied the mathematical principles behind traditional geometric mural paintings, sand drawings, ornamentation in goldsmithery, weaving techniques, and many other cultural productions found across the five continents. He describes a good part of these productions as geometric algorithms, and recognises in them specific mathematical qualities that can contribute to current scientific and computational development. Bearing in mind how much every computational process is necessarily based on algorithms, and how much technological innovation is an instrument of power (AKAKPO, 2016), the works of Eglash and Gerdes are interesting as alternative ways of speaking about technological possibilities yet to be developed, and about the struggle for the representation of peripheral groups within current power relations. Gerdes's educational and scientific research proposal rests on what he calls the search for cultural intercommunicability (GERDES, 2004, p. 20).

This dissertation, however, discusses the question of algorithms above all through their qualities as a **representational model** — as a tool, or method, for creating images. It is an approach oriented towards artists — or rather, towards how one can express oneself through algorithms — and towards designers — in other words, how one can design with them. First, we seek to understand the specificity of algorithmic images within the history of representation: why they would amount to a paradigm shift in relation to earlier representational models, and their place within the convergence of art history — or better, the history of images — with the history of science. Second, we proceed from the principle that it is essential to discuss the tendencies of imagistic and technological development through an approach focused on cultural diversity, considering such an approach a political necessity in order to comment on and, as far as possible, influence the changes taking place at a global level.

According to Bonsiepe, "the sciences approach reality from the perspective of cognition, of what can be known, whereas the design disciplines approach reality from the perspective of projectability, of what can be designed" (BONSIEPE, 2006, p. 29). The representational models used in the history of art, architecture and design are thus understood as essential instruments for reflecting on the changes in the modes of creating objects and images. Knowing how much such models are grounded in the scientific conceptions of an era, this reflection consequently also implies understanding the impact of mathematical notions on everyday life. One of the main concepts present in this research is the "mathematization of the world", the process by which our way of understanding and interacting with reality comes to be increasingly anchored in an abstract mathematical understanding of it. Mathematics and science are understood in this work through the formulations of the German philosopher Ernst Cassirer, who is relevant for asserting himself as a philosopher of culture, seeking to connect different cognitive processes, such as science and art, in the construction of an epistemology. In Cassirer's philosophy, the particularity of mathematics in relation to other forms of knowledge lies in its drive to simplify the experience of the world in order to describe it in a generalising way, on the basis of relations between quantifiable structures (CASSIRER, 1974, p. 183–184).

The first chapter of the dissertation deals above all with the history of two-dimensional representation and the history of mathematics. Linear perspective, as developed during the Renaissance, is given great prominence in this chapter, since it was a step of major importance towards representing the world through

mathematical models, the moment when physical space came to be understood as a homogeneous and quantifiable continuum (PANOFSKY, 1991, p. 28). Linear perspective — or rather, the epistemological turn that took place during the Renaissance — would mark the separation between project and final object, as expressed in the paradigm formulated by the Renaissance thinker Leon Battista Alberti (CARPO, 2011, p. 44). According to one of the key theorists for the development of this dissertation, the contemporary Italian historian Mario Carpo (2008, 2011, 2016), the model of project development and representation instituted by linear perspective was only essentially transformed when the tools of algorithmic design reached a mature stage, around the 1990s, and came to be widely used by designers, artists and architects.

The chapter thus begins as a discussion of Ernst Cassirer's concept of symbolic forms, mistakenly used by the German art historian Erwin Panofsky at the beginning of the twentieth century to speak of linear perspective. It moves on to a synthesis of Cassirer's philosophy and a brief narrative of the development of mathematics over recent centuries. In the following section, drawing on the work of the contemporary German visual anthropologist Hans Belting (2008), the history of linear perspective is analysed once more, from the point of view of the cultural exchange that was necessary for its creation to be possible. In the last section of the chapter, taking up the questions raised by the contemporary Togolese philosopher Yaovi Akakpo (2016), the political aspects of technological innovation are discussed, and the contributions of ethnomathematics are presented in its endeavour to think science beyond the most current narratives.

The second chapter addresses the characteristics of the representational model based on algorithms. It begins with the reading by the contemporary theorist Lev Manovich (1998) of databases and algorithms as the twin axes of the cultural productions of the so-called Computer Age, and then presents a definition of algorithms and of aesthetic propositions based on them, as formulated by Gips and Stiny in the 1970s. There follows a brief history of computational design software, in order to describe its characteristics and specificities in relation to earlier models of representation. In the following section, Mario Carpo's considerations (2011) on the production models of the pre-industrial, industrial and digital eras serve as a starting point for discussing current technological possibilities according to the ideas of the American physicist Neil Gershenfeld (2005, 2017). The chapter ends by contextualising the discussion of algorithmic design within a reading of visual productions based on geometric algorithms predating the development of computers, taking the research of the ethnomathematicians as its reference. The proposals of Gershenfeld, Eglash and Gerdes on how technology and mathematics can contribute to the creation of a less unequal future are framed by the questions posed by Yaovi Akakpo.

The third and final chapter deals with case studies. One of Paulus Gerdes's most published research subjects, the sand drawings of the Cokwe people called "sona", is explained in greater depth. The aim is to elucidate how research on the traditional visual productions of diverse peoples can contribute to aesthetic and pedagogical proposals, or even to advanced research in the fields of mathematics and computing. Finally, a project developed by the author of this dissertation is presented, in collaboration with the architect specialised in algorithmic design Veronica Natividade and with the creative coder of Chinese origin Wenqi Li. The development of this project was essential to the theoretical maturation of this dissertation, since it combined precisely the algorithmic description of patterns inspired by diverse cultures with the production of an immersive installation created using digital fabrication techniques.

CHAPTER 1

Models of Representation and the History of Science and Culture

1.1 Symbolic Forms, Art History and the History of Science



Fig. 4: Piero della Francesca (c. 1415–1492), panel *The Flagellation of Christ* (c. 1455–1460).
Source: Wiki Commons

In 1927² one of the most important texts in the history of art and in image studies was published: the essay "Perspective as Symbolic Form" by the German scholar Erwin Panofsky (1991). The book's main idea is to demonstrate how different the modes of perception of each historical culture are, each constituting a worldview of its own, expressed in its models of imagistic representation. Our own modernity would be characterised by a particular mathematical expression of the concept of infinity, within a notion of continuous and homogeneous space (Ibidem, p. 28). The author focuses mainly on the particularity of Renaissance linear perspective in relation to earlier representational models, considering this technique the most visible expression of a profound transformation in the understanding of space and of human vision — and consequently of how we perceive reality and the place of the human being in the world (Fig. 4).

Immersing oneself in Panofsky's reading, one sees how much linear perspective was based on an objectified, measurable understanding of physical space, grounded in a rationalist worldview with the individual at its centre. According to this analysis, one could even say that this perspective, with its vanishing point, is a more artificial form of representation when compared with the paintings of antiquity, since the latter would be closer to the spherical vision of our retina, which always sees dimensions in a somewhat

2. This same text had already appeared in the internal publication of the Warburg Library in 1924, but its best-known and most widely circulated publication was issued in 1927 (ALLOA, 2015, p. 51).

curved way even where there are straight lines in a given space. If Panofsky recognises the depth-drawing of Antiquity as also presenting a perspective, it is a different one from that elaborated in the Renaissance. More than a model of representation, linear perspective in Panofsky's interpretation is a mediating instrument between subject and reality, between architectural drawing and final construction; it is the affirmation of a homogeneous space that can be organised in coordinates — mathematical abstraction applied as a technology of representation (NATIVIDADE, 2010, p. 45). It has become a commonplace in art history, following Panofsky's own trail (1991, p. 66), to associate the rationalist philosophy and the Cartesian plane of the seventeenth century with the perspective of the fifteenth and sixteenth, counterposing them to the theocentric models of the European Middle Ages.

Perspective, however, was presented by the Renaissance masters as if it were natural, as if it corresponded to seeing through a glass window. Panofsky's intention is to show in what way our vision of space, our understanding of the world, is mediated by what are called "symbolic forms" — a concept of the German philosopher Ernst Cassirer that could be described as one in which "spiritual meaning is attached to a concrete, material sign and intrinsically given to this sign" (PANOFSKY, 1991, p. 41). Panofsky's essay became extremely well known within image studies, being one of the most influential writings in twentieth-century art history — among other reasons because of Panofsky's prominence³ and because of the clarity of the essay itself. Yet it is also the author's most criticised and contested text, mainly on account of its terminology and methodology — for, after all, Panofsky defines the expression "symbolic form" very precariously and never really returns to it in the development of his later theoretical corpus, however clear the essay's influence may be on the field of study in which Panofsky became the best-known name: iconology.

3. Panofsky went into exile in the United States from the 1940s onward, where he became a researcher at the Institute for Advanced Study in Princeton at the same period when Albert Einstein and other great European intellectuals also became resident researchers — that is, just when the institution came to enjoy extreme international prestige. Cassirer went to Yale, also in the 1940s, but died prematurely in 1945. Panofsky became very well known and influential in the American and international academic milieu connected to image studies, whereas Cassirer is seen as relevant mainly to the history of science, though he has received much renewed attention in recent years, chiefly through the work of the Stanford professor Michael Friedman.



Fig. 5: Fresco by Pietro Perugino (c. 1448–1523), *Delivery of the Keys*, 1481–1482, Sistine Chapel, Rome.
Source: Wikimedia Commons

Symbolic ideology, as Germain Bazin points out in his book *Histoire de l'histoire de l'art* (1989, p. 180), passed between Cassirer and Warburg — the latter with his visual anthropology, which studied images and symbols from the Pueblo Indians to the influence of pagan texts during the Renaissance — and from these two to Panofsky, who was also a personal friend of both. The term 'iconology' began to be used by Warburg in 1908 and was consecrated with an article by Panofsky (1972) in 1932, being described as the third and final stage of the analysis of a work of art, revealing underlying principles present in an image that attest to the values of a nation, a period, a class, a religious or philosophical outlook, shaped by the personality of the artist and condensed in the work. The term 'symbolic form', by contrast, was used erroneously by Panofsky. In an article published in the *Journal of Aesthetics and Phenomenology* by Emmanuel Alloa (2015), one finds the various arguments of other weighty theorists within the field of iconology itself, such as Ernst Gombrich, who set the analysis of linear perspective made in the famous essay against the philosophy of Ernst Cassirer — who referred to symbolic forms not as a specific production, such as oil painting, the portrait or linear perspective, but as manifestations of human consciousness in which a characteristic logical structure can be recognised. Art, or rather aesthetic perception, would be a particular way of perceiving, grasping sensations in a more intense mode, whereas mathematics would seek to simplify the experience of the world in order to describe it in a generalising way (CASSIRER, 1974, p. 183–184).

On moving in 1919 to take up the chair of full professor at the newly created University of Hamburg, Cassirer began to frequent the circle of intellectuals organised around the Library of Cultural Sciences founded by Aby Warburg (FRIEDMAN, 2016). It was at this time that the newcomer developed the theory of symbolic forms, asserting that the transformations of mathematical theories of space and time could not be adequately understood without attending to their most concrete and intuitive symbolic manifestations — for example, how a space is represented in a cultural context (CASSIRER, 1996b, p. 20). Aby Warburg, like

Cassirer, has also experienced a moment of 'rediscovery' in the last decade, as some of the most influential contemporary researchers on the image decided to study his work in depth — a list that includes the French scholar Georges Didi-Huberman (2007) and the Italian Giorgio Agamben (1999). Even considering how much iconological studies gained strength in the twentieth century, Warburg's successors, principally Panofsky, altered his proposals so much that they would have been almost forgotten, the exception being the Italian historian Carlo Ginzburg with his publications in the 1960s (LISSOVSKY, 2014). In Warburg's understanding, symbols are the intermediaries that make it possible to overcome the opposition between history, as the study of "conscious expression", and anthropology, as the study of "unconscious conditions" (AGAMBEN, 1999, p. 99).

Cassirer's book *The Philosophy of Symbolic Forms*, divided into three volumes, was published gradually from 1923 to 1929 — that is, at the very time when Panofsky began to use the concept. The first volume is devoted to language, the second to myth, and the third to the phenomenology of knowledge — or, one could also say, to the question of scientific thought. A fourth volume was published posthumously, an unfinished work on the metaphysics of symbolic forms. Myth, language and science would be relatable to the functions of consciousness: the *Ausdrucksfunktion* (expressive function), the *Darstellungsfunktion* (representational function) and the *Bedeutungsfunktion* (conceptual function) (CASSIRER, 1996a). Science would be directly linked to the *Bedeutungsfunktion*, but neither could one say that science dispenses with, or is uninfluenced by, the other functions. In the philosopher's narrative, we could see in human history a developmental sequence in the direction of a predominance first of the first, then of the second, and lastly of the third function, however much all of them interrelate and are present in the current moment. In Cassirer's words in the introduction to the first volume, his pursuit takes the shape of an epistemology, an attempt to understand the transformations of the human apprehension of reality (Ibidem, p. 69).

Cassirer's essay "The Problem of the Symbol and Its Place in the System of Philosophy" (1972), first published in 1927, is particularly good for synthesising his theory within the purposes of this dissertation. The essay argues that the human interpretation of a line drawing on white paper always signifies something beyond the drawing itself: it might be an expressive drawing, a mathematical representation, a religious insignia, a sign differentiating that sheet of paper from others — even in recognising the shape as a drawing we would already be associating it with a concept that lies beyond the mere existence of a random mark on a surface (Ibidem, p. 414). The drawing would have a completely distinct character in each of these alternatives, and would be perceived as a whole. Symbols are grounded in the attribution of a meaning beyond the thing itself; they are of the essence of human communication. But when we speak of the symbol in mathematics, as when we speak of painting, we refer to specific processes of abstraction in relation to perceived reality — for example, the development of notation systems. According to the Dutch historian and mathematician Dirk Jan Struik (1987) in his book *A Concise History of Mathematics*, first published in 1948, much of scientific development, especially in mathematics, was directly linked to the refinement of notation systems. At more than one point in the book, especially when speaking of the algebra of Descartes, Newton and Euler, Struik defends the thesis that as notations are refined, it becomes possible to ease the understanding of a theory and even to go deeper into its ramifications.

Where the aesthetic observer and connoisseur devotes himself to the perception of pure form and where the form reveals a mystical meaning to the religious person, the form which is visible to the eye can also offer itself to thought as an example of a purely logical conceptual structure. As Plato said, the constellations themselves mean nothing to the calculating astronomer but serve him only as "paradigms" through which he becomes conscious of the purely mathematical nature of their movement and the timeless ideal essences of "faster" and "slower." Thus for the mathematical mind the line drawing becomes nothing but a visible graphic representation of a certain functional development. He perceives something in its immediately given form that is completely beyond perception as such. He sees in it the illustration of a law, a form of ideal relation, which is the last foundation of all mathematical thought. (CASSIRER, 1978, p. 415)

It would not be possible to separate the sensation of the phenomenon from the meaning attributed to it; there are, however, diverse symbolic forms of understanding this phenomenon–meaning unity. The role of philosophy, as proposed in the books that make up the *Philosophy of Symbolic Forms*, is to understand these different points of view and their correlations — to understand each of them according to its constitutive principles. Critical philosophy could not schematise the processes of perception and attribution of meaning into one generalised form, for fundamental aspects of human cultural consciousness would be lost (CASSIRER, 1978, p. 416). Each symbolic form would have a certain orientation, a way in which it develops from simpler to more complex forms. They would be present in the three functions of human consciousness and would be influenced by humanity's epistemological upheavals. In this sense, as Alloa stresses in criticising Panofsky's work, perspective is not a symbolic form in itself, but stands at the meeting point of science and aesthetic perception, which reflected and stimulated the imagistic beliefs of a specific cultural moment (ALLOA, 2015, p. 66). Which does not mean that linear perspective is any less revealing or paradigmatic for not fitting precisely into the category of symbolic form as formulated by Cassirer.

Ernst Cassirer began his studies with the master of sociology Georg Simmel, later becoming a student of Hermann Cohen, the great name of the Neo-Kantian group of the Marburg School (FRIEDMAN, 2016). The Marburg thinkers practised a genetic reading of Kantian philosophy, refuting Kant's a priori categories such as space and time in favour of an understanding of them based on the constructions of intellectual history — for, after all, the geometry on which Kant relied had become outdated with the developments of nineteenth-century mathematics (Idem, 2008). Cassirer's first published work, in 1902, dealt with Leibniz's philosophy and its scientific foundations, using his doctoral thesis on Descartes as an introduction. In this book the philosopher develops a detailed reading of the scientific revolution, arguing that the application of mathematical theories to nature — the so-called mathematization of the world — was the central and most fruitful achievement of the age (Ibidem). According to Michael Friedman (2008) and Thomas Oberdan (2014), Cassirer began to distinguish himself from the Marburg School after absorbing Bertrand Russell's direct criticism of Cohen⁴ regarding the latter's interpretation of the differential calculus. As a consequence, the still young philosopher attempted to reconcile Neo-Kantian ideas with the scientific upheaval resulting from the publication of Einstein's theory of relativity.

4. According to Oberdan (2014), Russell devoted a chapter of one of his most famous books to explaining why Cohen's interpretation of the differential calculus was mistaken. Considering that Cohen structured his philosophy of science upon the definitions of the calculus, Russell's challenge was quite grave for the project of the Neo-Kantians of the Marburg School. Cassirer thereafter placed Einstein's physics at the axis of his interpretation of the genetic reading of the history of science.

In 1921 Cassirer published *Zur Einstein'schen Relativitätstheorie. Erkenntnistheoretische Betrachtungen* (On Einstein's Theory of Relativity: Epistemological Considerations), one of the first philosophical works on the theory of relativity, in which the German thinker discusses the need for physicists and philosophers to work out together the theoretical consequences of relativity (2017). It is in this publication that the project of the philosophy of symbolic forms is announced for the first time, conceived as a philosophical extension of the general postulate of relativity. As Friedman emphasises, influenced by the idea of relativity, a series of possible frames of reference or coordinate systems come to be seen as equally pertinent representations of physical reality; and in a similar way, the totality of symbolic forms (aesthetic, ethical, religious and scientific, among other possibilities) comes to be understood by Cassirer as relevant and interrelated (FRIEDMAN, 2008, p. 77). It would thus not be possible to separate changes in the mathematical understanding of time and space from the broader transformations in the cultural field of a civilisation (CASSIRER, 1996b, p. 11). Cassirer is also avowedly one of the greatest influences on those who went on to found the History of Science: Edwin Burt, E. J. Dijksterhuis and Alexandre Koyré (FRIEDMAN, 2008, p. 71).

In the genetic reading Cassirer made of spatial conceptions, it is not enough to speak of the theory of relativity without contextualising it within the historical process of mathematical and cultural thought. Descartes, for example, is particularly revealing for thinking about the transformation of mathematical concepts in the Western world, as well as about the place of the individual and the understanding of his cognitive capacity. He was a geometer who believed in a logic of space that could be grounded in rigid mathematical necessities (STRUİK, 1987), and that is why Panofsky associates him with the representational model of linear perspective, however sceptical Descartes himself may have been about the knowledge attainable through images (PANOFSKY, 1991). In the book *Computational Geometry: An Introduction* (PREPARATA; SHAMOS, 2012, p. 2), Descartes is identified as the first to formulate an original geometry since Euclid's *Elements*, on account of his description of space through coordinates, which allowed geometric problems to be described through algebra. Descartes's coordinates were the basis of Newton's and Leibniz's calculus; they built a bridge between two great areas of mathematics, leading to a renaissance in constructivist thought. It then became possible to construct new geometric objects by solving the algebraic equations associated with them (Ibidem, p. 2).

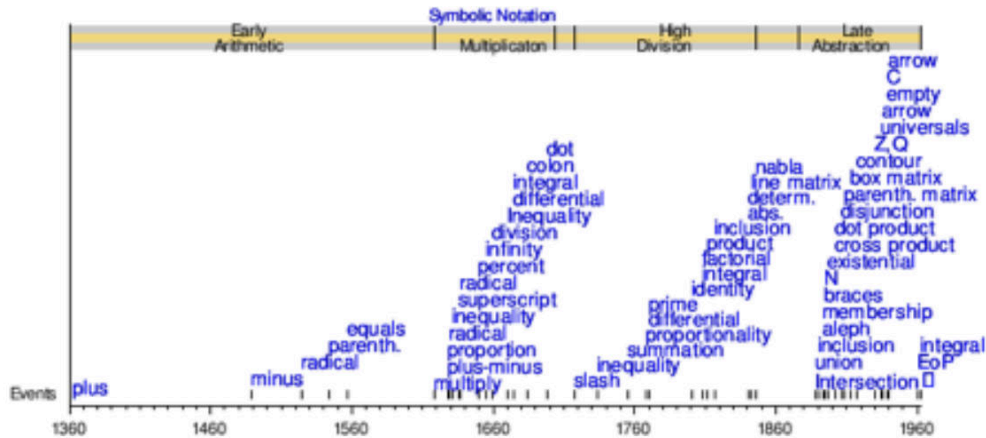


Fig. 6: Symbols by date of introduction into the field of mathematics. Source: Wiki Commons

The long history of refinement of mathematical notations that culminated in Descartes's system made it possible to dispense with numbers or words in explaining the relations between the elements of a function (Fig. 6). The nineteenth century, in turn, is particularly rich in theories of space, inaugurating the so-called non-Euclidean geometries, including topology, as well as describing infinity in a form considered unprecedented, through the work of the German Georg Cantor (STRUİK, 1987, p. 161). Yet the question of symbols, in particular of notation systems, is also of extreme relevance in the mathematics of that century. In 1854 the Briton George Boole published "An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities", showing that true-or-false logical propositions could be written and analysed using algebraic equations (GERSHENFELD, 2005, p. 230). In the twentieth century, thinkers such as Bertrand Russell, inspired by George Boole, sought the identity between mathematics and logical philosophy, holding that every mathematical postulate can be described by means of logic and vice versa (CASSIRER, 1978, p. 421). They were seeking precisely a solid foundation for mathematics, fundamental concepts anchored in demonstrable axioms, immersing themselves in the attempt to go beyond the ambiguities of human subjectivity (Ibidem, p. 422).

In modern philosophy it was most of all Leibniz who first realized this [symbol-mathematics] fundamental relationship and pursued it in every direction. It is well known how this, his basic view, like his demand for a "universal characteristic" that should always accompany logic, proved itself in his work and showed its productive capacity in the creation of the algorithm of the infinitesimal calculus. In fact it is not saying too much to maintain that the entire scientific development of logic and mathematics as it was carried out in the nineteenth century stood under its influence. The continuing development of Leibniz's leading idea has led, on the one hand, to Hermann Grassmann's calculus of extension just as, on the other hand, the effort to found symbolic logic with Boole, Peano, and Russell has depended on it. (CASSIRER, 1978, p. 412)

This tendency would prove so strong and dominant that its influence transformed the very understanding of the object of mathematics: from the end of the nineteenth century onwards, that object ceases to be numbers or quantities and turns instead to signs, set theory, the relations between elements (CASSIRER, 1978, p. 413). Although he studied extensively the mathematical theories of the nineteenth and early twentieth centuries, especially the logicians, Cassirer retained a wariness towards the enthusiasm of thinkers such as

Russell for treating mathematics as a plane almost independent of reality. After all, there would always be the shadow of the cultural influences upon scientific thought, in addition to our physical limitations, such as our very anatomy, in the attempt to seize reality through axioms (Idem, 1974). The power of scientific thought would lie precisely in its synthesis, in its impact on our relations with the physical world (Idem, 1978, p. 423).

Now, that universal language or *Lingua universalis* is developing, as already foreseen by Descartes and Leibniz, as a necessary organ for the progress of scientific thought. Only today in modern mathematics and symbolic logic, which serves as its foundation, has it reached its true cultivation and consummation. With this, language has completely left the area in which it is originally rooted. It has resolutely cast aside everything that is merely expressive. Nothing adheres to the signs of the symbolic language of mathematics and logic that in any way includes a relationship to the "subject" or to the individual world of feeling and sensation. They serve exclusively to represent the most general, objective, and necessary states of affair. (CASSIRER, 1978, p. 421)

One of the most consequential fruits of these changes within mathematics is the development of the first digital machines. According to Neil Gershenfeld (2005, p. 230), Claude Shannon, while working on a large analogue computer project at MIT in the 1930s, realised that Boole's logical statements could have a physical counterpart, using the presence or absence of voltage in a circuit to correspond to a true or false statement, allowing the circuit to solve the equations of logic — which gave the machines' ability to compute information far more applications and speed. At only 22 years of age, in 1937, Shannon wrote his master's thesis, "A Symbolic Analysis of Relay and Switching Circuits", which, together with his studies on communication and error checking plus the work of von Neumann, became the foundations of the digital technology that followed (Ibidem, p. 234).

It is clear that the understanding of mathematical conceptions of space and of models of imagistic representation underwent great transformations after the invention of linear perspective and the Cartesian plane. Yet in terms of representational model and conceptual process for the creation of images, the impact of such theories on areas like the visual arts, and principally architecture, remained considerably restricted until the development of graphic interfaces and their software for two- and three-dimensional editing and modelling. Photography itself was still based on the principle of the camera obscura, which is similar to the one that grounded the Renaissance gaze (BELTING, 2008, p. 54). Antoine Picon (2011), in debating the relevance of what he calls digital architectures, stresses how much highly influential nineteenth-century intellectuals in architecture distanced themselves from mathematics as an inspiration for their productions: since the diffusion of differential calculus at the end of the eighteenth century, it had become difficult to associate the new mathematical conceptions with questions of modelling, composition and design. Calculus came to be used mainly for studies of the behaviour of materials, elasticity, hydraulics; it was of more use to engineers, but its curve-rich geometries offered few practical tools for the drawing of architectural projects — at least not in any form that was practical to reproduce three-dimensionally (Ibidem, p. 35). In terms of project conception, two-dimensional drawing based on Gaspard Monge's descriptive geometry remained the rule, and even concepts such as vectors did not yet have a very practical use (CARPO; LEMERLE, 2008, p. 10).

I do not need to enter any further into all of these matters; here they should only serve as a confirmation of how the "symbols" in which the modern physicist describes the process of nature have in fact taken the last

decisive step and left the area of observation and representation for the province of pure signification. The newer mathematics and physics have not gone this way by chance. They have been led to it by the peculiar character of their methods and their objects. Of this there can be no serious doubt. But the kind of intellectual symbolic formation that is effected here now stands out all the more sharply from the basic attitudes of other areas of thought. (CASSIRER, 1978, p. 423)

The science of the early twentieth century had become so abstract that it drifted away from the everyday life of the other fields of knowledge. And not only science: the social organisation resulting from mass industrialisation had as one of its great consequences the specialisation of the fields of knowledge (GERSHENFELD, 2005, p. 31). Piero della Francesca, one of the most important Renaissance painters, is recognised as a supreme example of the application of the mathematics of his time to representation — of how interlinked the symbolic forms of scientific thought and of art were in the Florence of the fifteenth and sixteenth centuries (CARPO; LEMERLE, 2008). However much artists like Picasso (Fig. 7), or more recent ones devoted to new media, took an interest in contemporary physical theories of space and vision, the Renaissance continues to be seen as a golden moment of the connection between art and science. Indeed, something curious to observe when reading theorists of digital images and of the technological revolutions of our era is how frequently the Renaissance is held up as a counterpoint to the industrial age. Neil Gershenfeld (2005) argues that the Renaissance model of work organisation, in which different workshops and intellectual groups were in close proximity, is the ideal we should pursue today as we develop digital manufacturing. Mario Carpo (2011) bases much of his narrative of the history of the production of objects, and of architecture, on the analysis of the work of Leon Battista Alberti, going so far as to claim that the Renaissance master was closer to the conceptions of image and project of the Digital Age than the great names of Modernism (Idem, 2016, p. 29).

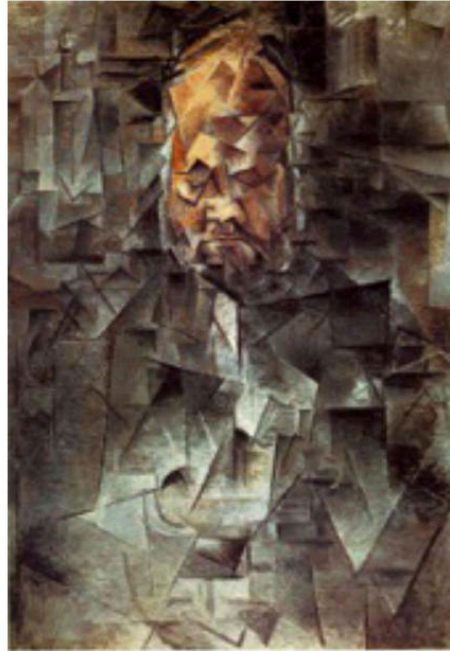


Fig. 7: Pablo Picasso, *Portrait of Ambroise Vollard*, oil on canvas, 65 × 92 cm, Pushkin Museum, Moscow, Russia.
Source: Wiki Commons

In art history there is a consensus on the relevance of the Renaissance, and of the representational model marked by Florentine linear perspective, in the unfolding of the imagistic production of the era of frescoes and easel painting, its influence persisting even in the framing of television (BELTING, 2008, p. 54). Linear perspective, then, following the ideas of Panofsky (1991) and the many thinkers who came after him (ALLOA, 2015), established a conception of continuous space, something essential to modernity; it was the fruit of, and at the same time inaugurated, a new kind of rationalism and individualism, describing the world from the point of view (and under the control) of individual perception. It was above all an epistemological shift concerning space and representation that remained substantially unshaken in terms of the conception of projects, of design, even through centuries of rapid changes in artistic styles and scientific developments. Could the many cultural transformations narrated here concerning the description of reality through mathematics not generate an upheaval as momentous as what perspective once signified?

1.2 Thinking Linear Perspective beyond the Eurocentric Narrative

In the essay discussed at the beginning of the previous section, one of Panofsky's main points is how essential the Middle Ages were to the Renaissance elaboration of a conception of continuous space — a new spatial notion emerging only after the three-dimensional representation of classical antiquity had been abandoned and, later, somehow retrieved during the Italian Renaissance, when it was already impossible to return to the worldview of the classical age. Panofsky defends the theory that in the history of art, when within a given cultural complex a type of representation seems to bear no further significant fruit, it tends to be rejected by the new generation, which seeks in earlier models and experiences references for other paths to follow. The Middle Ages would have rejected Antiquity, and the Renaissance would have returned to Antiquity for inspiration, but with an already altered gaze (PANOFSKY, 1991, p. 47).

Panofsky analyses the empirical development of the vanishing point through the work of artists such as Giotto and Duccio, and above all Van Eyck at the close of the Middle Ages. According to his narrative, Brunelleschi went further by formalising the representation of space through mathematical laws and an understanding of optics, while Alberti was essential in making perspective accessible to artists, introducing a more practical method for painters to use (Ibidem, p. 56–62). According to Mario Carpo, Alberti also helped develop other, less illusionistic types of representation for architectural projects — for linear perspective as technical drawing was not sufficiently precise, though it was excellent for project conception (CARPO; LEMERLE, 2008). For researchers of Renaissance culture, Alberti was the most fruitful intellectual of the age in terms of theories of the image; it is in his honour that one speaks of the 'Albertian paradigm' in architecture: the separation between the act of drawing and the act of building (CARPO, 2011, p. 44). In Alberti's logic, architectural creation took place in its project phase; one should build only what one could draw, and the drawing — or rather, the rigorous technical notation of the project — was the original work of art. The subsequent construction was a mere copy; differences between reality and project were imperfections (Ibidem, p. 26). For Renaissance artists, perspective was essential because it elevated art to the category of science, which improved its social status. Hence Leonardo da Vinci's famous aphorism "La pittura è cosa mentale" (Painting is a mental thing) (DA VINCI, 2011, p. 1).

The thesis of the contemporary German historian and visual anthropologist Hans Belting, however, in his book *Florenz und Bagdad: eine westöstliche Geschichte des Blicks* ("Florence and Baghdad: An East–West History of the Gaze", 2008), holds that Renaissance linear perspective — or rather, the change in the understanding of space and of the act of looking that occurred in Europe during the fifteenth and sixteenth centuries — was directly linked to the West's cultural exchanges with the Arab world. The German author introduces "Florence and Baghdad" by stating that the subject of his text is the History of the Gaze, which had initially been confined to Western culture. The mention of Florence in the title refers to the Renaissance in Italy, when the West's most important notion of the image took shape — that is, once again, linear perspective. Baghdad refers above all to Arab science, the city as the symbolic and political centre of the vast Arab world. One of the central points of Belting's book is to prove that the art of perspective is grounded in Arab studies,

in which mathematical theories of sightlines and of the geometry of light created a very particular kind of optical research. Belting emphasises the importance of going beyond the History of Art and engaging directly with the History of Science.



Fig. 8: Alberti's winged eye, 16th century. Source: <http://www.thewingedeye.eu/>

At this point we can cross Belting's discussion with Cassirer's contribution: if a symbolic form such as art can manifest a different understanding of space, then, by the philosopher's logic, it would also be anchored in scientific conceptions — notions of how we perceive space — divergent from those that previously prevailed. The purpose of this section is to make a brief genetic reading, anchored mainly in Belting's contribution, of the ideas about mathematics and representation that led up to the effervescence of the Florentine Renaissance, before we begin to discuss the relevance of the images of the computer age in global terms today — and by 'global' is also meant an approach that respects cultural diversity. Following Belting's argument, innovative mathematical conceptions, coming from a cultural context with characteristics distinct from Europe's, were essential in changing basic notions about the place of the human being in the world (Ibidem). Would this justify regarding linear perspective as part of a process of epistemological/representational rupture, rather than the Antiquity/Middle Ages opposition raised by Panofsky? From the standpoint of the History of Science, the contribution of the Arab world to the West seems far clearer than the one usually acknowledged in the History of Art.

A more comprehensive analysis of the Arab presence in European science can be found in the research of the Lebanese scholar George Saliba (2007a), who argues that there is a "classical" theory of the history of science that diminishes the originality of the Arab world by considering its scientific research a continuation

of the Greek world, with Persian and Indian influences. In interviews (Idem, 2007b) he recounts how the influence of Arab astronomy on the Western world began to gain visibility through an accident, when in 1957 the historian of science Otto E. Neugebauer found manuscripts by Ibn al-Shatir in the Bodleian Library in Oxford, England, and quickly connected those writings to the theories defended by Copernicus years after their production — a connection, according to Saliba, not previously discussed in European academia. In other words, Saliba questions how incipient the History of Science itself still is, and how much meticulous work by a large contingent of scholars will be needed to analyse the documents already held in European collections themselves, which have not even been translated from Latin or Arabic or made available to the public.

But let us return to the question of perspective, for it seems to be the best guiding thread in the discussion of images, mathematics and cultural transformations proposed here. Hans Belting's book, already mentioned, seeks to understand how mathematical principles of one culture could have been transformed into figurative images in another. Four chapters of the book are devoted to discussions of Arab optical theory, chiefly on the basis of the work of Ibn al-Haytham, known in the West as Alhazen, an Arab philosopher and mathematician who lived in Cairo during the eleventh century. What was at stake was a rationalisation of visual capacity through the use of geometry. Alhazen's work was already known in Europe during the Middle Ages; the very term "perspective" came into use after his seven-volume treatise on optics was translated into Latin in thirteenth-century Spain — in its first Latin translation the work received the title *Perspectiva* (also called *De Aspectibus*, rendered as aspect or appearance), which was changed in the modern printed edition of the sixteenth century to the "Book of Optics" (BELTING, 2008, p. 9).

Belting then presents an intellectual history of the reception of Alhazen's thought in the West from the time of its translation. The English Franciscan Roger Bacon was, in the thirteenth century, a great enthusiast of the Arab philosopher's theories, becoming famous for his defence of empirical knowledge and experimentation and giving great importance to the study of optics. Bacon and John Pecham, among others, came to be known as the 'perspectivists'. Other scholastics, however, vehemently opposed Bacon and the intellectuals who followed ideas similar to his; there was intense controversy over the extent to which vision could be a source of knowledge. For the theocentric world of the European Middle Ages such a proposition was blasphemy, since only divine inspiration or the biblical word of God could be a valid source of knowledge. Attributing to the sense of sight the dignity of a source of knowledge was part of an anthropological shift — something that, according to Belting, was already becoming visible in the growing naturalism of European art in the centuries preceding the invention of linear perspective (Ibidem, p. 150). This intellectual history of Alhazen's reception had already been described in the article by the American historian of science David C. Lindberg (1967), "Alhazen's Theory of Vision and Its Reception in the West", considered ever since a reference in the history of optics (TOSSATO, 2005). According to Lindberg, the book *Perspectiva* by Witelo, a compendium of optics gathering ideas from Euclid and Ptolemy to Al-Kindi, Alhazen and Bacon, had the status of the official textbook for the teaching of optics in the European universities of the fourteenth and fifteenth centuries (LINDBERG, 1976, p. 118).

According to the researcher Claudemir Roque Tossato (2005), the Greek theories were divided between anatomical (Galen), mathematical (Euclid and Ptolemy) and philosophical conceptions of vision; and within

the philosophical notion there was a further division between the ideas of intromission (the eye as receiver, as the atomists held) and emission (the eye as emitter of visual rays — Plato and Aristotle). Alhazen's contribution greatly developed the theory of intromission, demystifying the visual rays and discussing optics as a geometric construction of light, in addition to managing to combine the anatomical, philosophical and mathematical approaches in the same treatise (LINDBERG, 1967, p. 322). Belting's narrative distinguishes itself from most research on the history of optics by giving greater value to the originality of Arab thought, avoiding understanding it as a continuation of Greek ideas. Belting references the work of George Saliba, trying to be attentive precisely to the Arab cultural characteristics that proceeded from a cosmological notion distinct from that of the Greek world. Belting's thesis is that these characteristics — in particular the role of light as a transcendental element in the Arab world — allowed an interpretation of vision of its own, one that would not have been possible in Western culture, always so focused on images.

While preserving the fundamental aspects of the anatomy of the eye created by Galen and Euclid's notions of the visual cone, Alhazen understood vision more in terms of light than of bodies and objects (BELTING, 2008, p. 114). He asserted that light did indeed require some kind of physical body, but that it had properties peculiar to it among natural phenomena: it had its presence in the physical world through lines, through geometric rays of light, and these could be adequately measured mathematically. Through the reading of Hans Belting, but also of Arab theorists of art such as Seyyed Hossein Nasr (2015), it becomes evident that light has an essential transcendental character for Islamic culture: while sensible images carry the illusions of the transitory world, light reveals the divine harmony underlying reality. In Nasr's words (2015, p. 7), Islamic culture and art can only be understood through their specific spirituality, the fruit of a religion much more interested in the cosmological harmony of the world than in the detailed stories of men and saints so often narrated by Christian art.⁵

Alhazen was not so much interested in images as in points of light and how they could be measured (BELTING, 2008, p. 105). To this end he developed a series of apparatuses, instruments for testing the behaviour of light, such as camerae obscurae. He reached the conclusion that light passes through translucent bodies of low density, such as air and water, undergoing refraction between distinct densities, while it is reflected by opaque bodies, which are thereby illuminated and bounce back rays of light according to mathematical laws (Ibidem, p. 113). Our eyes would thus receive countless rays in a certain order, generating a sensory impression which is in turn interpreted as an image by our brain. One of the volumes of his treatise is devoted to the psychology of vision; it separates vision, light and the eye from visuality, which would be a psychological process — our brain constantly interpreting images so as not to be deceived, a process understood by the philosopher as both cultural and physiological (Ibidem, p. 124).⁶

The eye, then, would not emit any kind of light, but would be a receiver. Alhazen's theory recognises that our eye works in a manner similar to the camera obscura. This point is of great interest if we recall the

5. There are stories in the Quran, some even concerning biblical characters; however, they focus more on the spiritual meaning of events than on their details. Similarly, Nasr's analysis of Islamic artistic manifestations is directed mainly at their cosmological meanings.

6. These considerations of Alhazen's are commented on by Lindberg, Tossato and Belting; the first two, however, regard Alhazen's psychology of vision as a limitation of his theory, later superseded by Kepler, whereas Belting is interested in the cultural differences concerning images that show through Alhazen's ideas.

research of another influential contemporary theorist and historian specialised in the question of vision and the image, Jonathan Crary, who argues in his book *Techniques of the Observer* (1992) that the epistemological model of vision prevailing until the end of the eighteenth century in Europe was based on the metaphor of the eye as a camera obscura. In the nineteenth century the physiological and psychological aspects of vision began to receive a different kind of attention, which, according to Crary, influenced the development of various technical devices for deceiving the eye — among which photography would be one of the least three-dimensional.⁷ Therefore, following the logic of Belting's thesis, even though men like Kepler and Descartes made important contributions to the study of lenses and the anatomy of the eye, discovering the role of the retina, the model of vision upheld by Alhazen persisted.

What interests Belting, and our research, is principally the mathematization of the gaze and of light carried out by Alhazen, and how deeply that process was tied to his cultural context — to a world that attached far less importance to images than the West. Alhazen's treatise on optics was not used to create a visual space or to debate representational models; it was directed at the beauty found in understanding the behaviour of light in the natural world (BELTING, 2008, p. 39). Alhazen's theory of vision was the fruit of the same cultural milieu in which the physical appearance of architecture and art was dominated by mathematics. According to Belting, it is in the exchange between different civilisations that some of the greatest theoretical changes become possible. After all, in the very process of translating the treatise on optics a reinterpretation of the Arab philosopher took place, with various Latin terms of figurative meaning being employed to translate Arabic words that spoke of light, traces or marks, but not of images (Ibidem, p. 147).

Advancing in Hans Belting's narrative, in the fifteenth century we reach Biagio Pelacani of Parma, a theorist who was highly influential in proto-Renaissance circles, including the social milieu of Brunelleschi himself — a theorist whose work entitled "Questions of Perspective" was copied by a Florentine scribe in 1428. Pelacani was a great student of Alhazen but also of Vitruvius, who was beginning to be known among the Italians. His greatest contribution was to begin to consider that not only objects but the space between them, the void, should be quantifiable — this would be the first time empty space was introduced as an element of a theory of vision, which the German historian interprets as an incorporation of the abstract character of light from Alhazen's theories into the Western notion of the image (Ibidem, p. 162). According to Belting, this man's work was the foundation of the idea of mathematical space behind Brunelleschi's achievement, with vision exalted as an essential source of knowledge. From the great medieval controversies over the possibility of empirical knowledge, one arrives at the eye as the symbol of knowledge in the Renaissance. Belting debates this question mainly through the figure of Alberti, who had a winged eye in flames as his personal emblem. The German author also brings out how myths of antiquity were reinterpreted during the Renaissance, to the point of completely changing their endings: if in Antiquity the gaze was often something threatening in the stories, in the Renaissance it was empowering. In Alberti's own retelling of Narcissus, he does not drown gazing at his reflection, but finds in his image art itself. Narcissus becomes a

7. Photography became the preponderant method of its era, according to Crary, owing to its similarity to the representational models of painting. Panofsky (1991, first published in 1927), Berger (1972) and Belting (2008) likewise recognise photography as a continuation of the model of vision and enjoyment of the image already presented by linear perspective.

flower upon the water, and according to Alberti, in this tale we have a metaphor for the origin of painting (Ibidem, p. 247).

The Renaissance was greatly influenced both by classical texts and by its exchanges with other cultures, principally the Arab world, with which its diplomatic and religious relations were not always very warm. It was preferable, then, to differentiate oneself from the peoples seen as infidels by claiming to be the heir of Greco-Roman antiquity, even if this involved a free interpretation of it, including its paganism (Ibidem, p. 230). Several Italian artists of the sixteenth century claimed that Vitruvius had already referred to perspective, that it had existed in classical antiquity and they were merely reviving it — although Alberti acknowledged that it was not possible to infer the notions of linear perspective from the scant descriptions of architectural drawings in the Roman work *De Architectura* (BELTING, 2008 and CARPO, 2011). The interpretation of Vitruvius's notion of proportion also gained a prominent place in the relations between art and mathematics (PICON, 2011, p. 30).

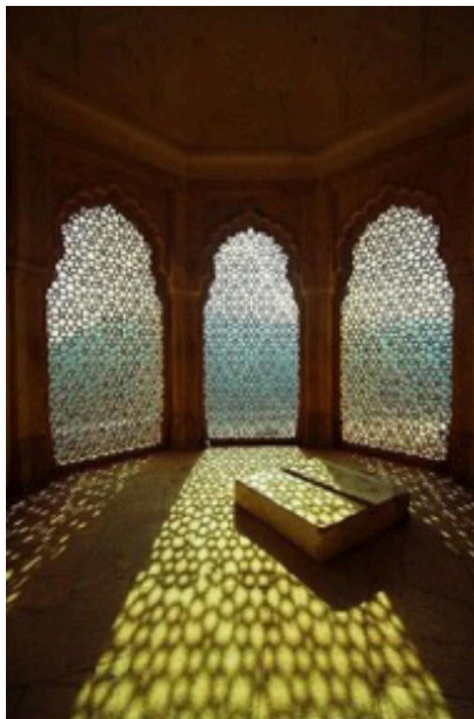


Fig. 9: Lattice window, part of the conservatory of Amer Fort, city of Amer, India, built in the 16th century. Source: Pinterest

Perspective would be above all a way of directing the gaze. In the Arab case, the lattice window had the purpose of staging light itself as a symbolic form⁸; the window of light would purify the gaze of the images of the outside world and protect the interior from external gazes — it would show light in its essential state.

8. Belting writes that he prefers to follow Cassirer's notion rather than Panofsky's with regard to the concept of "symbolic form", although he too does not seem very concerned with rigour in his use of the term. However, considering that Cassirer does not refer directly to linear perspective, according to Belting it would be Renaissance art — which comes to be marked by the representational model of perspective — that could be interpreted as a symbol of modernity (BELTING, 2008, p. 229).

Hans Belting contrasts the Eastern window of light with the Western window of the (outward) gaze. West: the subject becomes active through the gaze; a representational geometry is created with perspective. East: one sees and experiences light, a power that goes beyond the individual, like a cosmic spectacle; art and architecture are based on a representation of geometry itself. Differentiating cultures becomes an occasion for interpretation, for recognising what we take as obvious but which stems from a particular cultural baggage (BELTING, 2008, p. 272).

Various other strands of research make contributions similar to Hans Belting's, like the intellectuals who create other narrative axes by focusing on the importance of exchanges between diverse civilisations and groups as the driving force of the cultural transformations of each period. One can narrate centuries through the Silk Roads (ELISSEEFF, 1998 and FRANKOPAN, 2016), or explain capitalism and the techniques of surveillance through the experience of the great plantations in the American colonies, as does Nicholas Mirzoeff, a leading contemporary exponent of Visual Studies in the United States (MIRZOEFF, 2011). According to Saliba (2007b), what allowed the accelerated growth of the European sciences from the sixteenth century onwards was not so much characteristics intrinsic to Europe as a model of civilisation, nor its religious choices after the Renaissance (and on this point Saliba recalls the Inquisition and the Counter-Reformation), but the enormous flow of gold and goods from the Americas, which gave Europe significant advantages over not only the Arab world but all other civilisations for four centuries. For the Lebanese researcher, one of the most complex consequences of this phenomenon is that today's Arab academic institutions themselves are more sceptical about their own history than the public of the great American university centres, for the theories of the nineteenth-century European orientalists convinced the Arabs of a particular historical narrative about themselves (SALIBA, 2007a, 2007b).

This is important to bear in mind because — considering that the twenty-first century announces itself as an age in which European economic supremacy is already a thing of the past, and even the USA is no longer an uncontested leader of global power relations, and that the great powers become ever more multicultural with migratory flows — if we wish to attend to the current changes led by the logic of the digital, it would be important not to keep our eyes confined to narratives so geographically specific. In Cassirer's theory (1996b), broad cultural transformations are essential to the development of the scientific theories that end up creating new models for describing reality. How, then, can we not adapt our narrative, and the reference to the Renaissance, to the much more complex context of exchanges among various civilisations? Especially now that the institutions devoted to research and technological development are integrated into global information-sharing networks. The great universities necessarily become international, even in their teaching staff, but principally in the growing proportion of foreign students, coming from ever more countries.

It is thus maintained in this research that, in order to account for the symbolism underlying our representational models and methods, or our conceptions of space, it is also important to know how to contextualise the theoretical references that lie at the foundations of our academic training and our critical gaze upon the world. After all, taking the example of art history: how much of its theoretical corpus was developed largely by particular groups of Europeans, above all German speakers? The history of the history of art as an academic discipline, on which there is more consensus among researchers today (BAZIN, 1989), starts from the figure of the German archaeologist in love with ancient Greece (like almost every German

intellectual of the eighteenth century) named Winckelmann, deepened later by the Swiss Jacob Burckhardt with his notion of the modern individual as something nascent during the Renaissance, followed by the also Swiss Heinrich Wölfflin and his extreme formalism, which was countered by the theories of the Austrian Alois Riegl. Warburg and Cassirer opposed the theories of Burckhardt and Wölfflin, and both, like other intellectuals of their time, came to be influenced by modern anthropological studies of the European colonies, both having taken an interest in Amerindian cultures and African languages in developing their theories — on this point, Belting shows continuity with the approach and intellectual tendency of his predecessors in visual anthropology.

Discussing Belting's work once more: he narrates a curious phenomenon that occurred in the nineteenth century. While European artists were rebelling against representation in perspective as art and seeking in the East, principally in Japanese art, an inspiration for their freedom of expression, the opposite movement was taking place in other parts of the world — in the Arab world of the Ottoman Empire, in the Far East, almost everywhere. Realist representation came to be identified with modernity, and many peoples voluntarily adopted linear perspective as a representational model (BELTING, 2008, p. 52). With the incorporation of television and photography — which are, precisely, a continuation of the camera obscura model of vision — the adoption of the Western model of vision and image becomes truly global (Ibidem, p. 54). One could even say it seems evident how much looking at a two-dimensional surface and seeing images simulating depth with a vanishing point is understood almost as natural vision — or at the very least as something simply everyday in any corner of the globalised world.

1.3 Mathematics and Cultural Diversity

In his book *Science et reconnaissance* ("Science and Recognition"), the Togolese philosopher Yaovi Akakpo (2016) reflects on the duality of the power of technoscience in contemporary societies: technoscience would have the power both of liberation and of subjection.⁹ Drawing on the Belgian philosopher Gilbert Hottois, Akakpo defines technoscience as "a complex in movement of machines, networks, operations, powers, systems; it is a set of actions, processes, procedures that make it possible to intervene in nature and in the human condition in a manner radically different from magic, religion, literature, stoicism or Zen Buddhism"¹⁰ (Ibidem, p. 11). Unlike the symbol or the sign, technical science is centred on an objective and effective operative power; its goal is 'doing', intervening directly upon reality. In the understanding of Akakpo and Hottois, modernity and its scientific ideal are fundamentally distinct from traditional symbolic systems, in that they confront the questions of humanity's future increasingly in physical and not merely symbolic terms (Ibidem, p. 13). The notion of symbol for Akakpo seems above all related to a traditional belief system, to something whose power is based on a representational system of religious character regarding the universe, whereas rationality grounds itself through logic, empiricism, controlled and verifiable experimentation. This notion, therefore, has nothing to do with Cassirer's debate on symbols, discussed in the first section of this chapter; but both Akakpo and Cassirer agree that science has a perception of, and an impact on, reality essentially distinct from religious understanding, and both place science as the great tendency in the epistemological development of human history.

Modern science would pose new questions to the great religions and would itself be the object of continual self-questioning; it would thus mark a particular way of thinking. Through its power to intervene in reality, technoscience has become principally an instrument of decision and of control. It therefore exercises a colonising power over all societies, principally as an element of economic relations and of globalisation (Ibidem, p. 17). According to Akakpo, more than the colonisation of one society by another, in the contemporary world no aspect of social space escapes the rationality of science; society itself is transformed into a kind of technostucture through the modelling of technical products.

The ramifications of the possibilities of technoscience, however, are not obvious. Akakpo insists that it has the power both to bring liberation to peoples, by making them more autonomous, or at least giving them more power of choice, and of subjection, by making certain groups even more dependent on and vulnerable to hierarchical structures of power. In the Togolese philosopher's thesis, technoscience's vocation to colonise and subject flourishes above all in what he calls "disarticulated societies". This social disarticulation is linked to a deficit of a "functional end" of the social system, which is explained in the book by following the notion of "social organisation" of the German philosopher Max Weber. Thus, in analysing contemporary African societies, Akakpo has in mind a certain dissonance between traditional social structures, oriented towards the

9. "Assujettissement" in the original; rendered here as 'subjection'. — Trans.

10. "...est un complexe en mouvement de machines, de réseaux, d'opérations, de pouvoirs, de systèmes; elle est un ensemble d'actions, de processus, de procédures, qui permettent d'intervenir dans la nature et dans la condition humaine d'une manière radicalement différente de la magie, de la religion, de la littérature, du stoïcisme ou du bouddhisme zen." (AKAKPO, 2016, p. 11) — Translated from the French.

maintenance of a particular symbolic system, and the scientific ideal of modernity, which is deeply based on the idea, and the need, of constant innovation — on history as progress (Ibidem).

Mr. Weber, in effect, opposes the organisational models of modern societies to those of traditional societies. Traditional society negotiates its identity within the community by holding ethical and spiritual values in honour. Modern society, characterised by the disenchantment of the world, relegates sublime values to fraternal circles and is built around instrumental, strategic rationality. Modern society honours an organisational paradigm that is the incarnation of rationality in relation to an end. Speaking of the differentiation of social structures, Weber emphasises that the capitalist enterprise and the bureaucracy that embody it interpenetrate one another from a functional point of view. A modern society is structured and evolves in relation to a functional end.¹¹ (AKAKPO, 2016, p. 20–21)

This notion of science — or rather, of modernity as an understanding of the world based on a rationality dissociated from religious symbolism — has its genesis in Kantian philosophy. In Cassirer's words: "It is well known how this basic thought has proven itself in the 'Copernican revolution' that Kant executed in his attempt to lay the foundations of cognition" (CASSIRER, 1978, p. 425). As Michael Friedman explains (2008, p. 240), Kant inaugurates a philosophical basis for science out of his interpretation of Newton's physics. Whereas Newton understood his postulates as divine laws, as if physics itself were a religious activity, Kant starts from the question of what it is possible to know, and then posits the categories of time and space as natural a priori upon which our reason could build knowledge. Reason becomes a value in itself, untethered from the understanding of the divine. Kantian philosophy was intimately bound to Newtonian physics; it therefore needed to be revised when the non-Euclidean geometries of the nineteenth century challenged the hitherto accepted notions of physical space — which is why the Marburg School developed the genetic reading of the history of science and presented itself as Neo-Kantian philosophy. When Einstein's theory of relativity was published at the beginning of the twentieth century, it presented a notion of space and time incompatible with Newtonian physics, thereby casting doubt on the a priori categories of reason and on the whole tradition of philosophy of science and epistemology.

This discussion is relevant because it reveals that the already mentioned process of "mathematization" of the world likewise cannot be seen as a constant line. The end of the nineteenth and beginning of the twentieth century represent a moment of intellectual revolution on several levels, as demonstrated by the modernist movement in the arts and the great creations both of pure science — within which Einstein is one example among several — and of applied science, with countless transformations in engineering. That era is also particularly important as a turning point in the understanding of the contribution of other cultural systems to humanity's knowledge. In art this becomes visible with the work of Picasso and the Cubists (ARGAN, 1992); in art history with Germans such as Aby Warburg (WARBURG, 2003) and Carl Einstein (2008), who for the first time placed non-Western visual productions on the same level as the great classical works; but it is in the

11. "M. Weber, en effet, oppose les modèles d'organisation des sociétés modernes à ceux des sociétés traditionnelles. La société traditionnelle négocie son identité dans la communauté en mettant en honneur des valeurs éthiques et spirituelles. La société moderne qui se caractérise par le désenchantement du monde, relègue les valeurs sublimes dans les cercles de fraternité, et se construit autour de la rationalité instrumentale, stratégique. La société moderne met en honneur un paradigme d'organisation qui est l'incarnation de la rationalité par rapport à une fin. Parlant de la différenciation des structures sociales, Weber souligne que l'entreprise capitaliste et la bureaucratie qui en sont l'incarnation, s'interpénètrent, d'un point de vue fonctionnel. Une société moderne se structure et évolue par rapport à une fin fonctionnelle." (AKAKPO, 2016, p. 20-21) — Translated from the French.

anthropology of icons such as the Frenchman Marcel Mauss (1968) and the German-American Franz Boas (1945) that this change of valuation in relation to other peoples becomes most distinct.

Claudia Zaslavsky recounts that the European colonising experience of the nineteenth century required the production of a series of studies of local numerical systems, as well as explanations of ethical and religious customs, dress, and so forth. "How else could the colonial administrations employ Africans for wages and levy taxes upon them? Much valuable descriptive material became available to European and American readers" (ZASLAVSKY, 1999, p. 9). Zaslavsky shows how the research of the nineteenth century was extremely prejudiced towards everything related to black Africa, attempting thereby to confirm evolutionist ideas and supposed white racial superiority. Yet precisely this profusion of information ended up leading European intellectuals to question their premises, however slow and problematic that process may have been. Intellectuals of non-Caucasian ethnic origins also began to gain more visibility, such as the figure of W. E. B. Du Bois (2010) in the United States. In Cassirer's philosophy (2013), too, there is a questioning of the depreciation of other peoples under the notion of primitives — so much so that in his essay "Form and Technology", first published in 1927, the German philosopher comments on how what is called magic, with its concern for exact formulas and sequences to obtain a given result, approaches scientific experimentation and stands opposed to religious belief.¹²

Another relevant example is the German philosopher Oswald Spengler (1880–1936), who sought "to propose a philosophy of history that tried to understand the West under a new focus, seeing culture as an organic whole" (D'AMBROSIO, 2013, p. 16). For the Brazilian mathematician and philosopher Ubiratan D'Ambrosio, what makes Spengler's work relevant to later contributions in the philosophy of science and culture is his thesis that there are countless possible mathematics, since mathematics is not merely a given scientific field with a defined object of study. He considered architectural forms to be petrified mathematics, thereby recognising the relevance of applied mathematics and not only of theoretical formalisation. Considering that historians of science had until then focused on formal mathematics as developed in ancient Greece, Spengler's work opened the way for the recognition of the sophistication and contribution of the abstract scientific thought of other cultures (Ibidem, p. 17). Spengler is a very curious case, having enjoyed an improbable public success: he published his first book just at the end of the First World War and within a year passed from near anonymity in the academic world to being one of the most widely read contemporary authors in Europe (FRYE, 1974, p. 1).¹³

But if, from the intellectual point of view, Eurocentrism began to be questioned within Europe itself, in practice, with the acceleration of the process of industrialisation across the globe, countless countries adopted

12. "In religious intuition, the human is subjected to violent foreign powers to which he entrusts his whole being. Here there is as yet no fixed natural course, by virtue of its own figure and its own power; instead, it is at the mercy of transcendent higher forces. It is, however, exactly this basic view against which the magical view of the world protests. The latter understands nature as a sequence of rigorously determined events and seeks to penetrate the essence of such determination. It knows no coincidence; rather, it attains the intuition of a strict uniformity of events. And in this way it achieves, in contrast to religion, the first stage of scientific knowledge of the world. It is true that magic differs from science in its result, but not in its principles and its problem [...] It is 'primitive' not in its form of thought, but in the extent and certainty of its content of knowledge." (CASSIRER, 2013, p. 287)

technical manuals and school curriculum reforms based on the dominant pedagogies of Central Europe and the United States (VITHAL; SKOVSMOSE, 1997). Like Hans Belting's example (2008, p. 54) of perspective as a representational model becoming truly worldwide through the dissemination of photography and cinema, the authors Renuka Vithal of South Africa and Ole Skovsmose of Denmark, in joint work (1997), comment on how much European-based formal mathematics came to be far more present internationally through curriculum reforms, and through the very implementation of the European-style school system, in the countless countries that began their industrialisation in the twentieth century.

According to Vithal and Skovsmose, in the 1950s a forceful economic development began to take hold in the European and American contexts, which came to have as its intellectual counterpart an extremely optimistic theory of modernity linking economic and political development to technological development, understood as the great mark of civilisational progress. The authors stress how much interest was concentrated on the qualification of the future workforce, the question then being interpreted as a problem of education; a radical curriculum reform was therefore suggested. Agreeing with the British sociologist Anthony Giddens, Vithal and Skovsmose comment on how this theory of modernity, dominant in the post-war period, saw in industrialisation a liberating and progressive force, whose path would come from the illustrious model of the central Western societies, which the less developed countries should simply follow.

Parallel to a uniform conception of technology went a uniform conception of mathematics education. Thus, mathematics education became a central part of this modernisation theory.

This modernisation theory was applied to 'Third World' countries. To support progress in these countries, it became important to support their industrialisation, and this included support for education, with a special emphasis on subjects conceived as essential to an industrialisation. As a consequence, 'Third World' countries were flooded by (text)books from the USA and European countries. 'Westernisation' in education became a consequence of modernisation theory. (VITHAL; SKOVSMOSE, 1997, p. 132)

The troubled twentieth century witnessed industrialisation, along with the promotion of the secular nation-state — the latter based on a certain notion of progress and the rule of law (AKAKPO, 2016) — at a global level, principally after the Second World War, when the colonial empires collapsed and a different configuration of the world order took shape. All nations would then be interconnected in the global economy, but "disarticulated societies live within the system of technologies without being able to influence its process. [...] An emanation of another intellectual dynamic, high technology has no need of the service of the knowledge system of its host society" (Ibidem, p. 28).¹⁴ This is why Yaovi Akakpo dwells on how essential technoscience is today as political power, giving the example of North Korea as a nation that manages to invert traditional roles through military technological mastery (Ibidem, p. 53).

As a response to this uniformisation, carried out in accentuated fashion by the process of industrialisation and globalisation, various movements were organised at world and local level. One of the best known is so-

13. Ubiratan D'Ambrosio highlights Spengler's work as a precedent for the approach of Ethnomathematics, which will be discussed later in this section. It is worth noting, however, that despite Spengler's contribution to a less linear and more organic view of cultural processes in general, his positions, especially racial ones, were still sufficiently close to German Nazism that he was never even banned from the country in the 1930s, and his books oscillated between acceptance and the bonfire (FRYE, 1974, p. 9).

called postcolonialism, a field of research that proposes itself as epistemology, ethics and politics, led by intellectuals from the former colonies of the nineteenth-century European empires (CHILDS; WILLIAMS, 1997). Among the critical counterpositions to the Eurocentric model of knowledge that are more specifically concerned with scientific thought, we have the example of ethnomathematics, a concept introduced by the Brazilian Ubiratan D'Ambrosio in 1978, which recognises in the already cited work of Claudia Zaslavsky an essential predecessor. The word ethnomathematics is a combination of the terms *techné*, *máthema* and *ethno*, and it is proposed principally as a political necessity of recognising other ways of thinking — a blend of cultural anthropology, history of mathematics and cognitive theory. It thus encourages broad reflections on the nature of mathematical thought from the cognitive, historical, social and pedagogical points of view. D'Ambrosio (1985, p. 44), in one of his most influential articles,¹⁵ recalls that fields of research such as ethnobotany and ethnochemistry, among others, gained strength over the twentieth century, but mathematics tends to be considered a universal discipline, whose formal principles are too abstract to be conditioned by cultural specificities.

The recognition of intentionality in the formulation of abstract mathematical thought collides directly with the widespread determinist conception that regards various peoples as 'primitive', whereas the recognition of those peoples' understanding of plants does not have this same effect (EGLASH, 2006, p. 347). Therefore, recognising how much formal mathematics — and principally the most current history of the discipline — ignores of the structured knowledge that exists beyond its definitions and frontiers would be of relevant and necessary political character, besides being fertile ground for innovations within mathematics itself and in its applications. To this end, research programmes and teaching models have been proposed that highlight the contributions of many different cultures to mathematics as a discipline and validate a wide range of mathematical practices.

Methodologically, this programme recognises that in their adventure as a planetary species, man (the species *Homo sapiens sapiens*), as well as the other species that preceded it — the various hominids recognised from 5 million years before the present — have their behaviour fed by the acquisition of knowledge, of doing(s) and of knowing(s) that allow them to survive and transcend, through manners, modes, techniques, arts (*techné* or 'tics') of explaining, of knowing, of understanding, of dealing with, of living with (*máthema*) the natural and sociocultural reality (*ethno*) in which he, man, is embedded. (Idem, 2005, p. 112)¹⁶

D'Ambrosio (2013, p. 19) explains that the debates of the 1960s and 70s about the nature of scientific method and research strongly influenced his interest in developing ethnomathematics. Here he is referring to the famous confrontation between the then young American physicist Thomas Kuhn and the nearly retired Austrian-born philosopher Karl Popper, in 1965, whose aftermath, according to the historian Steve Fuller (2003), is almost universally recognised as the modern divide of the philosophy of science. Kuhn's relativist vision of shifting paradigms — which supposes that science is just another human activity, like art or

14. "Les sociétés désarticulées vivent dans le système des technologies sans pouvoir influencer leur procès." (p. 28) "Émanation d'une autre dynamique intellectuelle, les high tech n'ont pas besoin du service du système de savoir de leur société d'accueil." (AKAKPO, 2016, p. 28) — Translated from the French.

15. Considered the first comprehensive and theoretical treatise, in the English language, of the Ethnomathematics Programme (POWELL; FRANKENSTEIN, 1997, p. 13).

16. Quotation originally in Portuguese; translated by the translator.

philosophy, only more specialised — triumphed over Popper's more positivist notion of the revolutionary role of science in demystifying society's dogmas (Ibidem, p. 5). Kuhn's theory, even though he himself acknowledged the great influence of the Neo-Kantians on his work, stands opposed to Cassirer's intellectual project, for the latter sought to reconcile Einstein's physics with intellectual history up to that point, whereas for Kuhn scientific paradigms simply proceed from fundamentally distinct premises (FRIEDMAN, 2008, p. 239). It is precisely because the premises are irreconcilable that scientists are forced to change the paradigms that ground their scientific production — and they do so reluctantly, for in Kuhn's view scientists are far less innovative and questioning than is usually inferred (FULLER, 2003, p. 12).

Also important among the debates commented on by D'Ambrosio is the opposition between Imre Lakatos and Paul Feyerabend. The latter became known as a defender of epistemological anarchism on account of his continual criticisms of scientific method and of the social status of scientific activity. Although he had been a student much influenced by his master Karl Popper, tending towards scientific realism at the start of his career, Feyerabend — among other reasons — was marked by the student revolutions of the 1960s and by the projects for the racial democratisation of university education, which made the philosopher aware of the lack of diversity in the academic world and of how much it functioned as an arm of the imperialist powers (PRESTON, 2016). "His denunciations of aggressive Western imperialism, his critique of science itself, his conclusion that 'objectively' there may be nothing to choose between the claims of science and those of astrology, voodoo, and alternative medicine, as well as his concern for environmental issues ensured that he was a hero of the anti-technological counter-culture" (Ibidem). There would thus be no 'scientific method' capable of guaranteeing scientific progress. The Hungarian philosopher Imre Lakatos, in turn, also much influenced by Popper and a great friend of Feyerabend — however constantly he disagreed with and publicly opposed him — gained renown for his non-formalist vision of the philosophy of mathematics. He sought to explain the discontinuities and continuities of scientific theories: how, for example, Copernicus's theory came to be accepted by so many even though it was born in apparently refutable form, since necessary parts of his hypothesis could only be proven centuries later with the development of better telescopes and measuring techniques (MUSGRAVE; PIGDEN, 2016).

These debates were essential for raising questions about the place of science in society — about whether it could be an independent and progressive force or be simply destined to remain the technical wing of the military-industrial complex (FULLER, 2003). The contribution of ethnomathematics and of other proposals with a similar political engagement has been growing internationally in recent decades. According to the Jamaican sociologist Stuart Hall, "globalisation retains some aspects of Western global domination, but cultural identities everywhere are being relativised by the impact of space–time compression" (1992, p. 24).¹⁷ For Hall, one of the most striking examples of this aspect of globalisation is the phenomenon of migration, since global interdependence now operates in both directions: the outward movement (of goods, of images, of Western styles and consumerist identities) is matched by a movement of people from the peripheries to the centre, frequently owing to political-economic pressures, but also through cultural attraction. Today even the largest American universities have a substantial number of students coming from what Hall calls the 'periphery' of the world — students, professors, and even the very sponsors and patrons of the institutions —

17. Quotation cited by the author from the Brazilian edition; rendered here from the Portuguese.

making the relations of power and the exchange of knowledge considerably more complex.¹⁸ Great European and American universities, museums and research laboratories — that is, the most relevant knowledge-producing institutions of the West — have been opening branches in the Middle East and Asia as a way of expanding their influence and their research potential.¹⁹

According to the Dutch-born mathematician settled in Mozambique, Paulus Gerdes (*Ibidem*, p. 24), monolithic visions of what counts as valid knowledge are still dominant among educators, professional mathematicians and administrators. This raises the question of the responsibility of ethnomathematicians, historians and philosophers in trying to exercise more influence over their fields of research. One of the most urgent tasks would be the study of mathematical elements in cultural spheres under siege, in particular the cultures of the so-called "Third World", where there is great potential for the broadening and diversification of scientific thought. The aim, then, would be to manage to speak of global processes without maintaining narratives that prioritise power games and local or global hierarchies to the detriment of intercultural intelligibility.

In other words, ethnomathematical studies may broaden the (intercultural) understanding of what are mathematics, of what are mathematical ideas and activities. There cannot be a sole, unified view of mathematics. For a monolithic and dominant view there is no basis. At the same time, for the other extreme, a cultural relativism concerning mathematics, there is also no ground: intercultural intelligibility seems possible. (GERDES, 2004, p. 20)

18. Data supporting these claims can be found in the publications of the "Office of Planning, Evaluation and Policy Development, Office of the Under Secretary, U.S. Department of Education", such as the one cited in the bibliography of this work.

19. Examples: the opening of the Louvre's branch museum in Abu Dhabi: <https://www.louvreabudhabi.ae>

CHAPTER 2

Symbolic Forms and Algorithms — the Contemporary Image

2.1 The World of Software

In 1998 Lev Manovich published his article "Database as Symbolic Form", in which he argues that databases and algorithms are the two halves of the ontology of the world according to a computer. This implies, according to the author, that the computerisation of culture involves the projection of these two fundamental parts of software onto the cultural sphere. Following the text: if the novel was the great cultural production of the nineteenth century, and cinema of the twentieth, in the Computer Age the database would be the principal one. There is, however, the peculiarity that the database does not tell a story but presents collections of individual items, in which each one carries as much meaning as any other — and yet they are not simple collections of items either: they have their own diverse capacities for self-organisation, since they continually interact with algorithms.

Manovich compares the database with other cultural productions, saying that each presents a different model of what the world is. It is in this sense that he uses the concept of symbolic form as a reference, citing Erwin Panofsky's 1927 essay. Manovich does not go very deeply into the meaning of this concept in the cited article; he seems to use it as a strategy to give greater weight to his description of the distinct characteristics of each medium and his interpretations of their influences and configurations. Manovich's reference to Panofsky in the title of his article is not unusual — on the contrary, it places databases within a broader academic debate, for many theorists have paraphrased the famous essay "Perspective as Symbolic Form", writing articles with similar titles but replacing the word 'perspective' with their object of study, ranging from art to Law (COSKUN, 2007), sovereignty (BARTELSON, 2014), literature (BUESCH, 1973) and landscape (GILL, 2002), among many other possibilities. What matters for this research is to think the specificity principally of the algorithm as an instrument of representation of and interaction with the world, in relation to earlier representational models, or indeed methodologies for designing images or objects. Let us turn, then, to a definition of algorithm, as presented in the book *Algorithmic Aesthetics* (GIPS; STINY, 1978), which was praised at the time as a major contribution to the field of aesthetics; its purpose was described as an attempt "to unify the theories of criticism and design within a single metatheory based on the notion of algorithm" (HARRAH, 1979, p. 471).

An algorithm is a finite list of instructions that can be followed mechanically. An algorithm is applied to an input to produce an output. An algorithm has the following properties:

- (1) The input to an algorithm is a finite sequence of symbols. Any symbol occurring in an input to a particular algorithm is a member of a fixed, finite alphabet. For example, the alphabet for an algorithm could be the English alphabet, the numerals, and some punctuation marks. A possible input using this alphabet is some particular English text.
- (2) Each algorithm consists of a finite number of instructions. Each instruction is itself finite and can be executed or performed in a finite amount of time. Each instruction is unambiguous and rigorous in the actions it specifies. The instructions of an algorithm can be executed or followed mechanically by some computing agent, for example, a person with a pencil and paper.

(3) The output of an algorithm is a finite sequence of symbols. Any symbol occurring in an output of a particular algorithm is a member of a fixed, finite alphabet.

(4) For each given input, an algorithm produces a unique output. That is, each time an algorithm is given a particular input, the algorithm always produces the same output. Further, the sequence of instructions executed when a particular algorithm is applied to a given input is always the same, as is the result of executing each instruction. (GIPS; STINY, 1978, p. 205)

In the cited book, Gips and Stiny proposed algorithmic models capable not only of producing criticism of works of art but also of creating them. They used mainly abstract paintings as examples, but also speak of musical, textual and sculptural works, going so far as to postulate that an algorithmic creation or interpretation of a painting such as Raphael's *School of Athens* would be theoretically possible — even while recognising what an absurdly difficult task it would be with the technologies available when the text was written (GIPS; STINY, 1978, p. 6). They proposed a structure for criticism algorithms and design algorithms in which a variety of approaches to art could be represented. Thus, from a fixed structure, one could experiment with various styles of art, provided they could be described through a finite list of instructions. They started from the principle that the very exercise of describing an image algorithmically was already a rich experience for understanding works of art better, attaining an awareness of principles underlying a work — details that would not be visible to less rigorous methods. They further asserted that the exercise of representing ideas completely through algorithms provides a test of their consistency and pertinence, as well as a way of identifying their consequences by simulating them on a computer²⁰ (Ibidem, p. 7).

The proposal of Gips and Stiny can be summarised in the following logic: a criticism algorithm has a receptor that produces a description of a given object. The description is analysed, interpreted and evaluated. Depending on the evaluation, an appropriate response is produced, which the authors compare to applause. A design algorithm is given the description of some initial condition and then uses various algorithms to find the best way of satisfying those conditions. In both criticism and design algorithms, there is an aesthetic system that produces interpretations and evaluations. Each aesthetic system itself consists of four algorithms, the two central ones being an interpretation algorithm and an evaluation algorithm. There are two kinds of interpretation: the kind in which an object is understood in terms of how it can be constructed, and the kind in which an object is understood in terms of the ideas it evokes. In all this there is one principal idea: that the system operates not on objects but on descriptions of objects, and that every object is discriminable by a finite description (Ibidem, p. 110). The concepts developed by Stiny and Gips, in particular the idea of 'shape grammar', ended up having a greater influence on the field of computational design in architecture than in painting, although the authors had originally focused on examples of generative abstract paintings.²¹ Stiny and Gips place themselves within a broader discussion of visual aesthetics, in dialogue with twentieth-century authors such as the American philosopher Monroe Curtis Beardsley, the first to publish a philosophy of art and aesthetic system within the tradition of analytic philosophy,²² who proposed for aesthetics an approach

20. "In order to specify an algorithm that performs some process, it is necessary to specify unambiguously and deterministically every step in the process. Nothing about the process can be left to chance or the imagination of the computing agent. Because an algorithm must be specified in such detail, just the attempt to construct an algorithm for a given process provides an excellent means of exploring the process in all its aspects and features." (GIPS; STINY, 1978, p. 7)

21. More information at <http://www.shapegrammar.org/intro.html>

similar to what was then being discussed about language by Wittgenstein and Noam Chomsky (WREEN, 2014).

Philosophical proposals beyond analytic philosophy present quite distinct visions of the specificities of contemporary images. Some theorists, such as Flusser (2008) and Couchot (1993), see in bits, or in pixels — and not in databases or algorithms — the principal axis of the change in the ways of thinking and representing space, or language itself, in our day. Nevertheless, among the many approaches most discussed today, what can be perceived as a constant is the understanding of how much our way of representing and impacting reality is caught in a long tendency of becoming more complex and abstract, ever more based on mathematical-scientific thought. Lev Manovich argues that to understand the specificity of the contemporary digital phenomenon and its ways of representing and impacting reality it is necessary to study software itself (2012), for each piece of software follows distinct principles, both from the point of view of the mathematical choices that sustain it and of the objectives and social understandings of those who create it. Software could thus be thought of as conceptual premises, for the chosen algorithms reflect particular cultural conceptions, such as: which parameters are considered relevant, which data are used, which operations are performed, how the analysis of results and impact is carried out, and so on.

Mathematical concepts such as vectors — which had existed in physics since antiquity but underwent great development from the seventeenth century onwards (STRUİK, 1987, p. 54) — were used from the very first computer drawing software. When Ivan Sutherland created the Sketchpad software in 1963, it used a coordinate system to orient the points of the object and had its lines based on vectors. It already presented an interaction between an original drawing, as master drawing, and countless copies that would be altered automatically if one decided to make any change to the original (SUTHERLAND, 2003, p. 27). Ivan Sutherland (Fig. 10) argued that computer graphics could be used for both artistic and technical purposes, through his software, which constituted a new method of interaction between humans and computers (Ibidem, p. 3). From the beginnings of software it was clear that just as one can record the two dimensions of a given point, one could record three; thus the centuries-old limitation that two-dimensional drawings on paper imposed on architectural, design and engineering projects became obsolete, even if it still took time for graphic interfaces to truly unleash that capacity (CARPO, 2011, p. 33). In the logic of architectural design according to the great Renaissance theorist Leon Battista Alberti, one could only build what one could draw on paper, two-dimensionally; and even during the great debates of modernism, the tools for project conception still obeyed those limitations (Ibidem, p. 32).

22. Gips and Stiny's proposal to divide the operations of algorithmic criticism into description, interpretation and evaluation is directly based on Beardsley's aesthetics, which in turn closely resembles Wittgenstein's work on the analysis of language, according to the analysis of Michael Wreen (2014).



Fig. 10: Ivan Sutherland and the Sketchpad. Source: Wiki Commons

Software is based on countless mathematical concepts, and its performance is sustained precisely by the functions and geometries that constitute it. The American researcher Neil Gershenfeld (2005, p. 124) points out how the mathematical representation of logic by George Boole — discussed in the first chapter as the basis for the development of digital technologies — was also transformed into a series of digital editing operations known as 'Boolean transformations'. The same happened with the theories of mathematicians such as the German Euler of the eighteenth century and the Frenchman Bézier of the twentieth, which came to be present in various three-dimensional modelling programs as tools, the software Rhinoceros being one of the best-known examples (GUPTA; NEUMANN, 2016, p. 20). Another great contribution of the software used in what came to be called CAD (Computer Aided Design) is its parametric capacities. It specifies the essential aspects of a design in a way that allows them to be modified later — not merely because a person decides to alter each piece of information about the design, but because they are based more on the behaviour of functions than on specific quantities for their vertices (GERSHENFELD, 2005, p. 127). The design is created from associative relations. CAD programs can be integrated with CAM (Computer Aided Manufacture) systems, providing a continuity of production (Ibidem, p. 40) that is nowadays called digital fabrication.

Rhinoceros, widely used for parametric design projects, is based on NURBS (Non-uniform rational basis splines) (McNEEL & ASSOCIATES, 2017), a mathematical system for generating and representing curves and surfaces, which was initially inspired by the tools used by artisans to achieve more organic and free curves in the construction of boats and other objects (SCHNEIDER, 2014). These were curves that could not be adequately represented in projects drawn two-dimensionally with the traditional tools of compass, set squares and rulers — at least not in a form practical to reproduce physically (CARPO, 2011). Rhinoceros has as a plugin Grasshopper (DAVIDSON, 2017), a graphic editor of generative algorithms, which works by

connecting the various algorithms already offered by the program, using a format that makes them look like diagrams. The user defines the reference parameters for the algorithmic functions, and also has the power to import data to feed the functions. In interactive design projects — normally much smaller and more flexible than architectural ones — one can connect a sensor to a microprocessor so that it feeds algorithmic functions that make the structure, or the images, sounds, etc., of the final product responsive (GLYNN, 2014).

The 3D perspective visualisation, or the frontal and lateral sections, that Rhinoceros generates from the code assembled by the Grasshopper diagrams works as real-time feedback. This is an advantage of this software over others also associated with algorithmic design, such as Melscript, Rhinoscript or Processing (NATIVIDADE, 2018). In terms of processes, for a draughtsman the difference between designing with Grasshopper and developing a two-dimensional drawing is quite marked, requiring far deeper knowledge of descriptive geometry in order to know how to connect the functions of the diagrams. If the designer understands programming and calculus, their ability to play with the possibilities of Grasshopper grows considerably (GLYNN, 2014).

A key distinction for understanding the potential of tools such as Grasshopper was defined by Kostas Terzidis in his book *Algorithmic Architecture* (2006): there is computerisation and there is computation. Computerisation implies the act of entering, editing and storing information in a computer; it refers to automation, mechanisation, digitisation. The conception would already be complete; it would be like transferring the information of projects drawn by traditional methods, or formulated following their logic, into a computer. As part of a design process, the introduction of computerisation is said to have taken place in the French automobile industry of the 1960s, with inventions such as the Bézier curve. Computation, on the other hand, means the exploration of indeterminate, vague and undefined processes: "[...] computation aims at emulating or extending the human intellect. It is about rationalization, reasoning, logic, algorithm, deduction, induction, extrapolation, exploration, and estimation. In its manifold implications, it involves problem solving, mental structures, cognition, simulation, and rule-based intelligence, to name a few" (Ibidem, p. 11).

Another relevant point raised by Neil Gershenfeld concerns the ability of some software systems to model behaviour as much as the parts of the form. The mathematical technique FEA (finite element analysis) makes it possible to check how a design will respond to almost any situation: forces, vibrations, heat, fluids, electric and magnetic fields can all be simulated (GERSHENFELD, 2005, p. 128). The structure imagined or reproduced in the digital simulations is covered with points — this is called a mesh — and the relevant characteristic is that all the points are associated in a network while holding specific information (Ibidem, p. 129). Or, in the case of BIM models (Building Information Modelling), one can have information about the various phases and dimensions (time, materials, spatial and financial relations) of production constituting the project drawing itself — articulating, for example, the construction schedule with the purchase order for a given material and the total budget of the project (NATIVIDADE, 2010, p. 172). The mathematization of contemporary representation thus goes far beyond the question of space.

The Italian architectural theorist Mario Carpo makes clear in his writings (2011, 2016) that the computer programs of the last 50 years represent a methodological rupture in the conception of architectural projects, and of design in general, comparable only to the Renaissance. The programs known as CAD enabled

designers to draw non-Euclidean objects with the help of the repertoire of NURBS lines and polygonal modelling, among other techniques of three-dimensional representation, as well as to describe complex forms geometrically through topology (which would replace descriptive geometry), and allowed the holistic manipulation of architectural objects by means of parametrisations (NATIVIDADE, 2010, p. 113). According to Manuel Castells (1989, apud NATIVIDADE, 2010, p. 63), a fundamental characteristic of technological revolutions is that the main effects of technical innovations fall upon processes and not upon products. For Mario Carpo (2011), Schumacher (2008) and Kolarevic (2003), although the history of graphic interfaces began in the 1960s, it was only in the 90s that a maturity with regard to digital architectures was reached. Even so, in many schools of architecture and even of design, it is the didactics of the modernists that prevails at this start of the twenty-first century: software is treated as accessory, as tools; the theory taught has changed little, and knowledge of mathematics tends to be considerably limited — the computer constantly veiling the presence of the mathematics behind its tools (PICON, 2011, p. 35).

We thus face an impasse in contemporary society: while there is a tendency towards the financial and symbolic valorisation of scientific innovation and its technologies, there is also a tendency towards specialisation, with engineers, designers, artists and architects being required to show very different kinds of knowledge and results, both in schools and in the workplace. Innovation and specialisation do not necessarily work well together. Researchers like Neil Gershenfeld (2005, p. 31–34) admire and defend the model of the Renaissance man, a universal man endowed with countless abilities, like the painters/architects/engineers of the sixteenth century. With today's most common educational model, with its focus on specialisation, this becomes a Herculean task. Thinkers like Vilém Flusser (2008) ponder the need to understand the mathematical lines that make up the 'black box' — a metaphor that began to be used in reference to photography, but has now become a generalising term for the world of technologies governing our daily interaction. Whether or not one has access to, and interest in knowing, the codes of the black boxes, what is clear is how much the expectations placed on the designer's work have changed with the diffusion of digital technologies, forcing everyone to remain in constant learning.

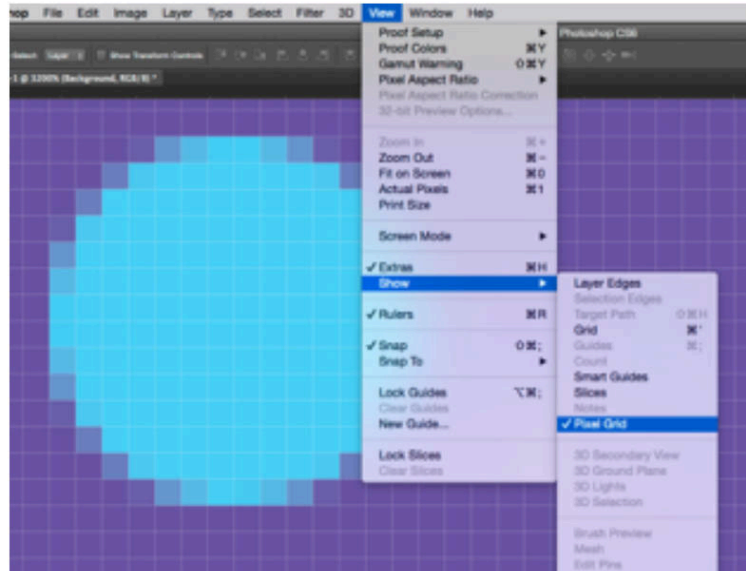


Fig. 11: The pixel grid in the Photoshop program. Source: Wiki Commons



Fig. 12: Photo of London's City Hall, a project by Foster and Partners. This building is famous for being based on parametric design; its form was conceived according to analyses of the optimisation of fluid behaviour and temperatures. Source: Foster and Partners

2.2 The Algorithmic Image, Design and Models of Production

The book *The Alphabet and the Algorithm* by the Italian researcher Mario Carpo (2011) begins with a discussion of the recognition of value attributed to monetary symbols throughout history. If for millennia the value of a type of coin was understood through the visual, as well as material, similarity the coins maintained with one another, after the industrial revolution value came to be based on identity, since the products of the industrial era were radically identical, at least at the moment of their production. In the digital era, by contrast, the form of the object matters very little: recognition occurs through the digital reading of individualised codes. Value lies in the code, which is usually unique yet still belongs to a series of codes — a series practically infinite in its varieties, which nonetheless retain some logical resemblance to one another. The question of money is used as a starting point to speak of the changes in the modes of production of these three periods and the countless cultural consequences of those processes.

According to Carpo, European culture had been living a tendency towards mechanical reproducibility since the proto-Renaissance (Ibidem, p. 19). The invention and diffusion of the movable-type press was one of the most significant milestones of this process; Alberti's drawings and his methods of architectural notation belonged to the same tendency. The Albertian paradigm, more than a conception of image and of physical reality, had above all social consequences, bringing the notion of authorship into focus while devaluing the manual work of the builders. The large-scale implementation of this paradigm only really occurred in the eighteenth century, with what we call the first industrial revolution — for, as Alberti sadly realised, the master builders were not willing to blindly follow every instruction of the great author, the architect, especially if he was not present on site (Ibidem, p. 22). It took a still more significant improvement of technical project drawings, a serial productive logic based on the identity between products, and a great change in social roles for a well-defined separation between designing and building to be achieved; but only with CAD/CAM systems did we reach the final consequences of Alberti's idea of project and construction (Ibidem, p. 78).

In the book *Uma Introdução à História do Design* (An Introduction to the History of Design), the Brazilian art historian Rafael Cardoso (2008) argues that the moment customarily recognised as the beginning of the History of Design — the eighteenth century in England — had as its transforming impulse above all a change in the organisation of work, with designers responsible for ever clearer technical drawings. Following Mario Carpo's argument (2011), Alberti's theories already presented the founding ideas of the Industrial Revolution — but, impressively, they went even beyond them. Considering that Alberti lived just before the diffusion of the printing press, he avoided as much as possible reproducing images in his books: if the work of copyists already tended to introduce errors into written transcription, copies of images were not regarded as reliable at all (Ibidem, p. 24–26). The most relevant book on the architecture of antiquity, the treatise of Vitruvius, likewise contained no images and had to confine itself to verbal descriptions; there was a tradition of books about images that contained no images — but Alberti's work stood out for the systems he invented to supply imagistic information (Ibidem).

Alberti developed ways of describing images through kinds of numerical coordinates, and even of explaining three-dimensional objects by separating them into parts that could be assembled, fitted into one another, as well as a series of devices for directing the gaze and interpreting his systems. His inventions were, in the view of some of his contemporaries and principally for Carpo himself, clumsy — but no less interesting for that (Ibidem, p. 54–58). In trying to describe images through numerical systems, in trying to carry the importance he attached to technical notation to such a level of abstraction, Alberti approached the present-day logic of digital descriptions. A similar relevance is attributed to Albrecht Dürer, with his detailed studies of human proportions in diverse body types, which resemble the innumerable measurements that underpin three-dimensional modelling today (CARPO; LEMERLE, 2008). Or again, according to Hans Belting: "Western visual culture found its paradigm in the mathematical 'visual image' of perspective. Thus Piero [della Francesca] anticipates digital processes when he subjects the physical world to a rigorous mathematical deconstruction only to then reconstruct it in the disembodied medium of painting" (BELTING, 2011, p. 30). At this point, the German author compares Piero della Francesca to the contemporary animators who work with three-dimensional modelling.

Following Carpo's argument (2011), in the artisanal mode of production the energy expended to produce each object is high, and each final product has its particularities, even if they resemble one another. The social organisation around the production of great undertakings was usually based on a cooperative logic, since several participants could add something to the construction. In the industrial era, all objects become identical, and the cost is proportionally lower the more one produces. The energy expenditure of production in the industrial era lies in the making of the moulds; but after that it becomes preferable to produce hundreds of thousands of objects to be sold anywhere in the world rather than a few dozen. An economy of scale takes hold — one already foreshadowed in the great logistical enterprise developed in the extractive colonies of the Americas. The question of authorship is also sharply defined in the industrial model; principally in art and architecture projects, the 'author' acquires an almost religious status, creativity being an admirable quality of certain geniuses who would stand apart from society.²³ The equipment of digital logic, however, possesses an unprecedented integration between project and manufacture, as the CAD/CAM system or even a digital printer reveals — where there is no difference in cost or time whether we produce 1,000 identical pages or 1,000 different pages.

Deleuze's book (1988) on Leibniz and Baroque aesthetics was extremely influential for what Carpo considers a revolution in architecture, which occurred in the 1990s, when architects and designers began to use digital tools not merely as aids to their projects but as a founding part of their conception. The book in question, cited by Carpo as essential but also given prominence in the narrative of Branko Kolarevic (2003), introduces the notion of *objets éternels* (DELEUZE, 1988, p. 108), which became known as the *objectile*, and came to designate thinking not of the object in itself but of the series of similar yet varied objects that result from a function (CARPO, 2011, p. 99). With CAD/CAM integration, there would no longer be errors or difficulties between notation and execution, since they become one uninterrupted process. With digital

23. In his book *Mozart: The Sociology of a Genius*, the German sociologist Norbert Elias (1994) discusses the creation of the modern concept of genius during the eighteenth century, at a moment of change in the social organisation around art, principally with regard to its mode of enjoyment.

variability there is no longer any need to reproduce always the same model, however much the algorithms maintain a resemblance between the final products.

In Carpo's view (Ibidem, p. 113), while the revolution of the 90s was based mainly on private, controlled network environments, from the 2000s onwards one sees the appearance of the open source movements, of the notion of interactivity, of P2P sharing, of the bidirectionality of information — it becoming difficult to define who was the original author of an idea. Carpo thus connects the changes in the architecture of the 2000s with what is commonly called Web 2.0. This term was coined by Dale Dougherty, at the time an executive of the digital-media communication firm O'Reilly Media and later founder of Maker Media — both pioneering companies, the first in internet media and content, the second in the field of digital fabrication (GERSHENFELD et al., 2017, p. 130). Tim O'Reilly popularised the expression in a blog text of 2005, in which he presents arguments as to the ways in which the internet that flourished after the bursting of the dot-com financial bubble in 2001 would be more democratic than the network of the 90s (O'REILLY, 2017). This new internet presented sites as platforms for user interaction rather than fixed content; concepts such as user experience; data platforms growing with use and being seen as products; software adapted to different devices — in short, much of what marks the contemporary internet as coordinator of human social and productive activities began to take shape. Tim O'Reilly is also recognised as an agent who popularised the open source movement (Ibidem).

In the view of the American researcher Neil Gershenfeld (2017), if the invention of the internet was initially a revolution in communication, and personal computers and smartphones represented a revolution in the accessibility of computing, we are now at the beginning of a third revolution, led by digital fabrication techniques.²⁴ In a considerably prophetic tone, the MIT researcher and professor predicts that digital fabrication, once it becomes mass-scale and — like personal computers — spreads as personal fabrication, will bring the programmability of the invented digital world into the physical world we inhabit (GERSHENFELD, 2005, p. 17). In this way, there would be little distinction between the model of a thing and the thing itself. Like Carpo when speaking of the consequences of production models incorporating digital logic, Gershenfeld asserts that more than a technological revolution it would also be a social revolution, and argues that people would regain control over what they consume by having the means of production, and the possibility of altering designs, at their disposal. It is at this point that Gershenfeld voices his admiration for the Renaissance, with its workshops mixing artisans, artists and scientists — a golden age of human creativity, when there was sufficient distance from the medieval world for experimentation, but not yet the serial production and specialisation of the industrial revolution. Moreover, people were far more inventive in their daily lives before they could enjoy the infrastructure we benefit from today. Artistic experimentation should be an example for scientists and engineers (GERSHENFELD, 2005, p. 114).

"The universe is literally as well as metaphorically a computer. Atoms, molecules, bacteria, and billiard balls can all store and transform information" (GERSHENFELD, 2005, p. 4). Gershenfeld's quest is to be able

to deal with atoms in the same way we deal with and think about bits: with total control, so as to be able to fabricate anything imaginable. Digital fabrication would increasingly approach the way living beings reproduce — innovation composing the very material of which they are made, capable of reproducing themselves with a very small margin of error and capable of evolving through small alterations in their own code. Gershenfeld writes with a tone of admiration about the analysis of the functioning of the ribosome, the cellular organelle responsible for protein synthesis, which manages to follow genetic instructions to assemble the perfect sequence of amino acids (Ibidem, p. 238–240). The functioning of these organelles in terms of code interpretation is so perfect that their margin of error is negligible, even if the cell itself presents a series of imperfections. Inventors like von Neumann, one of the most influential pioneers in the history of computing, showed that digital coding allows imperfect circuits to compute perfect answers; hence, with digital fabrication, Gershenfeld hopes that imperfect machines will make perfect and extremely complex parts.

The information technologies that developed upon the work of von Neumann and Claude Shannon made possible engineering of unprecedented complexity (GERSHENFELD, 2005, p. 244). "However, the exponential improvements that we've come to rely on, such as processor speeds doubling every few years, must stop when current scaling trends run into basic physical limits. Circuits cannot have wires smaller than atoms, signals faster than light, or charge carriers less than an electron" (GERSHENFELD, 2000, p. 1). Thus, however rapidly current technologies are evolving, they are still bounded by the known laws of physics, and present-day uses of communications and computing in information manipulation already operate close to many physical limits.²⁵ More innovation also requires new scientific theories. New scientific theories, according to Kuhn, require paradigm shifts that only occur when previous theories reach exhaustion (FULLER, 2003). Or, according to Cassirer, changes in the general theories of science are tied to broader cultural transformations that alter the symbolic structures of a society (CASSIRER, 1996a).

In the quest to catalyse the development of innovative projects based on digital fabrication technologies, Gershenfeld came to the conclusion that it was important to provide access to such equipment in the most varied social contexts possible. After all, people with different origins and needs could understand and create functions for the still incipient digital fabrication that the students and faculty of great university centres would be incapable of conceiving (GERSHENFELD, 2005, p. 32). Moreover, in analysing the negative impacts of the incorporation of digital technologies into society, in terms of the increase in inequality between social groups, researchers like Gershenfeld and his brothers came to defend the democratisation of the new technologies as a political necessity (GERSHENFELD et al., 2017, p. 170). For these reasons, in 2002 the project known as fab lab was begun (GERSHENFELD, 2005, p. 11–13), which aims to create digital fabrication laboratories with at least a basic kit of machines and a structure of mentors capable of teaching

24. In the fields of architecture and the automotive industry, a more widely debated concept similar to what Gershenfeld calls the third digital revolution is the notion of "Industry 4.0". The insertion of steam engines in the eighteenth century is considered the beginning of modern industry; the division of specialised labour at the start of the twentieth century a second phase; the introduction of programmable logic controllers (PLCs) during the 1970s a third phase. The fourth phase, Industry 4.0, is still under development and is tied to communication between humans and machines through great networks within cyber-physical systems (CPS) (BRETTEL et al., 2014, p. 37).

25. Various solutions to these limitations are currently discussed — for example, the verticalisation of computational memory. These would be design improvements aimed at efficiency, and there would still be much room for development. However, they would cease to be exponential improvements (GERSHENFELD et al., 2017).

anyone to use them. These laboratories can be implemented anywhere in the world, but from the start of the project they focused on areas with material difficulties, such as scarce access to electricity. The presence of a local inhabitant with advanced technological knowledge who could bridge to MIT seems to have been essential in the first fab lab experiments in places like the interior of southern Ghana and a rural area of western India, at the Vigyan Ashram school. The tone of Gershenfeld's text when writing about his experiences in 2005 makes clear a hope that such projects could lead to impactful economic transformations — as might have been the case of the instrument developed in Ghana for refrigerating food through a system that captures sunlight without needing to transform it into electrical energy first. In a region where much food was wasted owing to difficulties of preservation, such an undertaking could have significant consequences if it really continues to be developed (Ibidem, p. 80–88).

More than 15 years after the fab lab project began, there are currently more than 1,000 fab labs spread across the world (FABFOUNDATION, 2016); the FabAcademy was founded as a non-fixed structure to support the learning needed to expand the fab labs (FABACADEMY, 2017); and projects such as the FabCity were created — ideally, cities capable of fabricating and recycling everything they consume, importing only data (FABCITY, 2017) — among others. Digital fabrication is still largely an experimental technology, principally from the point of view of the wider consuming public, although it has begun to be more accessible in recent years (GERSHENFELD et al., 2017). Gershenfeld's most recent publication (Ibidem) was made in collaboration with his two brothers, who counterbalance the MIT physicist's optimism with critical readings from the standpoint of the social sciences, attempting to analyse the current scene of digital fabrication laboratories, both fab labs and those designated by the broader terms Makerspaces or Hackerspaces. If the social impact of the fab labs still seems rather restricted, their growing influence among governments, companies and various social projects is significant (Ibidem, p. 10). In Gershenfeld's own view (2005), the impact of social fabrication will be truly relevant to the extent that it is connected to changes in social organisation, with control of production returning to the hands of consumers. For that, it is necessary to go far beyond purely technological questions — or rather, it is necessary to make the very debate around technology more diverse.

2.3 Algorithms and Pedagogical Proposals Oriented towards Multiculturality

According to Hans Belting's thesis (2008), owing to their cultural differences, East and West historically present distinct uses of geometry in art: in the West one finds 'representative geometry', such as linear perspective, used to create a supposedly realistic image of reality, while in the East — in this case the Arab world — a 'represented geometry' predominates: geometry itself is the object of artistic interest (Ibidem, p. 68). Arab mathematics and optics up to the fifteenth century spoke above all of a theory of light, whereas in the West it took on the character of the active gaze, of the representation of that gaze's visual perception. It is also during the Renaissance that the notions of applied art and of art as intellectual activity take shape, linking the latter to an idea of authorship and of representation/knowledge of reality. Arab ornamentation thus remained influential in the West, much copied by the Italians and even by Albrecht Dürer, but it was understood as a minor art, for the great art was representation — principally the representation of the great biblical scenes (Ibidem, p. 70).

Nevertheless, both are geometries; both affirm images constructed upon mathematical logics. Following the argument presented by Belting in "Florence and Baghdad", one could say that the Arab notion that the world of images is deceitful — and that the abstract and perfect harmony of geometry is therefore essential to 'purify' the gaze — was subverted by Renaissance Italy. Geometry becomes the very structure of the image with linear perspective, making the power of two-dimensional representation to confound the senses even more potent and convincing through mathematical rigour. In the context of digital images, we have this same process taken to another level: polygonal geometries²⁶ of great complexity are needed to present a three-dimensional model of reality. Mathematical descriptions of the world are the basis of today's complex of simulation and manipulation technologies — voice recognition through artificial intelligence techniques in a smartphone, rocket design, or any environment in Virtual Reality. The mathematics of illusion.

As discussed earlier in this work, for Manovich the logic of the computer is based on the interaction between database and algorithms. Every mathematical simulation that represents a given phenomenon will thus depend on the transformation of the initial input through that interaction. Hence the discussion of algorithmic aesthetics and of the use of algorithms in much software in frequent use today by architects, designers and animators. But algorithms, as appears even in the definition presented above by Gips and Stiny (1978), do not necessarily have to be executed by computational processes: a person with a pencil can perfectly well carry out a given sequence of instructions. Many human activities follow finite, ordered instructions — that is, aesthetic systems based on algorithmic processes have been present in human history since long before the Computer Age. We could thus speak of algorithms present in a variety of geometric art traditions across the globe, such as Arab ornamentation, among many other examples.

The Mozambican ethnomathematician Paulus Gerdes is responsible for a vast academic production on what he regards as geometric algorithms produced by diverse cultures. He describes his practice as a way of

²⁶. There are countless approaches to algorithmic images today. In the case of 3D computer graphics, one can speak of polygonal modelling, NURBS surfaces, subdivision surfaces, and equation-based representations used in ray tracers.

'defrosting' the mathematics of colonised peoples: if much of the knowledge of such groups was lost, it would still be possible to reconstruct, or 'defrost', the mathematical thought hidden in traditional techniques such as mural painting, weaving and diverse forms of ornamentation (GERDES, 2004). As a visiting researcher at the Musée de l'Homme in Paris, Gerdes compiled basket-weaving practices from across Africa, the Americas, Asia and Oceania, analysing the mathematical concepts behind them and their similarities and differences (GERDES, 2000). His book *Women, Art and Geometry in Southern Africa* (GERDES, 1998), in turn, presents productions traditionally made by women, calling into question the low representation of the female gender in the history of science while proving how inventive their design solutions are. In short, Gerdes devoted his life to organising a database of geometric algorithms of long-lived cultures spread across the globe, in addition to developing teaching methodologies based on that research.

Gerdes's work arose from the need, in a newly independent country — Mozambique in the 1980s — to train primary school teachers in various areas. If there was a shortage of teachers in every discipline in a country struggling to create its own public education structure, in mathematics the lack was even graver: not only is the discipline seen as strategic in a nation's industrialisation process, but the discipline itself was understood as something foreign to local culture (GERDES, 2010, p. 18). Students understood mathematics as a European subject, dissociated from their daily lives and their traditions. Developing a teaching methodology that started from traditional activities — from weaving to storytelling — was therefore a strategy to engage students and make them perceive the richness of their own history.

But Gerdes's work goes well beyond an appropriate didactics for understanding basic concepts. In his study of the sand drawings of the Cokwe of Angola, which we shall see in the next chapter, Gerdes manages to associate the traditional drawings with circulant matrices²⁷ and discusses the relevance of such matrices in research carried out during the 2000s on genetics (GERDES, 1999, 2000, 2010). Something similar appears when the author moves from the study of the weaving techniques of pyramidal baskets, used in Mozambique and Tanzania as funnels in the salt-production process, to the structure of the five regular polyhedra (Platonic solids), and shows how from such concepts and proposed exercises one can arrive at discussions of contemporary chemistry.

The artisans who created such baskets used a hexagonal weaving technique and introduced a few pentagonal holes in order to create curvature over the planar faces at the folds of the basket. Gerdes also adds the example of the millennia-old tradition of the ball of the Malaysian game 'sepak raga', in which all the holes are pentagonal, twelve in all (GERDES, 1999, p. 113). The author recounts that until 1985 only two crystalline forms of carbon were known — graphite and diamond — when Curl, Smalley and Kroto discovered a third form, C₆₀, named 'buckminsterfullerene' in homage to the avant-garde American architect R. Buckminster Fuller (1895–1983), famous for his geodesic dome designs. They conjectured that this molecule had the symmetrical structure of a truncated icosahedron — that is, of a 'sepak raga' ball. This theory was the reason the three scientists were distinguished, in 1996, with the Nobel Prize in Chemistry (GERDES, 1999, p. 119–125). Gerdes presents as an exercise for secondary school students an explanation of

27. In Gerdes's book (2010) published on the subject in Brazil, the term "matrizes cíclicas" (cyclic matrices) is used. However, the most common name for this type of matrix in Brazil is "matriz circulante" (circulant matrix), which is why this spelling was chosen.

the pyramidal basket-weaving technique, so that they then build polyhedra and finally compare those polyhedra to models of 'buckminsterfullerene' carbon molecules (Fig. 13).



Fig. 13: On the left, a representation of the carbon molecule known as 'buckminsterfullerene'. In the centre, a 'sepak raga' ball. On the right, a representation of the weaving technique and of the ball made by Ron Eglash's students following Gerdes's exercise. Source: <https://csdt.rpi.edu/culture/legacy/african/hex/intro.html>

Hexastrip models of fullerenes and related carbon structures are not only beautiful and relatively easy to make from inexpensive materials — and as such are attractive from a didactic viewpoint — but also they give a clear picture of the bonding situation [of the atoms]: the edges of the holes represent the single bonds and the folds (that is, where woven hexagons are adjacent) the double bonds. Given a hexastrip model, it is not difficult to determine the number counting of vertices. [...] It is interesting to note that I introduced hexastrip models in the early 1980's — before the discovery of fullerenes — during the exploration of possibilities of incorporating the hexagonal basket weaving technique into the teaching of geometry in Mozambique. (GERDES, 1999, p. 123–124)

Gerdes describes a series of possible exercises based on hexagonal weaving, suggesting questions to be put by teachers to their pupils, in an attempt to lead them to deduce properties of the assembled polygons and to ask themselves what other forms might also be possible through similar processes. This pedagogical approach — which manages to discuss and connect characteristics of materials, weaving techniques, and chemical and mathematical concepts — has similarities with Neil Gershenfeld's proposals for the fab labs. The American researcher comments several times on how humans in areas with scarce infrastructure are frequently more inventive than students in privileged conditions, and on how the connection between areas of knowledge and the understanding of all stages of the productive process — from the weaving of fabrics to the design of the final product — are advantages for good scientific research or design projects (GERSHENFELD, 2005, 2017).

At the end of the book Gershenfeld produced together with his brothers (GERSHENFELD et al., 2017, p. 199–201), there is a section with examples of activities carried out by members of the fab lab community and hypotheses for future impacts of the fab lab network on the world, among which stands out: "What if advanced technologies preserved and extended ancient cultures?" The story is then told of the indigenous Iñupiat community of Alaska, based in Anchorage, whose administrative council formed a partnership with a video game company, E-Line Media, to produce a commercial product that would share and celebrate their native culture for a global audience. Alan Gershenfeld, one of Neil's brothers who co-authors the cited book,

is one of the founders of E-Line Media, which presents itself as a pioneering company in the games sector for its focus on positive social and educational impact.²⁸ Adapting traditional stories and the aesthetics of the ivory carvings called scrimshaw, the video game *Never Alone* — or *Kisima Injitchuḡa* in the Iñupiat language — was created collaboratively with more than 30 indigenous elders, writers and storytellers, along with internationally renowned video game developers (Fig. 14–15). The commercial success of this venture, which generated more than 3 million downloads, financed further projects devoted to education and the development of new technologies for the community, including a fab lab. The Gershenfeld brothers then ask how projects similar to the fab labs could go deeper into the connection between traditional cultures and technological development in the quest to secure the future of those cultures. They describe a hypothetical scenario dated 2030 in which a network of fab labs is accessible to indigenous populations throughout Alaska, and traditional knowledge of local materials is used within the network to reduce dependence on external products and the inefficiencies of supply chains:

The design and operation of a fab lab is also now seamlessly integrated with ancient cultural traditions. The use of a laser cutter to produce scrimshaw is understood as part of a long progression from the use of stones, to metal knives, to electric Dremel tools in carving scrimshaw. Kids who first learned to design and carve scrimshaw on a laser cutter are also interested in mastering the use of older tools. This is not an accident. The design of the fab labs includes workspaces for both modern products and services, as well as traditional crafts and practices. A mix of similar practices are shared with other cultures around the world, from the most remote parts of the planet to teeming urban centers.

Beyond being the place where ancient crafts are appreciated and practiced, fab labs are where stories and traditions are passed down from generation to generation. [...] Ultimately, what is being fabricated in these labs is continuity in the culture itself. (GERSHENFELD et al., 2017, p. 201)



Fig. 14: Main characters of the game 'Never Alone'. Source: E-Line Media

28. More information about Alan Gershenfeld's company at <http://elinemedia.com/about/>



Fig. 15: The game 'Never Alone', graphics based on the traditional Alaskan technique known as 'scrimshaw'.
Source: E-Line Media

The educational proposals of the fab lab start from the principle that knowledge of the diversity of production techniques can facilitate the process of innovation. Similar ideas appear in the doctoral thesis of the American Nettrice Gaskins (2014), in which she describes her pedagogical experiences with students from under-represented ethnic groups — in her case principally Black students in the State of Georgia in the United States. Her pedagogical method draws on the techniques called 'art-based learning' and on 'techno-vernacular creativity' — for example, the inventive capacity of DJs and rappers in using discarded electronics to create sound systems. Working under the supervision of the American ethnomathematician Ron Eglash, Nettrice starts from simple operations of rotation and repetition to create patterns, then transcribes those operations with the students into algorithmic language, creating new patterns, and finally shows the students a series of possible applications of such patterns, from the pictorial projects of celebrated artists to augmented reality experiments over mural paintings. The students even had lectures and mentoring from artists during Nettrice's PhD, and were able to take part in the creation of augmented reality apps (Ibidem). Gaskins became director of a digital fabrication laboratory in Boston soon after finishing her PhD and is currently working for the FabFoundation (KLIMCZAK, 2014).

Ron Eglash is an ethnomathematician of special relevance: besides being very active as a researcher, owing to his collaborations with artists of considerable reputation (GASKINS, 2014, p. 44) and principally to his talk at TED Global in 2007 (TED, 2017), he is the ethnomathematician who has achieved the greatest visibility beyond academic frontiers at an international level. He also considers himself a representative of cybernetics, defined as the interdisciplinary study of the structure of control of human or societal behaviour by rules or constraints (BERT et al., 2009, p. 1). Like Gerdes, Eglash also seeks to work with complex mathematical concepts based on traditional African cultures, having focused initially on questions relating to fractal geometries (1999). It was from his research on the preponderance of fractals as a design theme in the cultural productions of various regions of Africa that Eglash began to develop the tools called "culturally situated design tools" (CSDTs).

Eglash moves from ethnomathematics towards what he calls ethnocomputing. The CSDTs are software tools accessible online that allow students to create simulations from the mathematical principles of diverse cultural productions, principally of cultural groups frequently seen as primitive. They are expressive computational media that open new opportunities to explore relations between identity and culture, or the cultural construction of mathematics and computing. The objective of Eglash and his collaborators is to create a culturally enriched computational medium that offers students new opportunities for identity self-construction, through contact with tools and critical perspectives on the social and technical domains, as well as the interrelations between the two (Ibidem, p. 352).²⁹ "Computational ethnography helps us make our own assumptions visible; we begin to see that some 'self evident' aspects of math or technology are actually choices that could have been otherwise." (EGLASH et al., 2006, p. 350).

Eglash contrasts the development of the CSDTs with what he regards as the pedagogy of "multicultural mathematics", which merely uses the already existing pedagogical structure and swaps a few names and examples. In his understanding, the CSDTs are distinguished, first, by recognising deep design themes — such as the grid system for many European productions and fractal geometries for African ones — in short, coherent structures of the knowledge systems of a given society. Second, they are an anti-primitivist form of representation: by revealing the mathematical sophistication of such cultural groups, ethnomathematics challenges epistemological stereotypes about ethnic minorities. Third, the tools seek to make a translation of the mathematical principles present in the various cultures and not merely analyse them from the Western point of view, recognising the intentionality and originality of the cultural groups in their development of abstract concepts. Fourth, the CSDTs present a dynamic rather than static conception of culture, and therefore seek to bring those cultural productions into the students' daily lives, in addition to recognising the contemporary practices of ethnic minorities, such as the culture of DJs, rappers, graffiti, street vendors and so on. To this end, several CSDTs were developed, each best suited to a particular exercise, many of them working in a manner similar to Grasshopper, the Rhinoceros plugin discussed in section 2.1, as graphic editors of algorithms (Ibidem, p. 349).

Eglash, like most ethnomathematicians, focuses on the cultural productions of groups frequently seen as 'primitive' precisely because of the scarcity of research and above all of recognition regarding the sophistication and intentionality in those peoples' formulation of abstract mathematical concepts. It is, therefore, a political confrontation with the widely diffused notions of cultural determinism, which in the educational milieu, for various ethnic minorities, end up becoming self-fulfilling prophecies (Ibidem, p. 355). The question of sub-Saharan Africa is of extreme relevance owing to the catastrophic consequences of the slave trade for the preservation of local cultures, and to the perception by other peoples of the culture of the enslaved as something inferior — which attempted to justify the continuation of slavery. Moreover, as Eglash and Gerdes show in their research, sub-Saharan Africa is extremely rich in mathematical techniques and concepts different from those found most regularly elsewhere. The book that originally brought repercussion to Eglash's research, *African Fractals* (1999), defends fractal geometry as something developed in Africa long

29. To learn more about the characteristics of the CSDTs and the assessment of their implementation in schools in the USA in recent years, the reading of "Standardizing Culturally Situated Design Tools" by Christina Bert, Keshia King, Aiyana Brown and Deanna Watkins, 2009, is recommended; see bibliography.

before, and in a manner distinct from, what was belatedly 'discovered' in Europe — where its principles only began to be discussed in formal mathematics at the end of the nineteenth century with Georg Cantor, and its formalisation, the name 'fractal geometry', appears only in the second half of the twentieth century with Benoît Mandelbrot. Writing about the development of the CSDTs, Eglash also comments on how some algorithms traditionally used by indigenous groups proved more efficient than their counterparts most used in the digital sphere today (EGLASH et al., 2006, p. 353). The study of ethnomathematics would thus also be relevant to technological innovation.

A grave issue regarding the necessity and difficulty of the large-scale democratisation of new technologies is that the continual need for innovation is a clear characteristic of the knowledge and presuppositions that sustain them. One needs an entire social structure capable of constantly developing innovations, and costly research institutes are still the most relevant agents in this process. Even in the case of ethnomathematics, its flourishing was tied to great university centres, and its development in the direction of ethnocomputing and new technologies is centred principally in the USA. According to Yaovi Akakpo (2016), the very logic of innovation as an end in itself comes into conflict with traditional social organisations, with productive models based on craftsmanship, which seem less and less able to satisfy their population's needs to adapt to the pace of world changes. "The choice of active inventiveness demands that the arts and crafts which serve traditional technologies be transformed. It also requires that the intellectual categories that produce traditional technologies be transformed."³⁰ (Ibidem, p. 71)

Akakpo analyses the European process of modernisation of artisans' workshops in the seventeenth and eighteenth centuries in order to discuss the difficulties of today's West African countries in keeping up with global transformations in production chains. The Togolese philosopher brings up the example of Francis Bacon, who suggested during the seventeenth century that the practices of artisans be studied scientifically and that the artisans themselves go to the school of science. Similar ideas were also defended by Descartes; Akakpo considers such thinkers the 'annunciating fathers' of the West's technological project, and briefly narrates the history of the "Schools of Arts and Crafts". In his argument, the industrial revolution in the West began with the birth of schools of engineering and professional training, which initially incorporated industry into the knowledge of the arts and crafts. He concludes that the lesson to be learned is how essential the binding of industry's destiny to intellectual work, to the arts and to the practices of artisans is for the sound development of industry itself.

Developing societies can appropriate technological rationality by improving the centuries-old practices that the arts and crafts offer. This can be understood by considering the numerical importance of the men and women who content themselves with traditional technologies for lack of anything better. But it is also true that the mere improvement of traditional practices is justified only when it satisfies economic and survival needs. That is why the future of traditional technologies must be negotiated with the political and economic destiny of nations.³¹ (AKAKPO, 2016, p. 72)

The philosopher argues, however, that it is also important that the new technologies chosen for development in developing countries be those that generate less alienating economic relations. For technology

cannot be a simple question; it is above all a weapon of economic struggles and of the recognition of the sovereignty and autonomy of nations (Ibidem, p. 66). Developing countries, unable to keep pace with technological innovations, remain in the position of consumers and suppliers of raw materials in the world economy, thereby maintaining — and at times accentuating — their dependence on the political interests of the central countries (Ibidem, p. 62). "It was necessary to indicate that technology is a political and economic instrument in order to underline why, although it has the vocation of being universalised, its choice by nations and economic groups is not made without stakes." And he later adds that "the holders of technology have no vocation to share their monopoly"³² (Ibidem, p. 60–61).

Even so, Akakpo recognises that the growing multiculturalism present in today's central countries themselves is a motivator for reflection on diversity and representation within the discourse and practices of technological development (Ibidem, p. 130). The ideas defended by the field of ethnomathematics seem to be in accord with Akakpo's notion of technologies capable of generating less alienating relations for the social milieu in which they are implemented. In a similar way, the Gershenfeld brothers argue for the necessity of making an effort as soon as possible so that the coming technological revolutions do not accentuate the already existing inequality of access to technology. To this end, they argue that the social sciences need to take a more active role as creators of paths and alternatives, since at present a good part of social scientists and of other related disciplines are more concerned with reacting to and dealing with the consequences of the implementation of technologies than with designing possible models for their development and integration into the social milieu (GERSHENFELD et al., 2017, p. 123). They argue that it is easier to have influence over the consequences of technologies while they are still experimental than to alter what has already become embedded in society and has already been captured by big capital (Ibidem, p. 125). We would thus need every possible effort to think technologies from their conceptual base in the direction of their possibilities for liberation and the democratisation of knowledge, in order to try to design a more egalitarian future.

31. "Les sociétés en développement peuvent s'approprier la rationalité technologique par une amélioration des pratiques centenaires qu'offrent les arts et métiers. Cela peut se comprendre en considérant l'importance numérique des hommes et des femmes qui se contentent des technologies traditionnelles faute de mieux. Mais il est aussi vrai que la simple amélioration des pratiques traditionnelles ne se justifie que lorsqu'elle donne satisfaction aux besoins économiques et de survie. C'est pourquoi le devenir des technologies traditionnelles doit se négocier avec le destin politique et économique des nations." (AKAKPO, 2016, p. 72) — Translated from the French.

30. "Le choix de l'inventivité active demande que les arts et les métiers que servent les technologies traditionnelles soient transformés. Cela requiert aussi que les catégories intellectuelles qui produisent les technologies traditionnelles soient transformées." (AKAKPO, 2016, p. 71) — Translated from the French.

32. "Il fallait indiquer que la technologie est un instrument politique et économique pour pouvoir souligner pourquoi, tout en ayant la vocation à s'universaliser, son choix par les nations et les groupes économiques ne se fait pas sans enjeux." ... "Les détenteurs de la technologie n'ont aucune vocation à partager leur monopole." (AKAKPO, 2016, p. 60–61) — Translated from the French.

CHAPTER 3

Exercises in Algorithmic Design

3.1 Research and Applications of Ethnomathematics: Paulus Gerdes and the "Sona"

The Cokwe people, originally from present-day Angola, have a tradition of gathering around a fire or in the shade of leafy trees to enjoy good conversation while illustrating it with drawings in the sand. One of these tales narrates the encounter between a hunter and a chicken in the bush. The chicken sees the man approaching and starts to run. In the story, the bird is a symbol of cunning and intelligence; so, although it is fairly close to the entrance of its home, it decides to run in wide zigzags — for if it ran in a straight line it would be caught, or the man would learn the location of its hiding place. Running in zigzags, however, is not enough: it would only take the chicken further from home, and the hunter would eventually understand its tactic and manage to catch it. So the chicken changes direction, runs in a narrower zigzag, changes once more, speeding up its run in a straight line. It gets close to home and sees that the hunter is still following, so it resumes the zigzag, parallel to the first stretch of the run. Once more it changes direction and accelerates, alternating between zigzags and fast straight lines, until it manages to return home while the hunter is left tired and confused.

This story is part of the book *Da etnomatemática a arte-design e matrizes cíclicas* (From Ethnomathematics to Art-Design and Cyclic Matrices) by Paulus Gerdes (2010, p. 34–38), but it also appears in other formats in several other publications by the same author (1999, 2004). In the first book cited, the researcher narrates in the first person his experience of telling this story to diverse audiences while having them follow the execution of a geometric algorithm that illustrates the chicken's saga. Here is the representation of the runs:

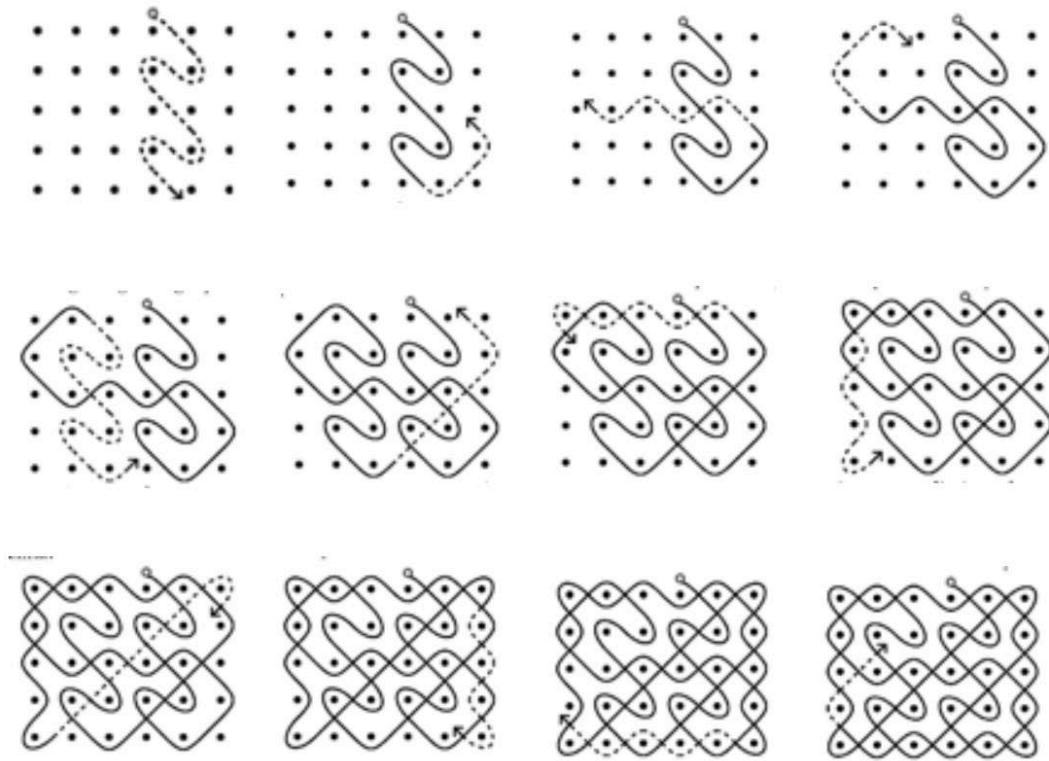


Fig. 16: Geometric algorithm of the 'chased chicken' type in execution. Source: GERDES, 2010, p. 35–38

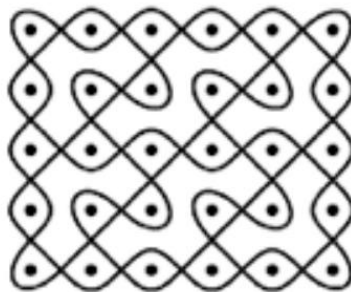


Fig. 17: Geometric algorithm of the 'chased chicken' type, completed. Source: GERDES, 2010, p. 38

According to Gerdes, hundreds of drawings like these — all geometric algorithms, methods for constructing what the Cokwe call "sona" drawings,³³ generally symmetrical and monolinear — "played an important role in the transmission of knowledge and wisdom from one generation to the next" (GERDES, 2010, p. 34). Gerdes begins his explanation of the "sona" with "the smallest rectangular mat that can be woven with a single strip", presenting a type of knot that corresponds to a 'sona' drawing. The plaited knot corresponds to the idea of friendship in various African cultures; it is a symbol of union. But "sona" drawings were also used to represent a whole series of animals, stories or actions, such as the chicken's run. Gerdes

³³. The spelling of the word "sona" is always presented between quotation marks, respecting the graphic emphasis given to the word in the cited publications of Paulus Gerdes. The same applies to other terms, such as 'Lunda-design'.

points to differences between the patterns of each type of representation: each follows a specific algorithm, which can vary according to the dimensions.

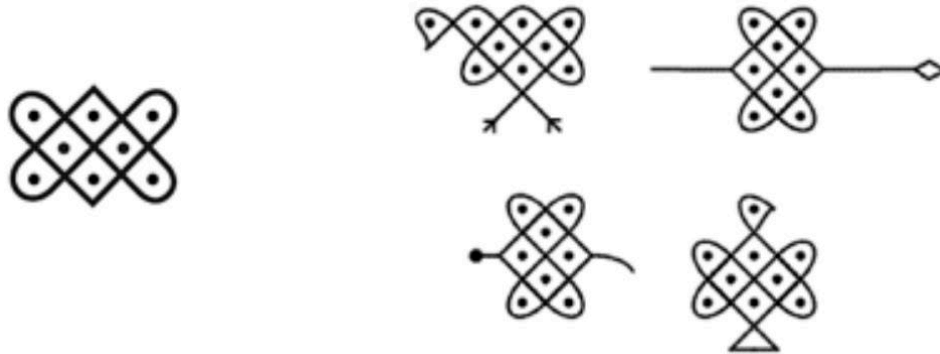


Fig. 18: "Sona" drawing representing friendship, on the left, and drawings of animals built from it by adding a few lines. Source: GERDES, 2010, p. 33–34

There is one algorithm for drawings of the 'chased chicken' type, another for drawings of the 'lion's stomach' type, or for telling the story of the fight between two rival chiefs; one algorithm makes drawings that represent fire, another, ant traps. The author comments on the work of the Portuguese ethnographer Fontinha, who managed to publish 287 different "sona" drawings, collected during the 1940s and 50s, when the culture of sand drawings was already in the process of disappearing (Idem, 1999, p. 159). The slave trade was responsible for the near eradication of the "sona", since they depended on the oral transmission of knowledge, passed only to men and fully mastered only by a few specialised elders who enjoyed great social prestige, called *akwa kuta sona*. The Ngangela and Luchazi peoples, located between Angola, Zambia, Congo and Zaire, also had similar productions. The Cokwe are known for the variety of their decorative art, which ranges from ornamentation on mats and baskets, ironwork, ceramics, sculpture and engravings on calabash to tattoos, paintings on house walls, and the sand drawings called 'sona', or 'lusona' (Idem, 2004, p. 10).

Fig. 19-21

Fig. 19–21: "Sona" drawings made through different algorithms. Each represents a story, action or animal/object. Source: GERDES, 1999, p. 162–166

Gerdes explains how the *akwa kuta sona* first cleaned and smoothed the sandy ground before marking with their fingers an orthogonal net of equidistant points. To ease the memorisation of their drawings, mnemonic systems were invented:

To facilitate memorizing their standardized drawings the *akwa kuta sona* invented the following mnemonic device: after cleaning and smoothing the ground, a drawing expert marks with the fingers of his right hand an orthogonal net (grid or array) of equidistant points or dots. Often an additional series of points is then marked in the centers of the unit squares of the first grid. After counting and checking the various points in each row, the drawer traces a line figure with the tip of his right index finger. Normally the drawing is composed of one or more lines that embrace the points of the net.

Both the number of rows and the number of columns of the array of points as well as the rules for executing drawings depend on the motif to be represented. By applying their method — an example of the use of a coordinate system — the akwa kuta sona reduced, in general, the memorization of a complete design to that of a few numbers (the dimension of the lattice of points) and of a geometric algorithm for the construction of the curve. (GERDES, 1999, p. 158)

The mathematician clarifies that most "sona" exhibited bilateral and/or rotational symmetries (90° or 180°), which made their execution easier. The drawings had to be executed continuously and with clean movements, for any hesitation would be seen by the audience as a lack of knowledge. Gerdes further informs us that the simpler drawings were made while a story was being told, whereas the more complex ones required moments of silence. Once the story and the drawing were finished, they were erased from the sand — an extremely ephemeral art.

Each algorithm has different properties, which Gerdes works through in the search for formulas of functions and tables to represent each one. Symmetry and monolinearity played an important role as cultural values. Monolinear means composed of only one line: one part of the line may cross another part, but no part of the line may ever touch another part. The Mozambican mathematician thus proposes to the reader mathematical exercises to be worked from the patterns. To this end he explains that the "sona" drawings work like the so-called 'mirror curves'.



Fig. 22: Representation of small double mirrors placed transversally between the points of the grid. Source: GERDES, 2010, p. 45

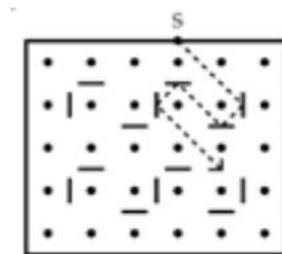


Fig. 23: Representation of the 'chased chicken' algorithm following the logic of mirror reflection. The rectangle enclosing the grid would also be mirrored. Source: GERDES, 2010, p. 44

Imagining that each line represents double mirrors, and that the rectangle around the net of points also functions as a mirror, one can understand the path made by the "sona" drawings as if it were a ray of light being continually reflected until it meets its point of origin again.

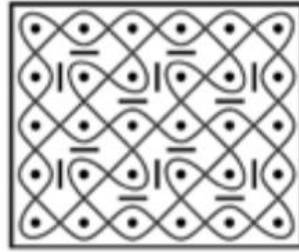


Fig. 24: Representation of the 'chased chicken' algorithm with its mirror curves. Source: GERDES, 2010, p. 45

Over his lifetime the mathematician was able to analyse similar geometric algorithms in ancient Egypt, monolinear images from ancient Mesopotamia, drawings of the Tamil women in India, Celtic knot decorations, sand drawings from the islands of Vanuatu (Oceania) and some pattern designs from other parts of Africa (GERDES, 2004, p. 13). He suggested that the methodology used in the reconstruction of the "sona" tradition be developed in the study of the other traditions. Marcia Ascher (2002), an American ethnomathematician and frequent collaborator of Gerdes, developed extensive research on similar drawings in the Tamil culture of southern India.

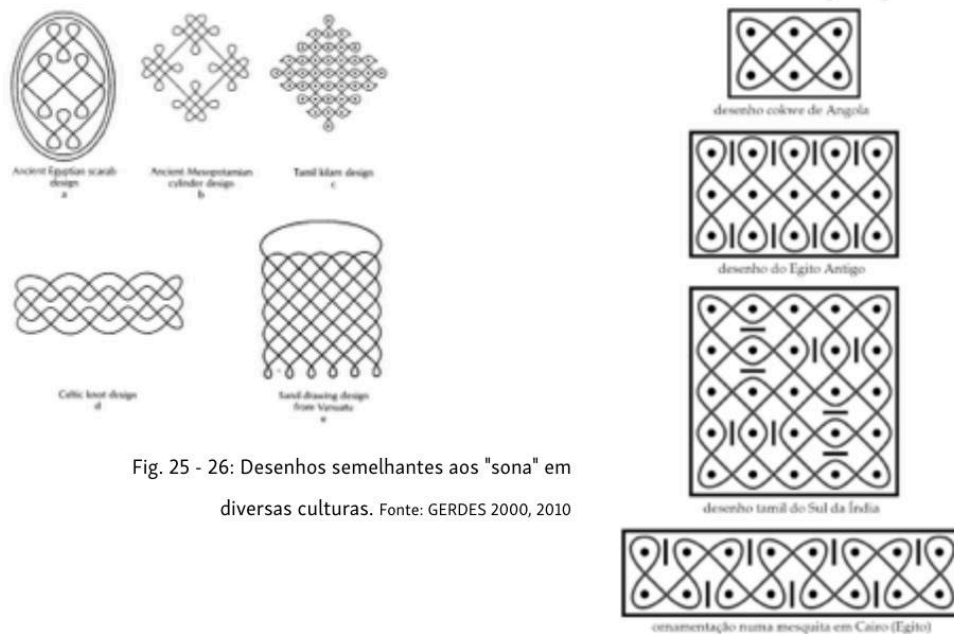


Fig. 25 - 26: Desenhos semelhantes aos "sona" em diversas culturas. Fonte: GERDES 2000, 2010

Fig. 25–26: Drawings similar to the "sona" in various cultures. Source: GERDES, 2000, 2010

Comparing the "sona" with Celtic designs, found in stone decoration, manuscripts and jewellery in the British Isles, Gerdes recognises different mirror curves, with another type of positioning of the small mirrors between the points of the grid, which the mathematician calls the 'generalised mirror curve', whereas most "sona" would be regular 'mirror curves' (GERDES, 2010, p. 60).



Fig. 27: Drawings of the 'generalised mirror-curve' type and the positioning of their mirrors. Source: GERDES, 2010, p. 60–61

Going deeper into the mathematical analysis of the algorithms, Gerdes proposes numbering the path made by the mirror curves of the "sona" drawings of the 'chased chicken' type, forming the following table.

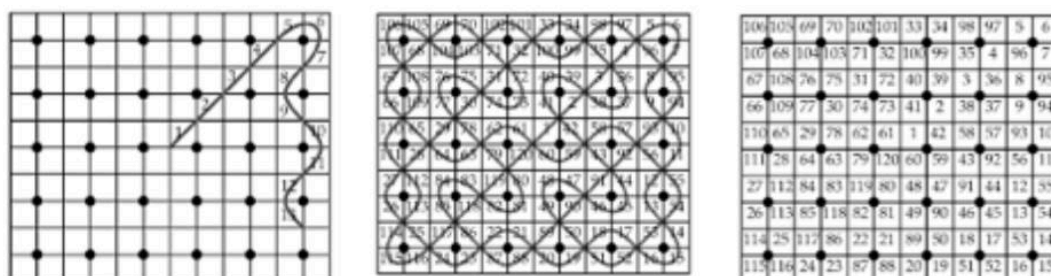


Fig. 28: Path of the line of the 'chased chicken' drawing being numbered and turned into a table. Source: GERDES, 2010, p. 50–51

"The first number of the first row, 106, corresponds to the last number of the last row, 15; the second number of the first row, 105, corresponds to the second-to-last number of the last row, 16. In both cases, the sum of the numbers of the two corresponding little squares is equal to 121" (GERDES, 2010, p. 51).³⁴ This happens with all the corresponding squares of the table, which Gerdes treats as a mathematical matrix. Gerdes then begins to play with the mathematical possibilities of the matrix presented, asking whether it could be a magic numerical rectangle — that is, one in which, for all rows and all columns, the sums of the numbers of their constituent squares are equal. Demonstrating that in this case not all rows and columns yield equal numbers, but that there are several repeated sums, Gerdes proposes the exercise of analysing the matrix through the application of a modulus. "Let m be any natural number greater than 1. Two whole numbers p and q are called equivalent, or equal modulo m , if $p - q$ is a multiple of m ." (Ibidem, p. 54)

34. Quotation originally in Portuguese; translated by the translator.



Fig. 29: Representation of the count of the sums of the small squares applying modulo 8, and representation of how a modulo 4 count works. Source: GERDES, 2010, p. 54–55

After presenting the numbering of the matrix using several numbers as possible moduli, Gerdes concludes that the numerical rectangle is 'magic' with modulo 8, since $28=20=36=4$ and $11=27=19=3$ modulo 8. Observing that in several cases four consecutive numbers are found around the same point of the grid, the mathematician proposes numbering the matrix according to modulo 4. In this way he demonstrates that with modulo 4 one finds the numbers 0, 1, 2 and 3 around all the points of the grid, alternating their arrangement clockwise and anticlockwise, and moreover that in each square formed by 4 neighbouring points equal numbers are found (Ibidem, p. 57).

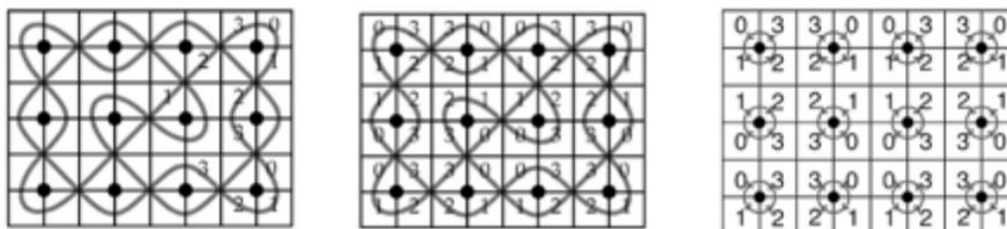


Fig. 30: Representation of the count of a 'chased chicken' algorithm using modulo 4. Source: GERDES, 2010, p. 56–57

In another part of his book *Da etnomatemática a arte-design e matrizes cíclicas* (Ibidem), Gerdes proposes the analysis of the matrix formed by a similar process applied to a 'mirror-curve' drawing of Tamil origin, from India, using modulo 2.

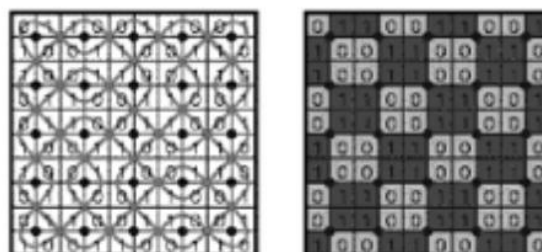


Fig. 31: Tamil drawing numbered with modulo 2. Light squares are placed for the number 0 and dark ones for 1, forming a pattern. Source: GERDES, 2010, p. 64

He concludes that "it can be demonstrated that all regular mirror curves generate similar matrices and dark-light patterns" (Ibidem, p. 64).

He then presents drawings that combine the regular and the generalised mirror curves, and applies modulo 2 to the matrix of such a drawing:

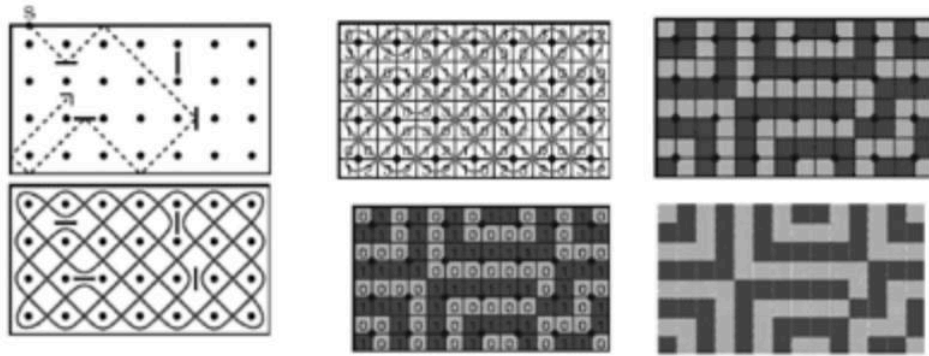


Fig. 32: Drawing mixing the types of mirror curve, creating a new kind of pattern. Source: GERDES, 2010, p. 65–66

Replacing the numbers 0 and 1 with different colours representing each, we see the appearance of a new type of pattern, which Gerdes names the 'Lunda-design', in homage to the region in northeastern Angola where the Cokwe live. The properties of the 'Lunda-designs' are:

- 1) In each row, the number of dark squares is equal to the number of light squares.
- 2) In each column, the number of dark squares is equal to the number of light squares.
- 3) Let M be any point on the border of the grid. The two squares against the edge that touch point M have different colours: one is light while the other is dark.
- 4) Of the four squares set between two (horizontally or vertically) neighbouring points of the grid, two are always light while the other two are dark.

(GERDES, 2010, p. 71–72)

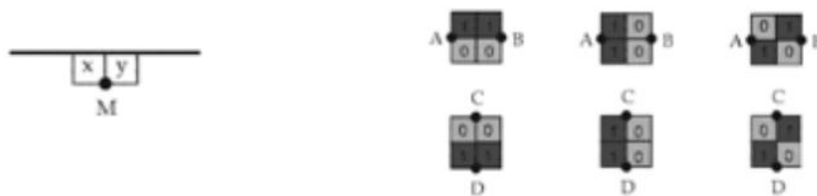


Fig. 33: Representation of the properties of 'Lunda-designs'. Source: GERDES, 2010, p. 72–73

It can happen that several mirror curves generate the same Lunda-design.

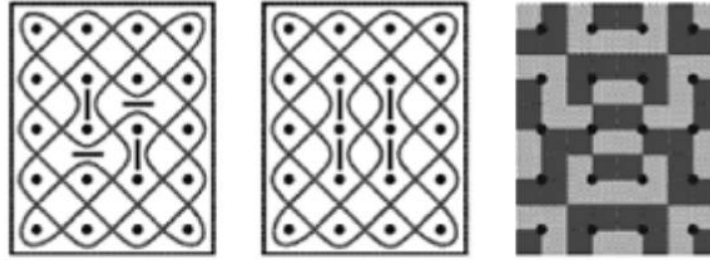


Fig. 34: Two mirror-curve drawings form the same 'Lunda-design'. Source: GERDES, 2010, p. 74

Gerdes experiments with fractals and with mosaic applications using the Lunda-designs:

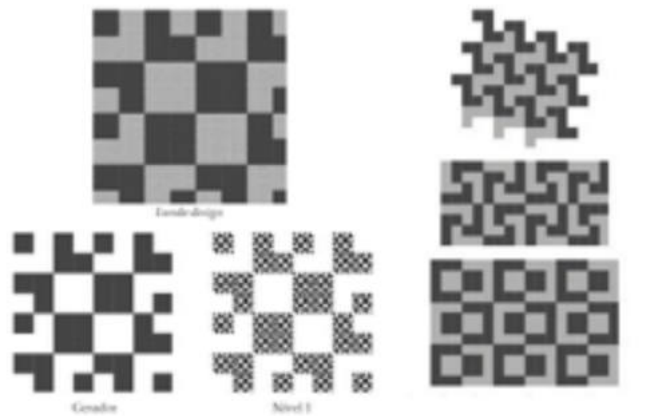


Fig. 35: Generation of fractals from Lunda-designs and mosaics. Source: GERDES, 2010, p. 84, 90

And he invents the Liki-designs, which are Lunda-designs with the following added restriction: "Of the four squares set between any two points, horizontally or vertically neighbouring on the grid, two adjacent squares are always light, while the other two are dark" (GERDES, 2010, p. 94).

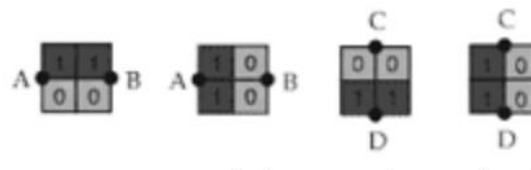


Fig. 36: Representation of the colour arrangement of Liki-designs. Source: GERDES, 2010, p. 94

Gerdes once again draws inspiration from the "sona" drawings to analyse the properties of the cyclic structures of the Liki-designs. He starts from a Cokwe drawing of a turtle, which is not monolinear: it presents three closed curves, to which the four legs and the head are then added. Each closed curve is considered a cycle. Each cycle will have two possible colourings. "In general, a Liki-design of dimensions $m \times m$ is composed of m cycles. There are 2^m Liki-designs of dimensions m by m " (Ibidem, p. 101).

After a series of operations and tests that we shall not delve into in this research, the Mozambican researcher arrives at the definition of circulant matrices and goes on to describe their properties. There is a brief elucidation of how circulant matrices constitute a new category in mathematical studies, giving as an example the research on genetic matrices by the Russian biologist Sergei Petoukhov and the Chinese-American mathematician Matthew He, published in the early 2000s (Ibidem, p. 149). According to Gerdes, several interesting properties of genetic matrices arise directly from the general properties of circulant matrices, which can use any rational, real or complex number. In his examples, Gerdes compares the positive and negative forms of the matrices not only to the "sona" drawings but also to the plaited mats made by Makwe women of northeastern Mozambique, since these always work on both sides of the material, each side being an inversion of the other, and their patterns exhibit behaviour similar to the matrices. According to the American physicist Steven G. Johnson (2017, p. 2), the multiplication of vectors by a circulant matrix is relevant to many applications, among them: digital signal processing, image compression, physics/engineering simulations, number theory and cryptography.

Deepening his experiments with the Lunda-designs, Gerdes (1999, p. 197–200) also presents hexagonal designs and designs with three colour symmetries, Lunda-polyhedral-designs and circular symmetric Lunda-designs, making clear the variety of graphic effects possible through play with these same principles.

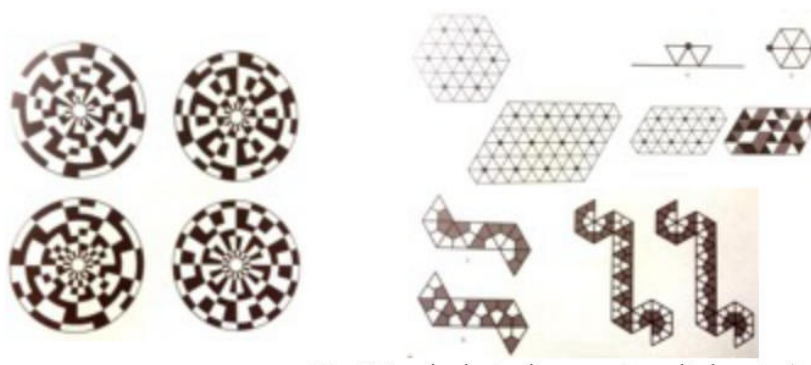


Fig. 39: Hexagonal, polyhedral, circular-symmetric, three-colour Lunda-designs. Source: GERDES, 1999, p. 197–200

Gerdes encourages his readers to use software such as Excel to make the calculation of the matrices faster, and so that one can experiment with deducing properties of the patterns and matrices presented. However, using computational languages and software more developed for graphic expression, such as Processing, it would be possible to create animations from the mathematical operations between the matrices behind the patterns, in addition to many other possibilities. Gerdes's analyses are extremely rich material for experiments in creative coding, such as those that will be presented in the next section of this chapter.

For Gerdes, the study of patterns like the "sona" is proof of mathematics as a universal activity, a defence of the historiographical thesis of the multilineal development of Mathematics, besides providing an approach to mathematics and education as socio-cultural processes (GERDES, 2010, p. 159–161). The presence of 'mirror-curve' drawings in various parts of the globe would show — as would be the case of ancient Egypt for

the Mozambican — that several distinct mathematical traditions can coexist in the same society, interrelated or not. In this sense, he asserts that it would become difficult, if not impossible, to speak of the mathematics of ancient Egypt, the mathematics of Greece, etc.

From an ethnomathematical perspective, mathematics becomes the product of all cultures, being the school mathematics experience of a researcher only one form of mathematical experience. Mathematics is not the product of a particular culture sphere, 'western', but a common human experience. In the process of studying mathematical ideas in diverse cultural contexts, the understanding of what mathematics is, or better of what constitutes mathematical activity, may be deepened. Mathematical thinking is only interculturally intelligible. And there lies a challenge for philosophy in general and for the philosophy of mathematics in particular: to contribute to the development of the means for intercultural intelligibility. (GERDES, 2004, p. 20)

3.2 Case Study: Infinity Game — Artistically Experimenting with Multicultural Algorithmic Design

This section is devoted to the analysis of an art exhibition developed from the study of ethnomathematics and algorithmic design. The project was entitled "Infinity Game" and is the fruit of a collaboration between the visual artist Vanessa Rosa, author of this dissertation, the architect specialised in parametric design Veronica Natividade, and the Chinese creative coder Wenqi Li. It was exhibited for the first time at the Babycastles gallery, a non-profit organisation devoted to experimentation with video games, located in Manhattan, NYC, in August 2017. The project can be seen at the link va2rosa.com/Infinity-Game, where there are photos and videos by the artists. Vanessa Rosa and Veronica Natividade had already developed other projects with a similar proposal. In 2016, Veronica's students created laser-cut stencils with parametric design tools, in particular Grasshopper. The stencils were used to make a collective mural painting inspired by traditional Yoruba mural painting, using the research of the Nigerian academic Bolaji Campbell as pictorial reference (CAMPBELL, 2008). This work was entitled the Sankofa Project³⁵ and aimed to claim political recognition for a quilombola group in Rio de Janeiro and to draw attention to the negligence of public and private authorities towards the Afro-descendant cultural and architectural heritage of the city's port zone. Considering that the project was carried out during the Rio de Janeiro 2016 Olympic Games, it was an essential moment to give visibility to dissident voices in relation to the political and urban project then being implemented in the city (Fig. 40–42). On the site va2rosa.com/Pedra-do-Sal-Visible-Memories there are photos by the artists and links to press reports on the Sankofa Project.



Fig. 40: Laser-cut stencils developed by Verônica's students. Source: Personal archive of Verônica Natividade

35. Sankofa is part of the system of pictograms called Adinkra, of the Ashanti culture. It represents a bird that looks backwards while moving forwards. This signifies learning from the past in order to build the future (GÁ; LARKIN, 2009, p. 40).

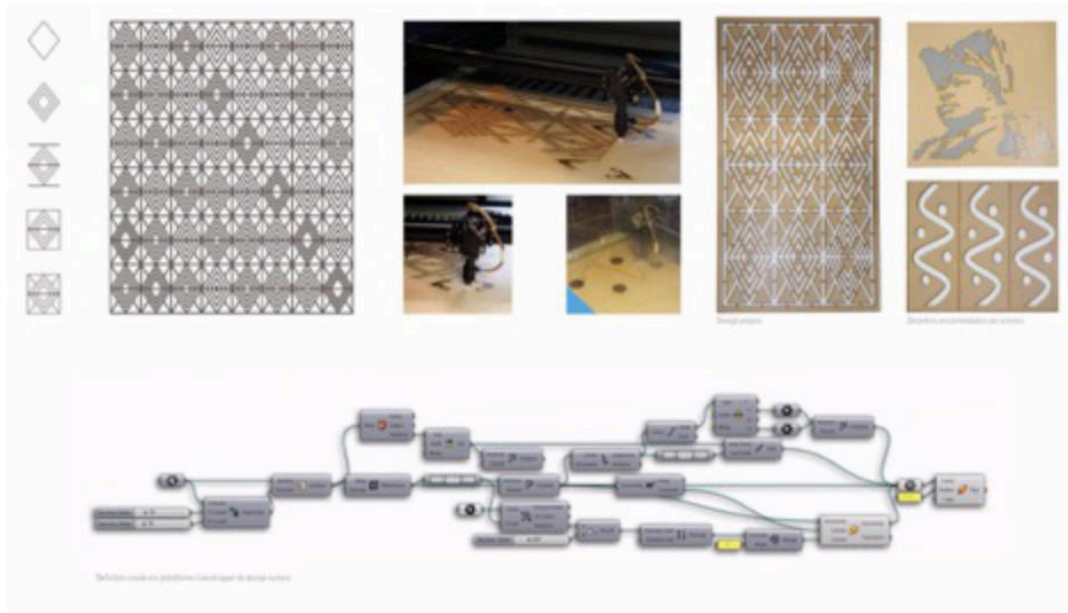


Fig. 41: Development of the fractal pattern from the Grasshopper code. On the right, two other stencils made for the project. Source: Personal archive of Verônica Natividade

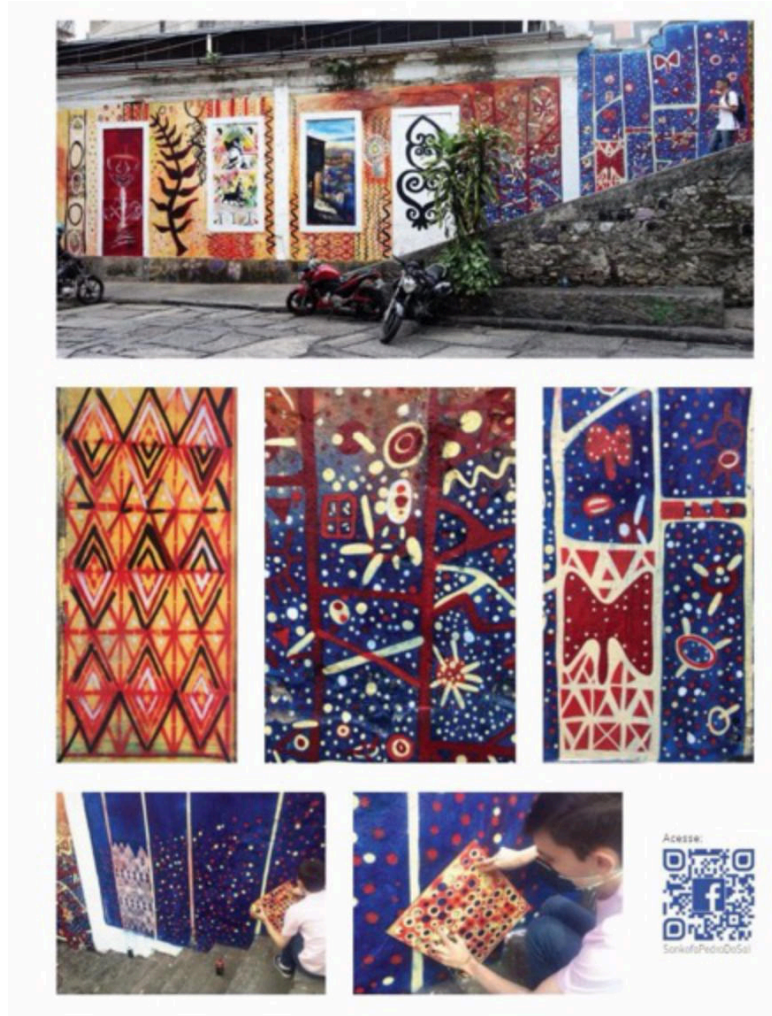


Fig. 42: Photos of the final mural painting and its process. Source: Personal archive of Verônica Natividade

Below is a technical commentary by Veronica Natividade on the design process developed with the students of the Faculty of Architecture of PUC Rio de Janeiro, written specifically for this dissertation:

The students studied and replicated the constitutive logics of the patterns by means of visual programming language with Grasshopper.

The pattern of concentric lozenges was created from a rectangular grid. The subdivision into cells is organised in rows and columns, numbered from 0 to n. Each element of the grid corresponds to a cell with four lines and four vertices, also numbered from 0 to 3. Grasshopper's "tree" organisation facilitates the repetition of operations on similar elements. Therefore, if the command "connect midpoint 0 to midpoint 1" is given, Grasshopper will repeat the operation on all the points named 0 and 1 that are on the same hierarchical level — or, as one says, on the same branch of the tree.

In the case of the lozenge pattern, the midpoint of each line was found first. Then the points were interconnected, generating a lozenge. Each lozenge was scaled x times from the centroid of each cell. The number of scaling operations performed in each cell obeyed a logic of proximity to a reference point — that is, the closer the cell's centroid was to the reference point, the more scaling operations were performed. Finally,

the midpoints of each side of the lozenge were again extracted and connected to the nearest vertex, generating a "cross" between the scaled lozenges.

For the monolinear pattern seen in the right-hand corner of Fig. 41, a sinusoid curve function was used, in which the amplitude and frequency variables were parametrised — that is, instead of establishing a fixed value, a component called a "slider" provided a set of values within a domain. On altering the value of the slider, the curve was automatically redrawn. The centres of the circles that complete the pattern were positioned as a function of the alignment between the maximum and minimum points of the curve obtained. (NATIVIDADE, 2018)

The "Infinity Game" project, however, had a broader approach, visually exploring the crossings between cultures spread all over the world. Through the combination of architectural forms and geometric patterns usually employed for ornamentation, interactive and surrealist representations were created; the intention was thereby to speak of the universality of mathematical language. A shared visual narrative was proposed — a reminder of how important free and open exchange of influence is for the formation of collective identities. The conception of the project began with the study of visual and numerical representations of infinity through the eyes of different cultures, proposing an exhibition that was also a game. A game made of paintings, programming, mapped projection and little infinity boxes. An immersive environment. Visitors were invited to look for numbers hidden in the exhibition space, inside the little infinity boxes. Each number was capable of transforming the projection-mapping installation in a unique way, and if visitors found them all, they won a prize: a laser-cut stencil and an e-mail with the research behind the project.

The interest of analysing this project for this dissertation lies in its proposal of cultural diversity through geometric patterns and in the algorithmic design exercises necessary for the development of the patterns. These were presented both as laser engravings on the little infinity boxes and in the stencils used for the paintings and in the programming of the Processing animations that composed the mapped projections. Besides the three artists mentioned, Veronica Natividade's supervisees at the Faculty of Architecture of PUC Rio de Janeiro took part, in particular Camila Rodriguez and Marcelo Chaves, who had also participated in the Sankofa Project.

The starting point for the development of "Infinity Game" was the choice of which cultures would be used as reference. A major reference for the project and for the choice of geometries was the book *African Fractals* by Ron Eglash (1999), since fractals are a visual representation of the concept of infinity, and Eglash's research revealed the sophistication of the abstract thought behind visual expressions of African origin. This reading inspired the creators of the project to ask how the concept of infinity might also be represented in other cultures, or how to blend the patterns of diverse origins. A survey was then carried out of patterns that could represent the concept, both for their symbolic meaning and for their geometric characteristics, favouring those that constituted fractals, such as the Ethiopian cross. The chosen cultures, roughly, were: Catholic Christianity, Chinese (Han), Ethiopian Orthodox Christianity, Cokwe, Yoruba, Hindu, and Islam (in particular the Nasrid Muslims and their patterns present in the Alhambra). From this choice, an image survey was conducted to be used as aesthetic reference, organised in an album on the Pinterest site under Vanessa Rosa's account (Fig. 43).

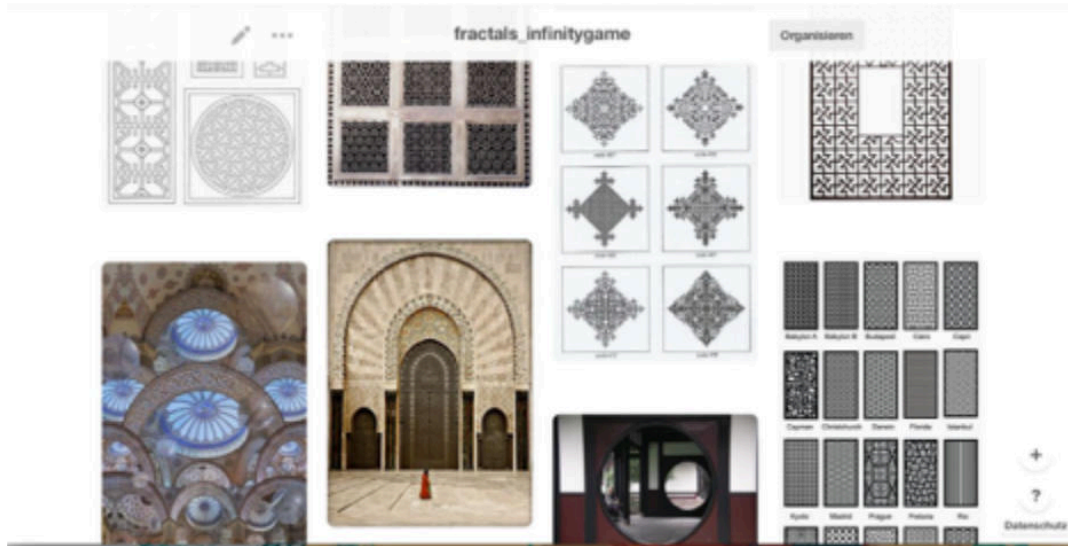


Fig. 43: Screenshot of the album on the social network Pinterest, showing part of the image references used in the project. Source: Personal archive of Vanessa Rosa

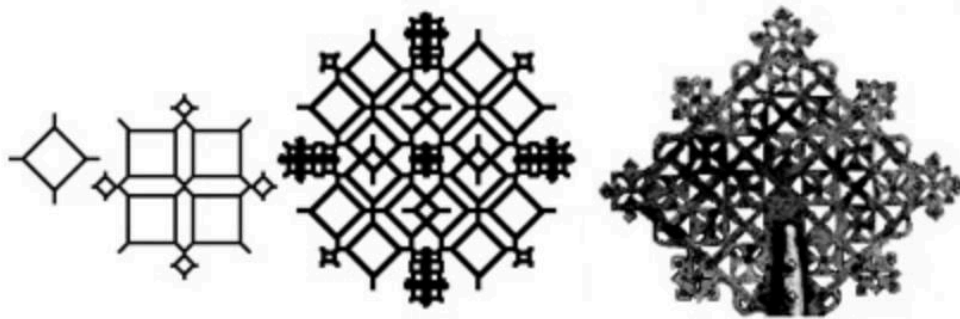


Fig. 44: Fractal logic of the Ethiopian Cross as presented in Ron Eglash's book *African Fractals*. Source: EGLASH, 1999, p. 136



Fig. 45: First pattern studies made in Rhinoceros 3D. One can see a mixture of patterns inspired by Christian, Islamic, Han (China), Ashanti (Ghana), Hindu (India), Cokwe (Angola) and Jewish cultures, and a playful nod to an illustration of an alien. Source: Personal archive of Vanessa Rosa

Several studies were carried out on how the patterns might interact. Fig. 46 presents some of the initial examples, which were not used because they did not present a harmonious interaction — one giving the impression of continuity — between the different geometries.

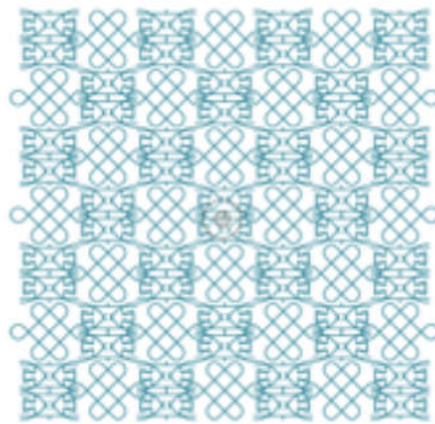


Fig. 46: Interaction of a pattern inspired by Ashanti culture with a Cokwe pattern. Source: Personal archive of Vanessa Rosa

The work then moved on to the exercise of composing fractals.

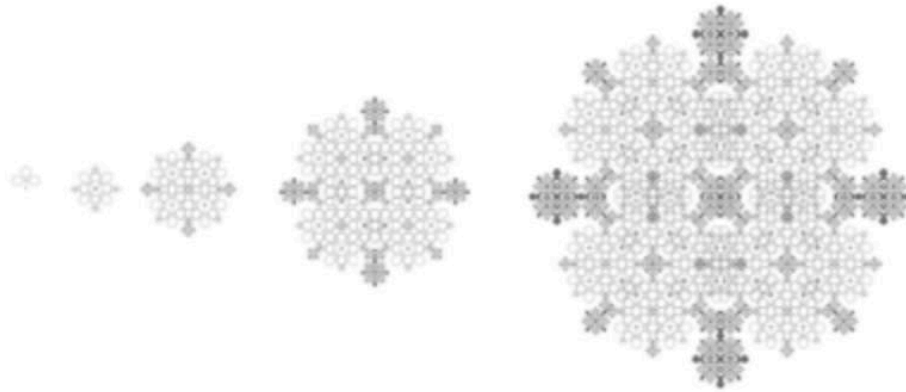


Fig. 47: Fractals developed from a clover-type pattern. Source: Personal archive of Vanessa Rosa

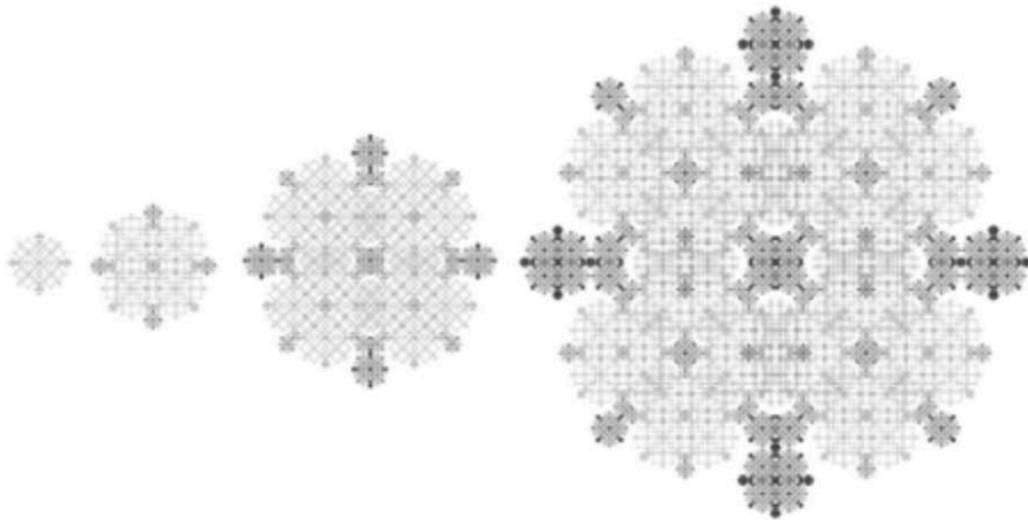


Fig. 48: Fractals developed from an Ethiopian-cross-type pattern. Source: Personal archive of Vanessa Rosa

Below, another commentary by Verônica regarding her design process for the "Infinity Game" exhibition.

The first stage of the work consisted in discovering the generative process behind each chosen pattern or symbol. It was necessary to identify and isolate each element of the drawing, as well as the combination of the parts in the formation of the whole. In some cases, such as the pattern generated from an interpretation of the Ethiopian cross illustrated in Fig. 48, a single element was scaled and repositioned in various ways so as to generate a pattern. In other cases, such as the Arab patterns, the prolongations and intersections of the segments of one figure generated the next figure, and so on successively.

In the first attempt, illustrated in Fig. 46, the basic constituent element of each pattern was drawn in isolation, and the elements were then combined in a standard grid. The procedure proved problematic, because it failed in the continuity between the elements, generating truncated and disharmonious patterns. From this first

experiment, it became clear that it would be more efficient and interesting to combine the generative logic of the patterns with one another, and not their elements. This decision, however, proved equally difficult to implement in Grasshopper. The patterns more grounded in geometric operations, as in the case of the Arab and Chinese patterns, proved more complicated to manipulate than those based on functions, such as the Indian one and the "sona", owing to the very limitations of the software. When working with strictly geometric operations, it is very common for certain "breaks" to occur in the parametrisation owing to loss of reference in the transition from Cartesian coordinates to parametric space. This problem is much easier to get around by using descriptions of the elements as mathematical functions rather than geometric descriptions. (NATIVIDADE, 2018)

These compositions were later used to decorate the little infinity boxes, with the following numbers added: 0, 2, 4, 11, 12, 42, 969, 1729 — chosen both for their mathematical properties and for the symbolic uses of each. 12, 4 and 2 represented constant symmetries in the description of the patterns; 1729 is the smallest number that can be written in two distinct ways as the sum of two cubes, according to the Indian mathematician Ramanujan — a mathematical curiosity. The number 42 was added as a joke, a reference to the popular fantasy-fiction book *The Hitchhiker's Guide to the Galaxy* (ADAMS, 2004), well known within the social context of video game players. 969 is the final age of Methuselah, the oldest biblical character of the Old Testament (Genesis 5:21–27), but it is also an important number for Buddhism, for which it represents the virtues of the Buddha, of Buddhist practices and of the Buddhist community (BOOKBINDER, 2013). Numbers and words that are palindromes are frequently used to represent infinity.

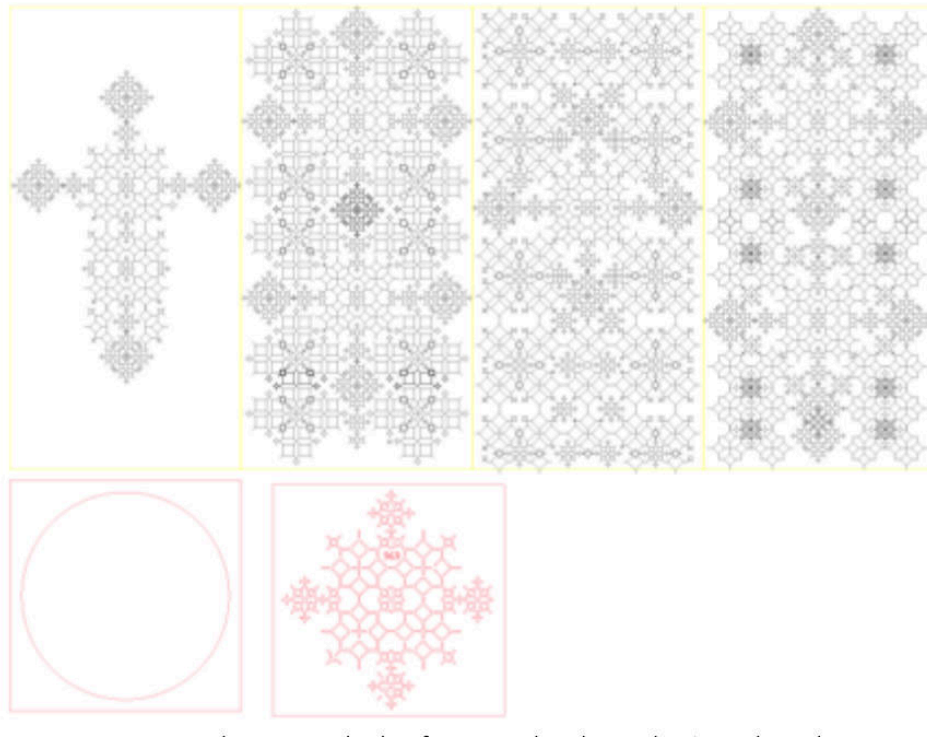


Fig. 49: Layout of a little infinity box, using the algorithmic design based on the Ethiopian cross. Source: Personal archive of Vanessa Rosa

The patterns of Fig. 49 were generated from various operations of scaling and rotation of the basic element of the Ethiopian cross. The method used respected the original logic: 1. Start from a square; 2. Rotate the square 45°; 3. Create a prolongation from each vertex the length of the side of the square, aligned with the corresponding quadrant; 4. Position an original element at half scale at the intersections of the prolongations; 5. Repeat the series. For the composition of the little infinity box, new rules of rotation and scaling were incorporated to generate new drawings. (NATIVIDADE, 2018)

The development of the visitors' interaction with the projections — that is, the game element of the exhibition — was carried out mainly by Wenqi Li. A custom software system was created based on the open source programming language for visual artists called Processing, as well as on the JavaScript language for the commands between server and tablet, and on the projection-mapping software Madmapper. JavaScript and Websocket are used — the latter being "a technology that enables bidirectional communication over full-duplex channels on a single Transmission Control Protocol (TCP) socket" (KAAZING, 2017) — to direct user interaction data to the server and check whether they are the correct answer for each stage of the game. The whole game process takes place on a server. Once the data complete the process, the server sends them to the Processing application. In the latter, various algorithms were used to draw all the patterns generatively, using some basic grid systems inspired by fractals from a given reference. The results were then combined with further generative patterns of different inspirations, using the same grid system in this mixture.



Fig. 50: Little infinity box, design inspired by the Ethiopian Cross. Source: Wenqi Li

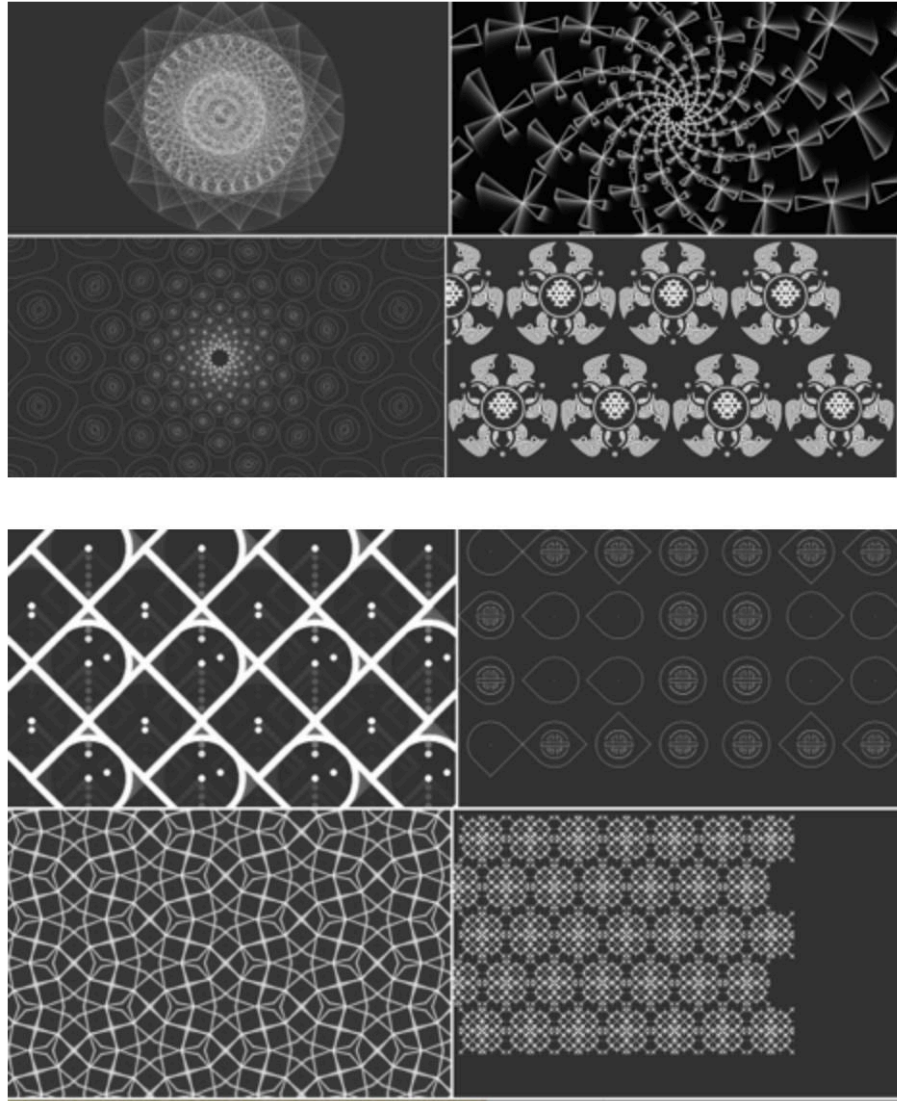


Fig. 51–52: Animations by Wenqi Li made in Processing. Each figure presents 4 different compositions that needed to be placed in sequences of 4 in the line of code to ease the entry of multiple images into the Madmapper software.

Source: Personal archive of Vanessa Rosa

It was possible to access the server and play from any device with internet access; however, a tablet was provided for visitors, simply to facilitate interaction. The game interface had a quite simple structure: it began with the question "Would you like to play the Infinity Game?", giving the option of pressing Y to answer affirmatively and N for the negative. Pressing Y brought up text explaining the rules of the game: find the numbers hidden in the little infinity boxes, type them in one by one, find them all and win a prize. The artists and the gallery staff were responsible for handing over the stencils as prizes. Visitors could also enter their e-mail on the device upon finishing the game in order to receive a PDF with explanations of the research.



Fig. 53: Installation of the "Infinity Game" exhibition. Source: Personal archive of Vanessa Rosa

As for the installation of paintings with mapped projection: painted canvases were nailed directly against the wall, without stretchers. The paintings were made in a mixture of freehand technique and the use of algorithmically designed, laser-cut stencils, which left their marks on the canvases when aerosol paint was applied. There were 5 canvases, each 2.3×1.20 m, arranged so that they appear to continue one another, totalling almost 6 m of paintings. Two IN124ST InFocus projectors were placed approximately 2.5 m from the canvases, fixed to the ceiling. The projectors were connected to Mac Mini computers running the customised software system.

The exhibition was understood by its creators as a successful prototype. The proposal is to continue developing the project, both at the level of historical research and algorithmic design and at the level of visual and interactive solutions. The game mechanic, above all, proved to be something to improve, since visitors did not quickly understand how they were supposed to interact with the projections, and someone was therefore needed to guide them. As for the algorithmic design, the creators judged it important to experiment with more ways of combining patterns from diverse cultures in conceptually interesting ways, going deeper into the possibilities of creating animations from the transitions between them, following a stricter logic between the geometric properties and similarities of those patterns.

In conceptual terms, the project was directly related to the readings and the connection of ideas elaborated in this dissertation. The counterposition between "representative geometry" and "represented geometry" discussed in section 1.2 on the basis of Hans Belting's work (2008, 2011) appears in the paintings on canvas and in the "little infinity boxes", with a dialogue arising between these elements when a number found in the boxes is typed into the tablet and the animated patterns are activated over the paintings. The "little infinity boxes", made of false mirrors with laser engravings, are close to the Arab notion of light as a transcendental element. Geometry would be a way of purifying light and revealing the underlying harmony of the world — a

mathematical and divine harmony. In the original version of the "little infinity boxes", not only would they create an infinity full of patterns enclosed in a small illuminated space, but they would also function like the traditional Arab lattice windows, since their backs are made of stencils. As can be seen in Fig. 56 (end of the chapter), the light passes through the back and creates new patterns on the floor, made only of lights and shadows that recreate the cut-out forms of the stencil with the effect of an infinity of reflections, the fruit of the mirrors. A 60W halogen light was chosen, for its luminous quality and for giving better results in keeping the shadows sharp. In the context of the "Infinity Game" exhibition, however, there was too much light in the space for this effect of the boxes to be visible.

The paintings, in turn, are an allusion to the idea of "representative geometry", putting the illusory effect of linear perspective into evidence. To this end, photographs were chosen of architectures belonging to the selected cultures — always images with pronounced depths, with strong counterposition between several planes (Fig. 55). To the drawing of the architectures were added countless patterns (belonging to the same cultural complex as the building) made from the stencils of patterns distorted in perspective, cut by laser. Many traditional architectures were covered in detailed ornamentation, but the interest of this practice lay in the expressive effect resulting from the intensification of the illusion of depth that these patterns in perspective provided. There is also the sensation that the architectures are built not of solid materials, but that the patterns constitute them.

If Renaissance paintings were thought of as windows onto another world, framing a scene — a theatrical play condensed into a single frozen image, in Alberti's vision (BELTING, 2008, p. 102) — in the case of this project, the paintings in perspective are portals to an undefined world. Perspective itself, the illusion of entering the canvas, is the focus of this practice of painting as representation, always with diffuse horizons at the end of the countless inclined planes. The mapped projection enters precisely at these diffuse horizons, filling them with more patterns — animated, alive, mutating and interactive.

The whole aesthetic of the project is based on religious symbols and centuries-old geometries present in temples. In working with the aesthetics of spaces of power — such as the temples and palaces made of materials noble enough to survive for many centuries until they could be photographed — one notes the centrality of religious symbolism. According to the Togolese philosopher Yaovi Akakpo, such symbolism, as a basic element of social organisation and cohesion, frequently stands in contrast to the notion of rationality, the "organisational paradigm that is the incarnation of rationality in relation to an end"³⁶ (AKAKPO, 2016, p. 20), an essential characteristic of modern societies. Another central attribute of such societies, also a fruit of this notion of rationality, is the power of technoscience as organiser of all activities — a process ever clearer with digital management and design tools, which are necessarily based on algorithms and databases. Algorithms can be created to describe a magical procedure or a pragmatic simulation of a given phenomenon of physical reality. Algorithms do not carry within themselves, as a method of describing operations, a worldview. The same can be said of mathematics, although it is constantly connected to a notion of universality disconnected from cultural particularities.

36. "La société moderne met en honneur un paradigme d'organisation qui est l'incarnation de la rationalité par rapport à une fin." (AKAKPO, 2016, p. 20–21) — Translated from the French.

Therefore, the combination of traditional techniques such as painting and light drawings based on the principles of the lattice windows, of sacred geometries and authorial creations inspired by such forms, with interactive software, projectors, tablets, the internet and the other devices necessary to create the "Infinity Game", was a way of bringing forth, in terms of images, this complex of symbols of human history. A history understood without a beginning, without an end, rich in its contradictions.

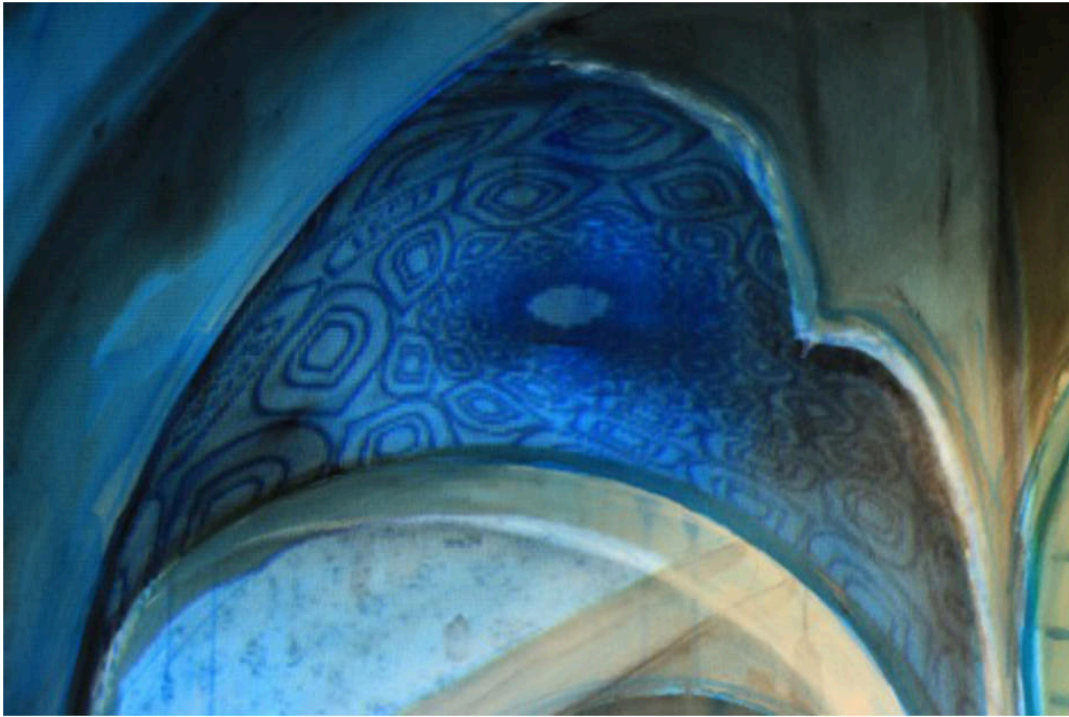


Fig. 54: Detail of the installation of the "Infinity Game" exhibition, painting with video mapping. Source: Personal archive of Vanessa Rosa



Fig. 55: Layout of the installation composed of paintings on canvas. Source: Personal archive of Vanessa Rosa

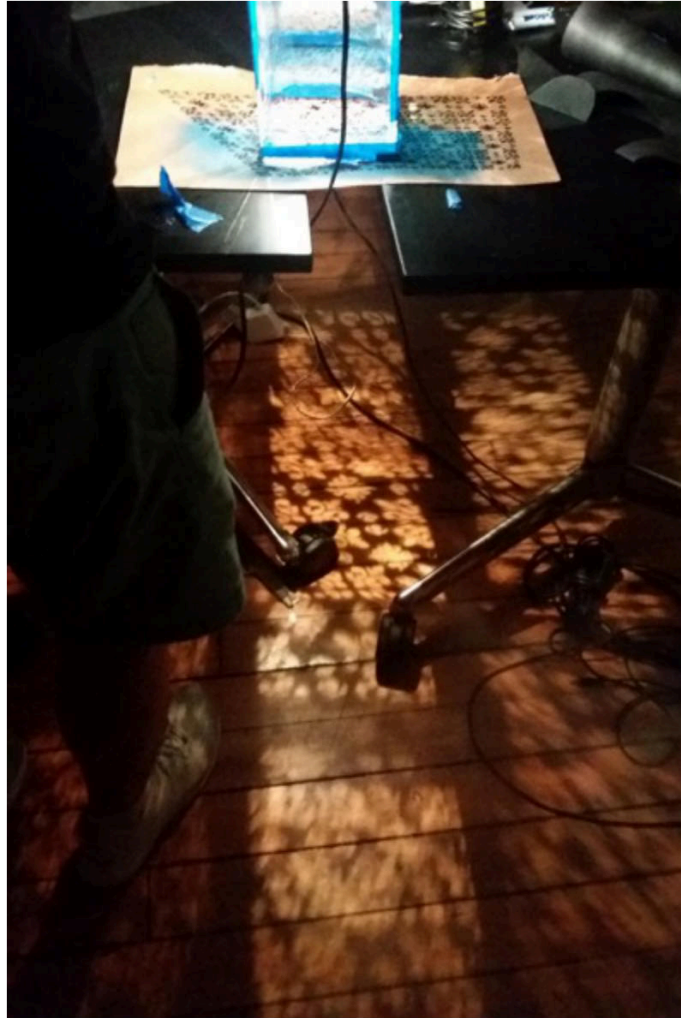


Fig. 56: Light tests for the "Little Infinity Boxes". Source: Personal archive of Vanessa Rosa

FINAL CONSIDERATIONS

In the book *The Second Digital Turn*, Mario Carpo (2016) argues that the alphabet is a system of information compression of great utility for the industrial model of production. Using only some 30 symbols as the elements of a typewriter or a keyboard was a considerable advantage over the ideogram systems of Eastern languages. However, to the extent that today's computational possibilities offer us ever less need for information compression in order to produce objects and content, Carpo asks whether linguistic systems with a greater diversity of constitutive symbols might present some kind of advantage in terms of expressive possibilities. The researcher gives as an example the thousands of emojis available on smartphones today, used in very inventive ways, principally by the young, for communication (Ibidem, p. 22).

What is relevant in Carpo's example for this dissertation is that it raises the question of how diverse models of communication, based on distinct cultures, interact with the implementation of new technologies. Each cultural system could bring distinct advantages or perspectives that generate other ways of thinking and of developing technologically. If Chinese ideograms still need to be transcribed into Western characters through the system called pinyin in order to be typed on a current keyboard, this need is transformed as new writing tools are developed, such as voice-to-text transcription or the refinement of touchscreen drawing tools. Carpo notes that systems of communication through gestures may also flourish anew in the digital medium (CARPO, 2016, p. 23). Ideograms, letters and gestures are simply distinct systems; each may present its advantages. The interesting move, then, would be to defend the diversity of models and tools as a way of stimulating the possibility of innovation. In consequence, this ends up encouraging challenges to the Eurocentric model of knowledge.

When we speak of the essential role of algorithms in the models of representation and projectivity of contemporary society, bringing forth the enormous cultural diversity behind algorithmic systems may be relevant not only in terms of a more egalitarian political representation, or of intercultural intelligibility in Gerdes's vision, but also in terms of potential for innovation. If we are living through a tendency towards the mathematization of the world, it becomes essential to be able to question, in every possible way, what this mathematization means and in what ways it may yet have an impact on our reality. It is of great relevance to think of technology as a human production within the expanded field of culture — one that does not exist autonomously, and is therefore always influenced by transformations of the social context.

Taking into consideration the writings of the Gershenfeld brothers (2017), we need as many people as possible engaged in playing an active role in the design of the future we desire, above all in the development and implementation of the technologies that will give shape to the social interactions to come. To think mathematics, or the processes that create images and objects, from the point of view of cultural diversity — in other words, of the diversity of narratives and premises existing in our multicultural society — becomes a defence of values of equality for humanity. We are living through a period of effervescence in the flourishing

of discourses on multiculturalism, which becomes clear when the richest nation in the world, the United States of America, shows, according to analyses of its population censuses,³⁷ a clear demographic tendency towards becoming a truly multi-ethnic country, without an absolute majority of Caucasian ethnicity (Fig. 57). This process, however, does not occur without struggles and without setbacks, as the positions of the current American government administration reveal through its policy of immigration restriction (JORDAN, 2017), or the recent visibility of the movements called "white supremacy" (NEWKIRK, 2017).

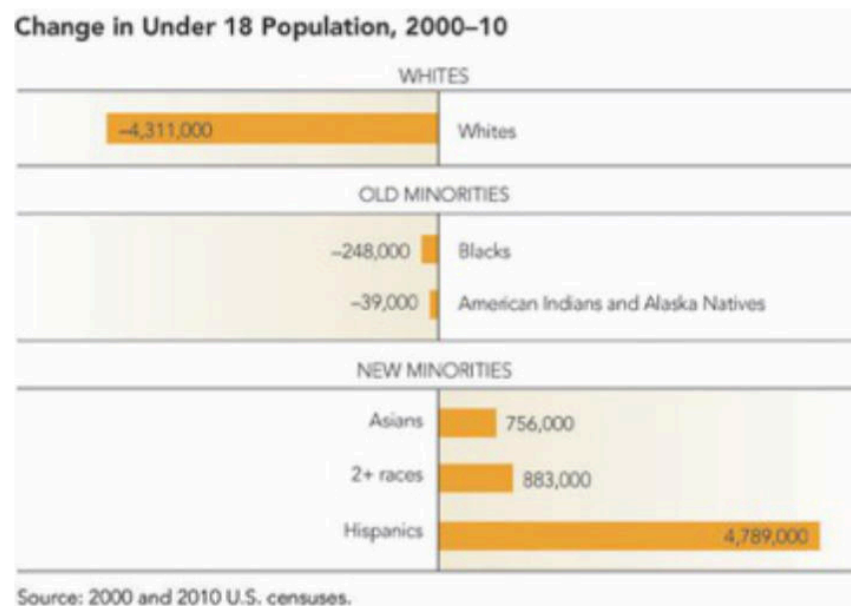


Fig. 57: Demographic changes in the USA. Source: <https://newrepublic.com/article/120370/five-graphics-show-why-post-white-america-already-here>

To conclude, we return to a work presented in the "Infinity Game" project (Fig. 58). A "little infinity box" whose sides were laser-engraved with algorithmically designed patterns based on the Ethiopian cross. Infinity is enclosed in an object, a finite space, observable from any angle around the object. The patterns stand out because the laser engravings reflect more light than the mirrored sides. The patterns themselves are a geometric representation of infinity. Light is one of the central expressive elements of the work. Given the cultural context of the patterns and the very composition of the work, its spiritual character is clear; but because it is presented together with other "little infinity boxes", each designed with the patterns of one religion, the notion of spirituality appears less as the affirmation of a specific religion than as a common quest present in diverse cultures. In some "boxes" there is also a combination of symbols of distinct religions, such as the Islamic with the Christian cross, in a reference to the historical and philosophical exchanges of those cultures.

37. References: 2010 Census Demographic Profile Summary File. U.S. Department of Commerce, Economics and Statistics Administration, U.S. Census Bureau. Articles analysing the results have been published (CNN WIRE STAFF, 2011). One of the points raised by the census and discussed in the cited CNN article is that the population growth of the white ethnic group in the United States rests on the contribution of Latino groups who identify as white. Whites of Latino origin, however, represent a political group quite distinct from whites of Central or Eastern European origin.

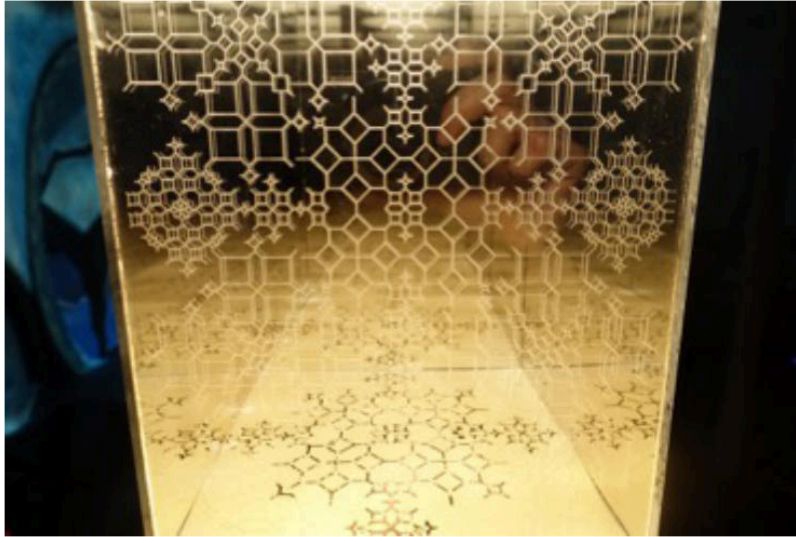


Fig. 58: Little infinity box, design inspired by the Ethiopian Cross. Source: Personal archive of Vanessa Rosa

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